

Elements of Value Exploratory Analysis

Research Report

Product Experience Research,
Samsung Electronics America

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Key Evidence-driven Takeaways from Exploratory Research

- LO owners (especially males) show the highest purchase intent of book-format (inner fold design) and some do not mind paying more. Durability is a more important driver than price as foldable purchase driver.
- Gen Z and Baby Boomers have the lowest purchase intent of foldables.
- LO female smartphone loyalty declines more rapidly with falling satisfaction. The primary drivers of dissatisfaction are camera, chassis material, size, refresh rate, consistent usability, battery capacity, one-hand use and fast charging.
- The top 10% drivers of smartphone satisfaction (out of 43 tested) are camera, consistent usability, storage capacity and look and feel.
- Samsung has been losing Gen Z to LO in Video Call, Messaging, Tracking, Smart Watches and Wireless Earbuds.
- LO native Health, iCloud, Maps and Tracking App (Find My) have a higher adoption for females compared to corresponding apps in Samsung and Android.
- Strengths for Samsung are Camera, Camcorder, Photo Apps, Gaming, Health/Fitness, Productivity (MS) and Smart TV for Samsung users (Samsung Smart TV is also highly rated by all brands).

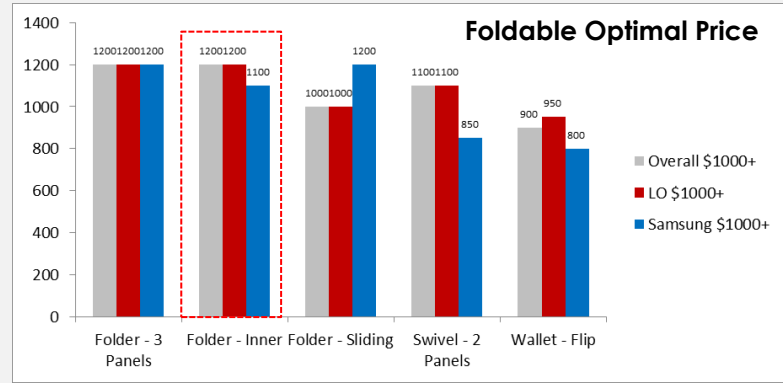
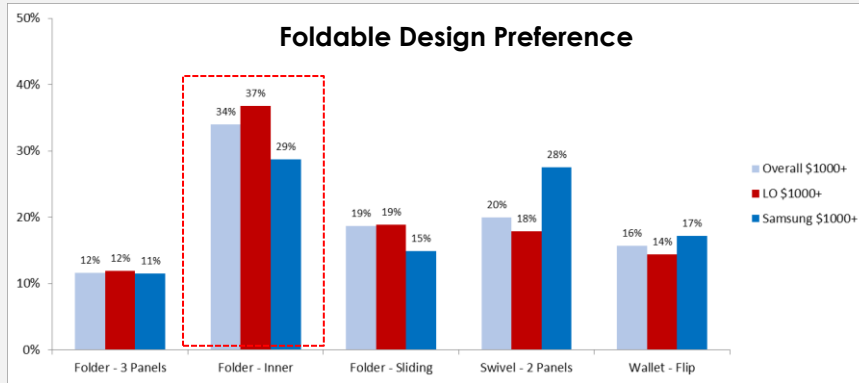
Foldable Designs and Pricing

Samsung should market book-format inner folder aggressively (similar to Galaxy Fold2 5G) to LO is Hi-premium and Hyper-premium tiers as it is popular with LO owners. They also are more inclined to pay more than Samsung owners for foldables with such design.

Book-format inner folder (similar to Galaxy Fold2 5G) is the most popular form factor among the different designs. Samsung high-premium and hyper-premium tier shows higher interest in Swivel form factor than LO while LO 1000+ tier shows more interest in book-format inner folder.

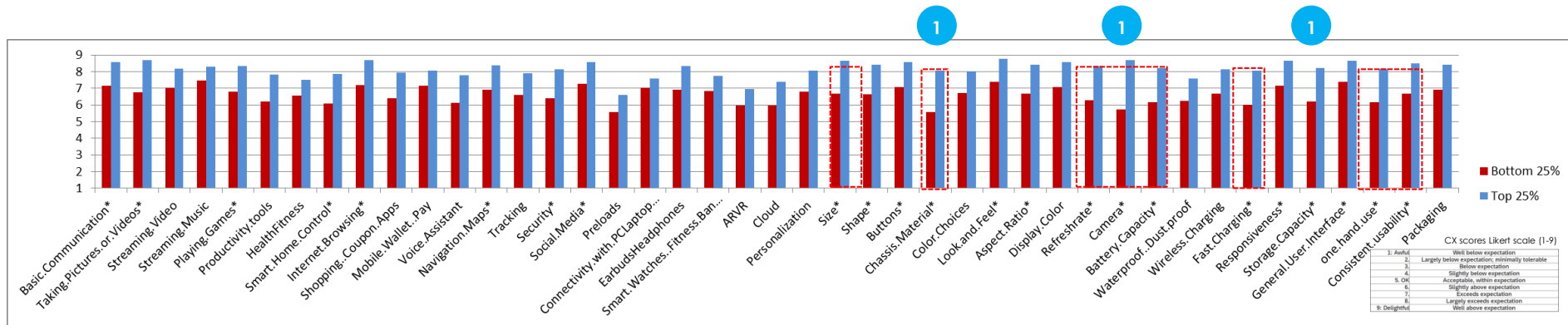
The optimum price of inner folder of LO 1000+ tier owners is higher than Samsung owners while Samsung owners may be willing to pay more for foldables with rolling/sliding display.

(Van Westendorf Pricing model)



Caution: small sample size with some of the less-preferred designs

Switching LO Female with low loyalty to Samsung



Top loyal vs Bottom loyal LO female

The satisfaction scores of 23 out of 43 apps, design, hardware, software and UI features from LO female (N=51) with top 25% of loyalty and smartphone satisfaction scores are significantly higher than LO female (N=25) with bottom 25% loyalty and satisfaction scores.

t-test unequal samples with Bonferroni correction)

Top Dissatisfaction Drivers

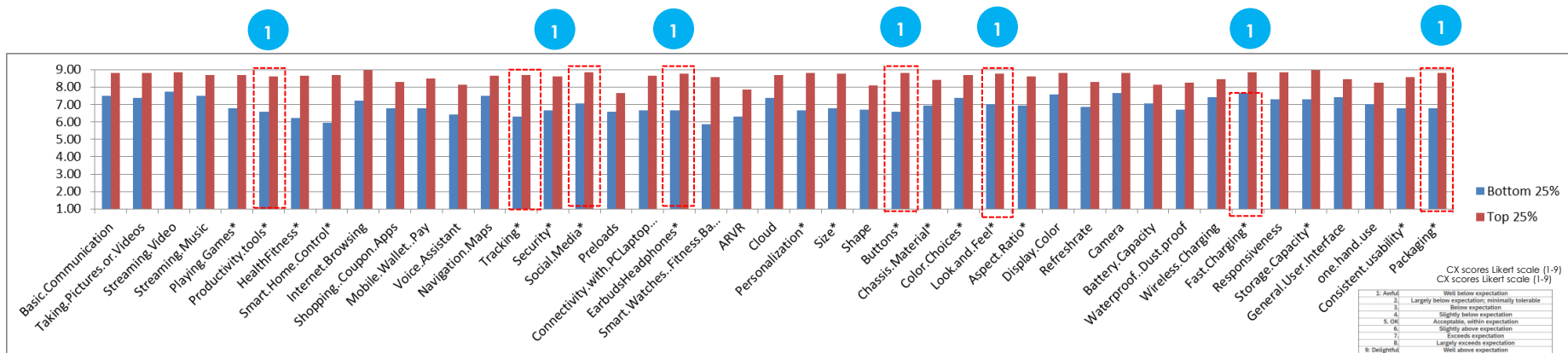
The top 8 drivers* are identified as Camera, Chassis Material, Size, Refresh Rate, Consistent Usability, Battery Capacity, One-Hand Use and Fast Charging

*statistically significant dissatisfaction drivers.

Recommendation

With iPhone 12 improvement of camera and chassis material and availability of different sizes, Samsung may want focus on **refresh rate and battery capacity** as marketing message to LO female.

Keep Samsung High and Hyper-Premium Tiers from Churning



Top loyal vs Bottom loyal Samsung 1000+

The satisfaction scores of 21 out of 43 apps, design, hardware, software and UI features from Samsung 1000+ (N=20) with top 25% of loyalty and smartphone satisfaction scores are significantly higher than those with bottom 25% loyalty and satisfaction scores (N=14).

t-test unequal samples with Bonferroni correction)

Top Dissatisfaction Drivers 1

The top 8 drivers* are identified as Packaging, Look & Feel, Buttons, Tracking, Health/Fitness, Earbuds/Headphones, Social Media and Productivity Tools.

*statistically significant dissatisfaction drivers.

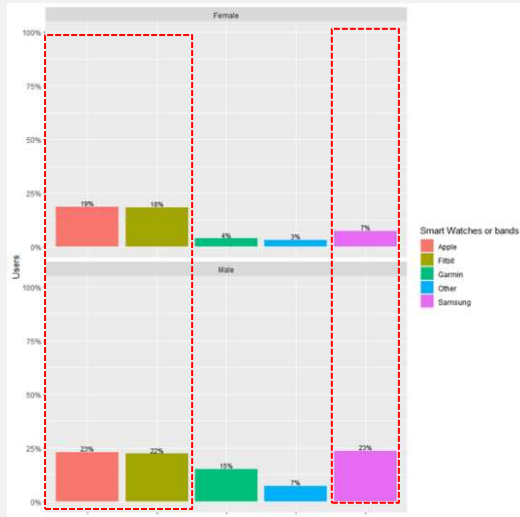
Recommendation

Packaging is also identified as top satisfaction driver for Productives. It seems to play a surprising role in driving satisfaction for high-tier users. Both Tracking and Earbuds have also been identified improvement areas for Gen Z and females and should receive priority in design improvement.

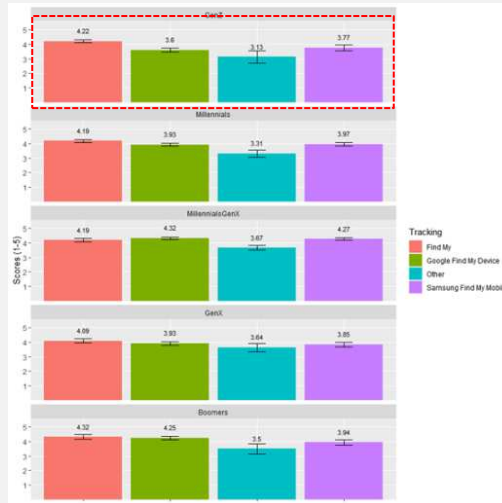
Design for Diversity to capture more females and Gen Z

Samsung can further **embrace design for diversity** to win more prospective female and Gen Z users, with holistic and emotional design of both hardware, software and native apps design with focus on **Video Call, Messaging, Cloud, Maps, Health, Tracking, Smart Watches, Wireless Earbuds.**

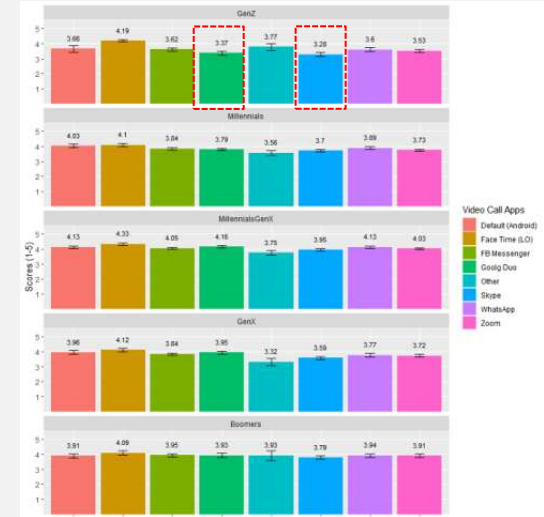
LO and Fitbit Female smart watches market penetration is substantially higher than Samsung



Gen Z rate CX of Samsung Find My Mobile substantially lower than LO Find My.



Gen Z rate CX of Google Duo and MS Skype second last and last



Overview and Recommendations

Foldables Purchase Interest

Group	Findings	Commentary
Brand	• LO owners show the highest purchase intent of foldables	• Foldables can be instrumental in switching LO users to Samsung (when foldables are perceived having similar reliability to slate form-factor smartphones).
Gender	• Males are substantially more likely to purchase foldables than Females	• One way to gain strategic advantage over LO in longer term is to create foldables that appeal more females
Age	• Both Baby boomers and Gen Z are less likely to purchase foldables	• The short-term goal of foldable can be focused on the increasing adoption of Millennials and Gen X.
Tier	• Hyper-premium tier show the highest purchase intent of foldables followed by high premium	• Generally, hyper-premium tier do not mind paying more for foldables.
Segment	• Productives have highest purchase intent than any other segments, followed by Expressers	• Productives in the hyper-premium tier are the natural targets for foldables.

**LO owners
(especially male)
have the highest
purchase
likelihood of
foldables and
some also expect
to spend more.**

"Because this will be featured with many new things with a bigger display." **Male iPhone 11 Pro, 35 - 40.**

"Because of the best innovation I will be willing to pay any amount of money." **Male, iPhone SE 2020, 35-40**

"I don't mind spending more for foldable phones because it's all in all. I can also use it as a slate smartphone." **Male, iPhone 11 Pro, 41-44**

"It's a latest smartphone, I am comfort with this, I like it." **Male, iPhone 11 Pro Max, 35-40**

"It's worth the cost." **Female, iPhone 11 Pro Max, 25-34**

"Because I believe it has more hardware inputs factored into it's foldability. As a user, it can serve as a tablet and phone; hence, it is worth it." **Male, iPhone XS Max, 25-34**

Many Gen Z users do not like or do not need foldable phones. Some also kept both smartphones and tablets for separate purposes.

"I feel like a tablet and smartphone being separated would be better to keep it more organized." **Female, iPhone 11, under 24**

"I have everything I need to cover my needs."
Male, X20 Ultra, under 24

"I use my tablet for work and sometimes need to be able to access multiple devices at once." **Female, iPhone 11 Pro, under 24**

"I don't like the folding design." **Female, Note 20 5G, under 24**

"Tablets are more convenient." **Male, iPhone XR, under 24**

"I like having dedicated devices and the ability to multitask with both at the same time for efficiency."
Male, iPhone 6, under 24

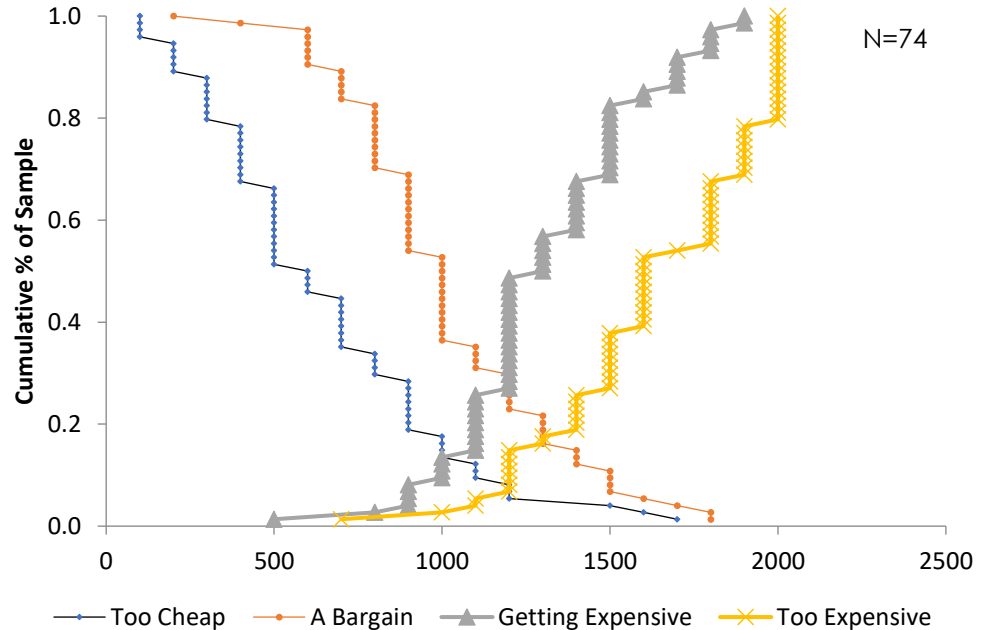
Inner Folder Pricing: LO High & Hyper Premium

If the price of foldable (with inner folder design) is over \$1,300, more than 20% LO high and hyper-premium tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 10%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,300
Optimum Price	\$1,200
Indifference Price	\$1,200

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



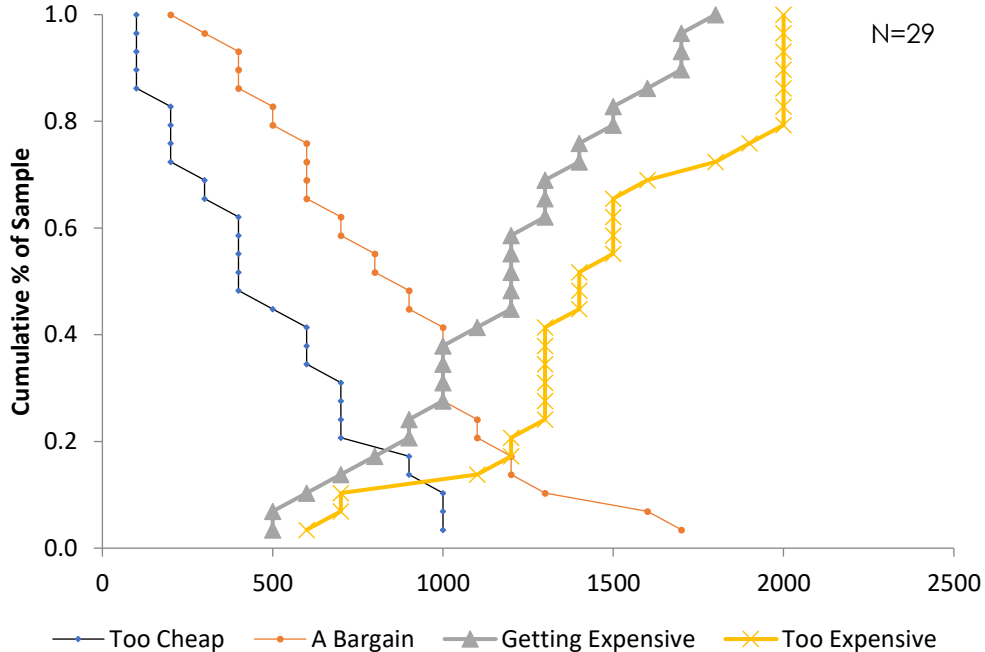
Wallet Foldable Pricing : LO High & Hyper Premium

If the price of foldable (with inner folder design) is over \$1,200, more than 15% LO high and hyper-premium tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 15%.

Price Sensivity Meter	
Acceptable Range	\$800 to \$1,200
Optimum Price	\$900
Indifference Price	\$1,000

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price – "too cheap" and "too expensive" intersection
 Indifference Price – "bargain" and "getting expensive" intersection



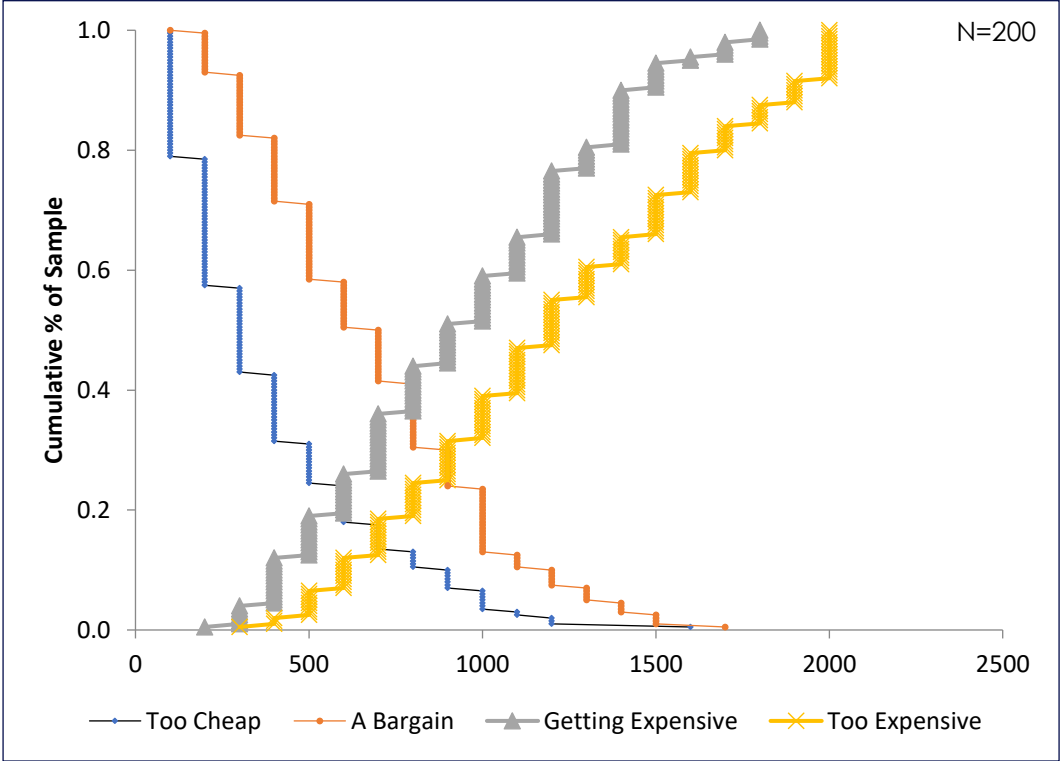
Foldable Pricing – Millennials

A base premium folder priced at around \$700 will likely have the best appeal to Millennials (and catch all for general market).

Price Sensitivity Meter	
Acceptable Range	\$600 to \$900
Optimum Price	\$700
Indifference Price	\$800

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price – "too cheap" and "too expensive" intersection
 Indifference Price – "bargain" and "getting expensive" intersection



Loyalty & Satisfaction: Findings & Commentary

Group	Findings	Commentary
<u>Brand</u>	LO loyalty is higher than Samsung and other Android owners at all satisfaction levels	This trend has persisted since the inception of smartphones and is unlikely to change substantially in the near future.
<u>Age</u>	Baby boomers are less loyal at almost all satisfaction levels.	Boomers can be deprioritized for experience design of high-tier smartphones.
<u>Tier</u>	- Loyalty of High Premium Tier declines more gradually than others with lower satisfaction scores - Loyalty of Value and Mid Tier are lower than other tiers across all satisfaction levels.	This trend is only observed with LO high-premium tier, Samsung high-premium loyalty decreases more rapidly with decreasing satisfaction. High Premium is probably one of the more vulnerable tier for Samsung. A possible approach is to get as many high-premium to migrate to hyper-premium and the rest to mid-premium.
<u>Segment</u>	- Productives have higher loyalty across most satisfaction levels .	Samsung may adopt a defensive approach to this segment, sustaining and improving the experience for this segment.

Loyalty & Satisfaction: Findings & Commentary

Group	Findings	Commentary
Brand Gender	• LO female loyalty drops steeply with decreasing satisfaction while LO male loyalty remains higher than other brands.	• Targeting LO female with low satisfaction may have better chance to convert LO users to Samsung
Brand Age	• Samsung loyalty is lower than LO with decreasing satisfaction in age groups 35 or above	• It is less likely for LO users to switch to Samsung in this age range. Samsung may want to adopt a defensive approach with these age groups.
Brand Tier	• LO High-Premium loyalty is most resistant to drops in satisfaction. • Samsung Hyper-Premium loyalty is lower than LO but drops less steeply than LO hyper-premium with decreasing satisfaction	• This can become a dangerous trend for high-premium Samsung smartphones should Samsung missteps. The same may apply to LO hyper-premium. Offering a variety of hyper-premium products (e.g., foldables) may help to combat these trends, moving some high-premium tier to hyper-premium.
Brand Segment	• LO Social Savvies Loyalty is higher compared to both Samsung and Android owners at different satisfaction levels.	• Focus on developing attractive experience specific to social savvies (e.g., selfie camera experience) will be needed to keep Samsung competitive with Social Savvies
Age Tier	• Gen Z Hyper-premium loyalty falls rapidly with dissatisfaction	• It can be challenging to satisfy the needs of Gen Z with their higher expectation than other age groups as reflected in the assessment of elements of value.

Satisfaction & Apps/HW/UI Customer Experience

Overall, the highest (top 10%) common apps/HW/UI features that contribute to overall satisfaction are:

- Camera
- Consistent Usability
- Storage Capacity
- Look and Feel

Relative contribution (%) to satisfaction of current smartphone

Features	Overall	Female	Male
ARVR	-0.97	-2.98	2.32
Aspect.Ratio	-0.17	-1.27	1.30
Basic.Communication	4.70	3.84	4.34
Battery.Capacity	2.15	3.73	-1.99
Buttons	1.27	0.30	1.77
Camera	7.25	7.82	3.24
Chassis.Material	1.88	1.74	1.55
Cloud	-0.01	1.53	-3.30
Color.Choices	3.90	3.40	2.67
Connectivity.with.PCLaptop..Tablet	-1.50	-2.66	0.89
Consistent.usability	6.67	6.38	4.45
Display.Color	4.61	2.40	6.24
EarbudsHeadphones	-0.65	-1.14	-0.06
Fast.Charging	1.90	4.07	-2.73
General.User.Interface	3.30	4.13	0.49
HealthFitness	-1.54	-2.42	0.24
Internet.Browsing	3.04	3.38	0.60
Look.and.Feel	4.94	2.52	6.94
Mobile.Wallet..Pay	-0.91	0.36	-2.81
Navigation.Maps	3.13	1.58	4.06
one.hand.use	-1.50	-0.82	-2.41
Packaging	2.92	1.06	4.27
Personalization	-0.01	-0.35	0.10
Playing.Games	1.18	1.06	1.03
Preloads	-0.67	1.16	-3.37
Productivity.tools	-0.57	-1.00	0.18
Refresh.rate	3.15	3.42	0.73
Responsiveness	4.23	4.13	2.73
Security	4.64	2.64	5.97
Shape	1.30	0.51	2.03
Shopping..Coupon.Apps	-0.93	1.07	-4.44
Size	3.25	2.36	3.25
Smart.Home.Control	-0.73	-0.27	-1.48
Smart.Watches..Fitness.Bands	-1.41	-2.43	0.53
Social.Media	3.15	2.81	1.95
Storage.Capacity	7.98	7.51	5.38
Streaming.Music	-1.66	-1.89	-1.18
Streaming.Video	0.79	0.65	0.53
Taking.Pictures.or.Videos	4.52	4.91	1.69
Tracking	-0.47	-0.59	-0.58
Voice.Assistant	-0.19	0.96	-2.47
Waterproof..Dust.proof	0.23	-0.29	0.75
Wireless.Charging	0.04	0.45	-0.96

Elements of Value: Most Important Values and Gaps

Group	Findings	Commentary
Overall	Top values: Connects, Variety, Simplifies, Fun, Aesthetics, Reduces Anxiety, Motivation, belonging, Actualization	Using rating 7 as an arbitrary baseline to gauge , “Makes Money” , “Reduce Cost” , “Reduce Anxiety” , “Nostalgia” , “Therapeutic” and “Heirloom” score lower than baseline .
Gender	Top functional, emotional and life-changing values similar to overall	Females select “Connects” and “Belonging” more often than males while Males select “Actualization” more often
Age	Top functional, emotional and life-changing values similar to overall	- Boomers value “Informs” more and “Hope” less than other generations “Aesthetics” is strongest with Gen Z
Brand	Top functional, emotional and life-changing values similar to overall	“Reduce Cost” and “Make Money” are selected less often by LO users and they value “Wellness” a bit more. “Actualization” is strongest with Samsung owners
Tier	Top functional, emotional and life-changing values similar to overall	“Reduce Effort” is higher and “Fun” is lower for Super Premium. “Badge Value” is lower with Value and Premium Lite Tier
Segment	Top functional, emotional and life-changing values similar to overall	“Transcendence” is higher for Productives while “Variety”, “Connects”, “Fun”, “Aesthetics” and “Belonging” are lowest

Basic Apps: Findings & Recommendations

App	Findings	Recommendation/Commentary
Video Call	Gen Z rates Google Duo and MS Skype lowest. - Facebook is the most widely used video chat app (74%) - CX scores for FB are consistent across all age groups with reasonably competitive scores	- Short Term: Work with Google to improve CX of Google Duo especially for Gen Z - Long Term: Consider forging strategic relationship with Facebook to integrate Facebook Messenger Video Call for wide penetration and usage of video call - Evaluate other emergent apps (e.g. Google Meet)
Messaging	- LO iMessage bests CX scores, followed by Android default Messages and FB Messenger FB Messenger is most popular across brands, age, product tier and segment	Integrate or preload FB Messenger into the Samsung mobile UX to reduce friction in messaging and improve UX across different platforms and demographics
Email	- Gmail tops CX scores vs. all other email apps - It also has the highest market penetration	- Consider deprioritizing Samsung email app - Improve UX if app is needed for strategic reasons
Browser	LO Safari has the highest CX scores, followed by Chrome. Gen Z rates Samsung Internet low. - Google Chrome has the highest market penetration (89%)	- Consider deprioritizing Samsung Internet app - Improve UX (for Gen Z, Gen X and Boomers) if app is needed for strategic reasons
Social Media	Facebook and Instagram have the highest CX scores and market penetration (86% and 72%)	Consider forging a more strategic relationship with FB to integrate their social media, messaging and video call experience to provide seamless and frictionless UX.

Photography & Videography: Findings & Recommendations

App	Findings	Recommendation/Commentary
Camera	• LO and Samsung Camera UX scores are comparable and outperform other Android smartphones rated by brand owners • Camera features usage and adoption is consistent across brands • CX scores of Macro, Night Mode and Zoom are especially low for value and mid-tier devices	• Focus improvement effort on night mode, macro and zoom performance
Camcorder	• CX scores of Stabilization, Night Mode and Special Effects are low among females. • Gen Z, Millennials and younger Gen X use many camcorder features heavily.	• Focus improvement effort on stabilization, night modes and special effects
Photo Apps	• Samsung Gallery is top-rated by Samsung users and tie with LO Photos by LO users. Google Photos follow closely • Samsung Gallery usage is very high and comparable to LO Photos (both over 90%) • Google Photos has a strong presence across platforms. • Usage of both Snapseed and VSCO increase rapidly with higher tiers	• To maintain competitiveness, especially for Premium and Super premium devices, Samsung should consider building features supported by 3rd party apps such as Snapseed (through collaboration with Google?) or VSCO into the Gallery app.

Entertainment Apps: Findings & Recommendations

App	Findings	Recommendation/Commentary
<u>Gaming</u>	• LO Arcade penetration decays rapidly with age. • MS Xbox subscription penetration is strong from Gen Z through Gen X • MS Xbox subscription CX scores are competitive with LO Arcade for all age groups	• Bring MS Xbox experience to future premium and “premium lite” devices to further capture the market.
<u>Streaming Music Services</u>	• Spotify has the highest CX scores, followed by LO, Amazon Music and Pandora. • Spotify has the highest adoption for younger age groups while YouTube Music and Pandora have strong presence for all age groups.	• Consider forging strategic relationships with music services beyond Spotify with the consideration of popularity of competitive music services with different age groups (especially for Gen X and Boomers where LO Music has a low penetration)
<u>Paid Streaming Video Services</u>	• Netflix, Amazon Prime Video, Hulu and Disney+ are the top 4 streaming video providers with high CX scores and adoption. • Subscriptions of niche video services increase with higher tiers	• Explore unique opportunities for bundling less known but attractive services to super premium tier to micro target prospective users
<u>Free Video Streaming</u>	• YouTube is by far the top free video streaming service with substantially higher CX score (4.21) and high market penetration (91%). • Need for free video streaming services is common across all tiers	• Leverage TV Plus service on Samsung Smart TV and provide better MDE CX between the mobile TV Plus service with Samsung Smart TV to promote cross sales opportunity

Practical Apps: Findings & Recommendations

App	Findings	Recommendation/Commentary
Mobile Pay	• LO Pay has the highest CX score, followed by Google Pay and Samsung Pay. • LO Pay has the highest adoption from Gen Z and Millennials while Samsung Pay is more widely adopted by Millennials & younger Gen X • The adoption of Google Pay is consistent across all tiers compared to LO Pay and Samsung Pay	• Expand Samsung Pay to entire product tiers to capture under-indexed Gen Z and Boomers
Health & Fitness	• LO and Samsung Health top CX, followed by FitBit Coach, MyFitnessPal Google Fit. • Samsung Health (and most other health apps) have a ratio highly skewed towards male (LO is much more balanced)	• Make the design and features of Samsung Health more appealing to females. Recent addition of female health features are big strides in that direction
Navigation, Maps	• Samsung users rate CX score of Android Auto substantially better than other Android users. • Google Maps has substantially higher CX scores than all other navigation maps. • LO Maps is popular with LO females	• Continue to improve Android Auto with Google
Cloud Storage	• Google Cloud, Amazon Cloud and Samsung cloud CX scores are comparable. • Both MS OneCloud and Google Cloud are more widely used by Gen Z, unlike Samsung Cloud • LO iCloud is popular with LO females	• Further collaborate with MS and/or Google to improve UX/CX of those cloud service, especially for Gen Z users.

Practical Apps: Findings & Recommendations

App	Findings	Recommendation/Commentary
<u>Productivity</u>	- MS Office and Google Drive and Docs have the high CX scores. As productivity tool, Samsung S Pen & apps have highest CX rating - Google Drive and Docs are widely adopted by all brands (60%+) MS Office and 365 Subscription adoption are also high across all brands	- Collaborate with MS to enable more integration of S Pen and apps with MS Office platform - Expand common productivity tool to lower tier devices since Windows PC and laptop have sizable share across all product smartphone tiers
<u>Tracking</u>	- Both LO, Google and Samsung tracking apps have comparable CX scores - LO Find My is popular with LO females High adoption of LO Find My is likely due to its integration of “Find My Friend” with “Find My Device”, making it very appealing for Gen Z.	Take a holistic approach to app/product design by taking the emotional and social needs (aka elements of value) of Gen Z (and other age groups). In this case, Samsung can plan on developing a tracking app that can integrate with devices, family, friends and pets etc.
<u>Personal, Family and Community Safety</u>	- Personal/Family/Community Safety Apps are more popular with Millennials and younger Gen X, possibly because of young children - Google Trusted Contacts and Life 360 are more widely used across brands. Apps related to pandemic are also used more widely.	- This class of apps generally have substantially lower CX ratings compared with other app categories. - With the pandemic, social unrest and more frequent occurrence of natural disasters, safety apps may become more important in meeting users’ emergent needs (e.g., to satisfy the need to “reduce anxiety” in the elements of value)

Ecosystem: Findings & Recommendations

Device	Findings	Recommendation/Commentary
Smart TV and Media Streamers	• Samsung TV has a strong market penetration and highest CX ratings across all brands, all age groups, gender, tiers and segments. • Samsung TV and Roku Media Streamers have the highest adoption, followed closely by Amazon Fire Sticks.	• Roku is the major competitor to watch out for with the value and quality of its TV (e.g., TCL Roku TV). • Samsung may need to consider developing or acquiring a streaming platform to compete with TCL/Roku and/or Amazon as media streamers and TV with built-in streaming platforms become main stream.
Smart Speakers & Displays	• Both Amazon and Google smart speakers have high CX scores. Third-party Google smart displays do not score well. • Ownership of LO HomePod by Samsung users (12% vs. 16% by LO owners) suggests that there are quite a few dual ecosystem users. • Gen Z rated LO HomePod lower than other age group, which is contrary to their typical ratings of LO products. Gen X also rate it lower.	• Amazon Echo products dominate the market across all demographic and segmentation breakouts.
Tablets	• Both LO and Samsung tablets receive high scores across brands, with LO tablets highest scores by LO and other Android users • Gen Z rated both Amazon Fire Tablet and Samsung Tablet substantially lower than LO tablets.	• Tablets ownership stretches across all demographic, brand and product tier breakouts. • One common theme identified is that Gen Z has a substantially low CX ratings for non-LO products. If this trend continues, Samsung and Android will lose this generation.

Ecosystem: Findings & Recommendations

Device	Findings	Recommendation/Commentary
PC, Laptop, 2-in-1	- MacBook family products CX scores are substantially higher with LO and other Android owners than Samsung owners. - Windows PC have the highest market share across brands (except with 27% owning MacBook family products).	Continue to bring mobile and PC MDE across all product tiers as Windows PC and laptop have sizeable share across all smartphone tiers even with value tiers. A smooth and seamless integration of mobile experience is important across all tiers for long term success.
Smart Watches & Fitness Bands	- Smart watches from LO are more widely adopted by Gen Z and Boomers compared to Samsung Smart Watches. More females use LO and Fitbit smart watches than Samsung (and Garmin) smartwatches.	Embrace design for diversity as best practice. The design for Samsung smartwatches is still not appealing enough to women. Design for Gen Z may be another area that Samsung may need to invest in with a more holistic approach spanning across both hardware and software.
Wireless Earbuds	- LO AirPods and Samsung Galaxy Buds receive higher CX scores by respective owners - LO has higher ownership of wireless earbuds but 11% of LO users use Samsung Galaxy Buds Gen Z has a substantially low ownership of Galaxy EarBuds compared to LO.	This is another area where “Design for Gen Z” matters.

Foldables – Purchase Likelihood & Van Westendorf Pricing

Likelihood to purchase foldables is lukewarm overall

Durability plays a more important role in likelihood of purchase than whether the cost of foldable is similar to combined cost of slate smartphone and tablet.

Agreement Scores (1)

5-Point Likert Scale

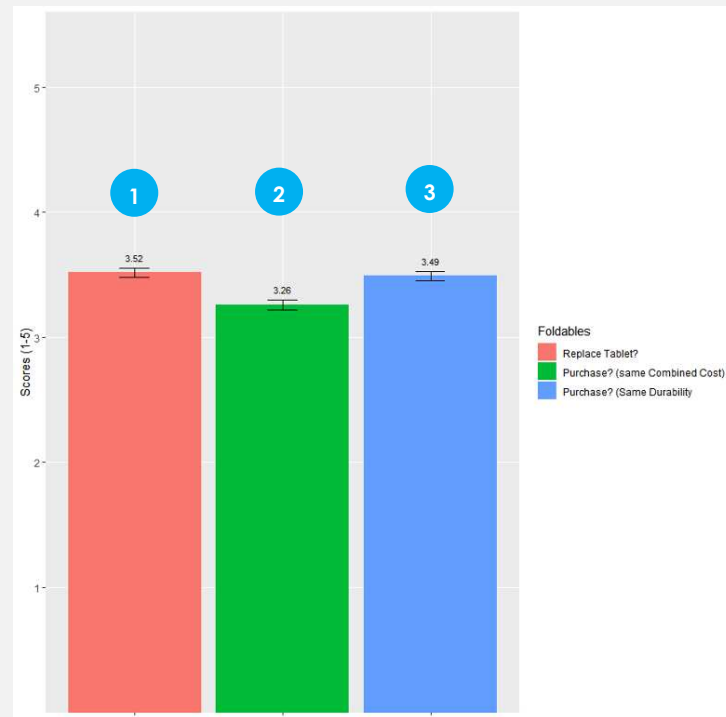
- 1 Strongly disagree
- 2 Disagree
- 3 Neither disagree nor agree
- 4 Agree
- 5 Strongly agree

Likelihood Scores (2 and 3)

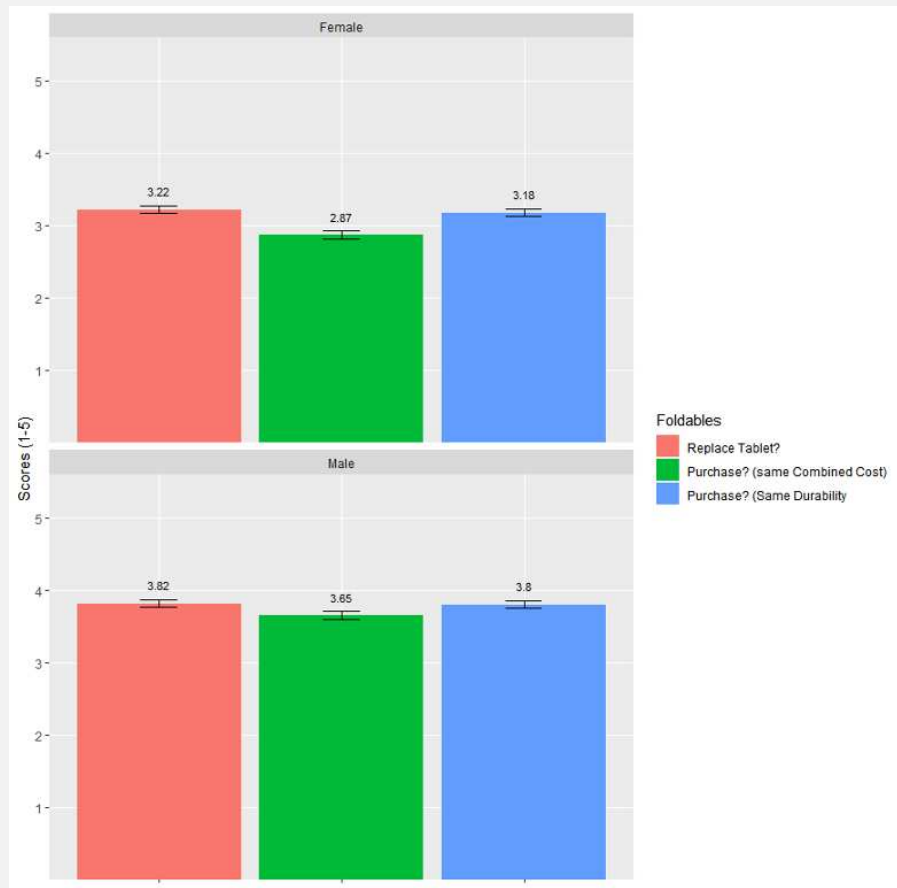
5-Point Likert Scale

- 1 Very unlikely
- 2 Unlikely
- 3 Neither unlikely nor likely
- 4 Likely
- 5 Very Likely

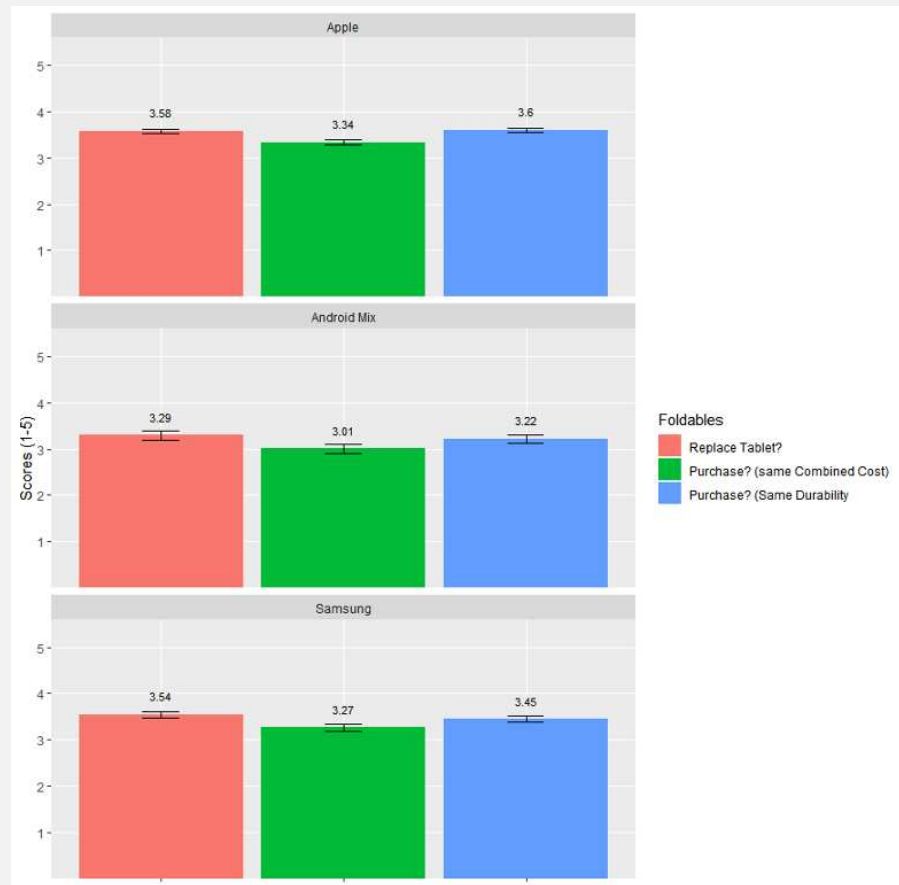
- 1 The foldable smartphone will replace tablet because you can use it as a smartphone when folded and as a tablet upon unfolding it.
- 2 How likely would you be to purchase a foldable smartphone as a replacement of tablet and smartphone if the combined cost of the tablet and a slate-type smartphone is similar to the foldable smartphone?
- 3 How likely would you be to purchase a foldable smartphone with same features and durability as a slate smartphone of the same class as your next smartphone?



Males are substantially more likely than females to purchase foldables



LO users have the highest likelihood to purchase foldables especially when foldables have similar reliability with slate smartphones.



LO owners
(especially male)
have the highest
purchase
likelihood of
foldables and
some also expect
to spend more.

"Because this will be featured with many new things with a bigger display." **Male iPhone 11 Pro, 35 - 40.**

"Because of the best innovation I will be willing to pay any amount of money." **Male, iPhone SE 2020, 35-40**

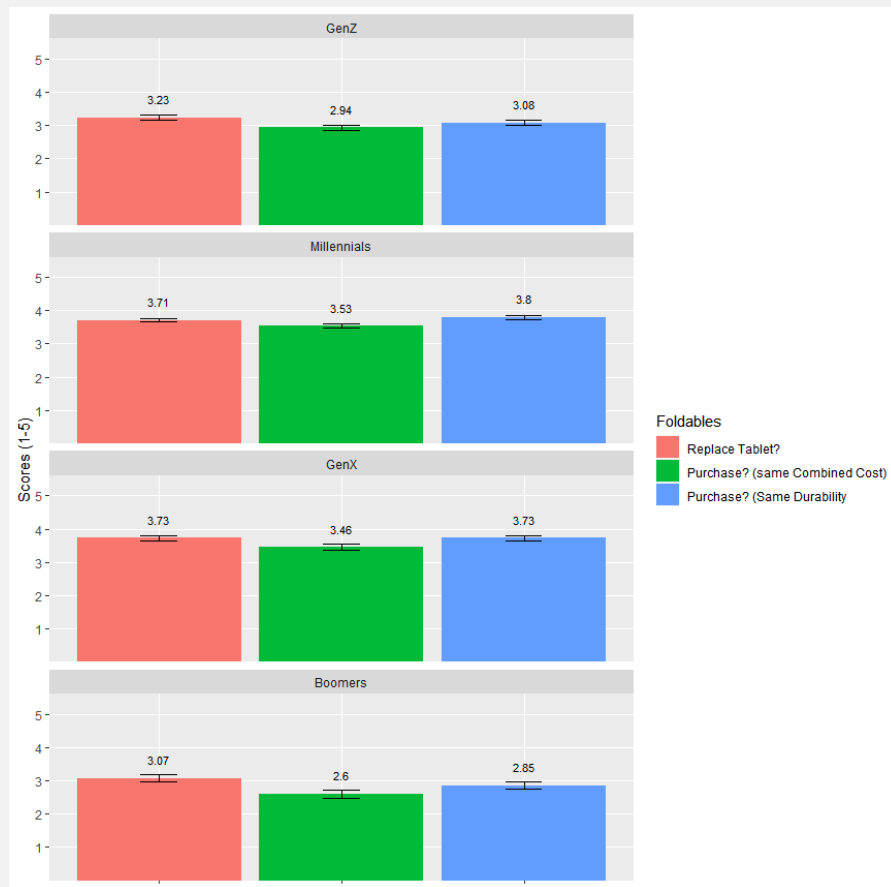
"I don't mind to spend more for foldable phone because it's all in all. I can also use it as a slate smartphone." **Male, iPhone 11 Pro, 41-44**

"It's a latest smartphone, I am comfort with this, I like it." **Male, iPhone 11 Pro Max, 35-40**

"It's worth the cost." **Female, iPhone 11 Pro Max, 25-34**

"Because I believe it has more hardware inputs factored into its foldability. As a user, it can serve as a tablet and phone; hence, it is worth it." **Male, iPhone XS Max, 25-34**

Boomers and Gen Z users have the lowest likelihood to purchase foldables compared to Millennials and Gen X.



Many Gen Z users do not like or do not need foldable phones. Some also kept both smartphones and tablets for separate purposes.

"I feel like a tablet and smartphone being separated would be better to keep more organized." **Female, iPhone 11, under 24**

"I have everything I need to cover my needs."
Male, X20 Ultra, under 24

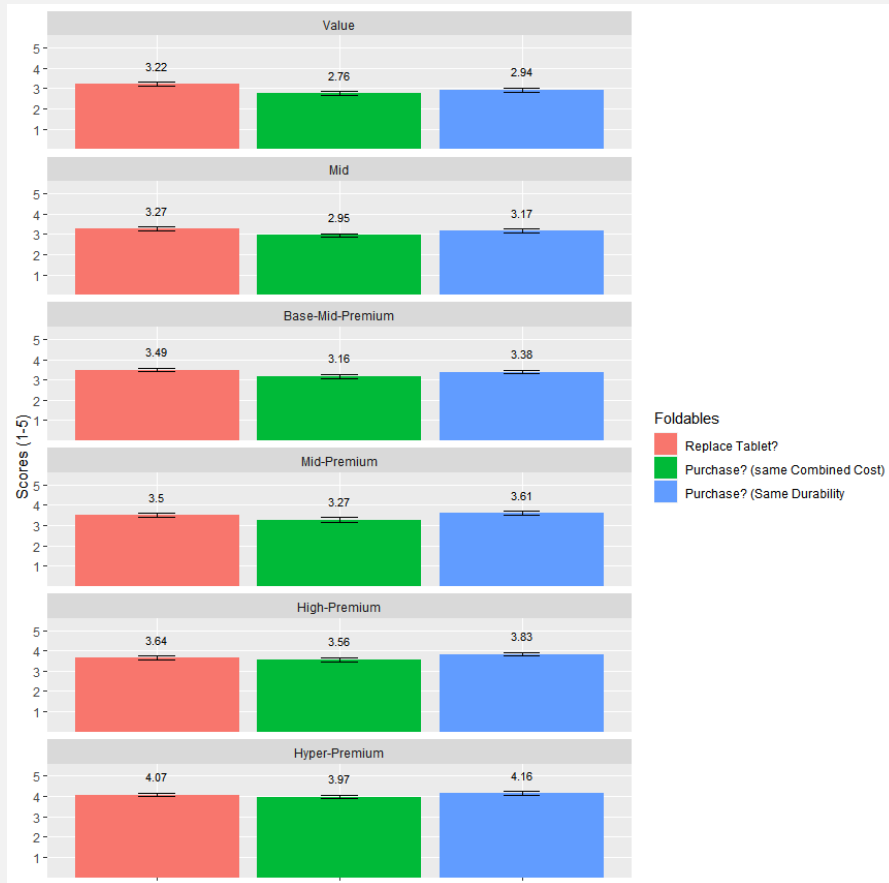
"I use my tablet for work and sometimes need to be able to access multiple devices at once." **Female, iPhone 11 Pro, under 24**

"I don't like the folding design." **Female, Note 20 5G, under 24**

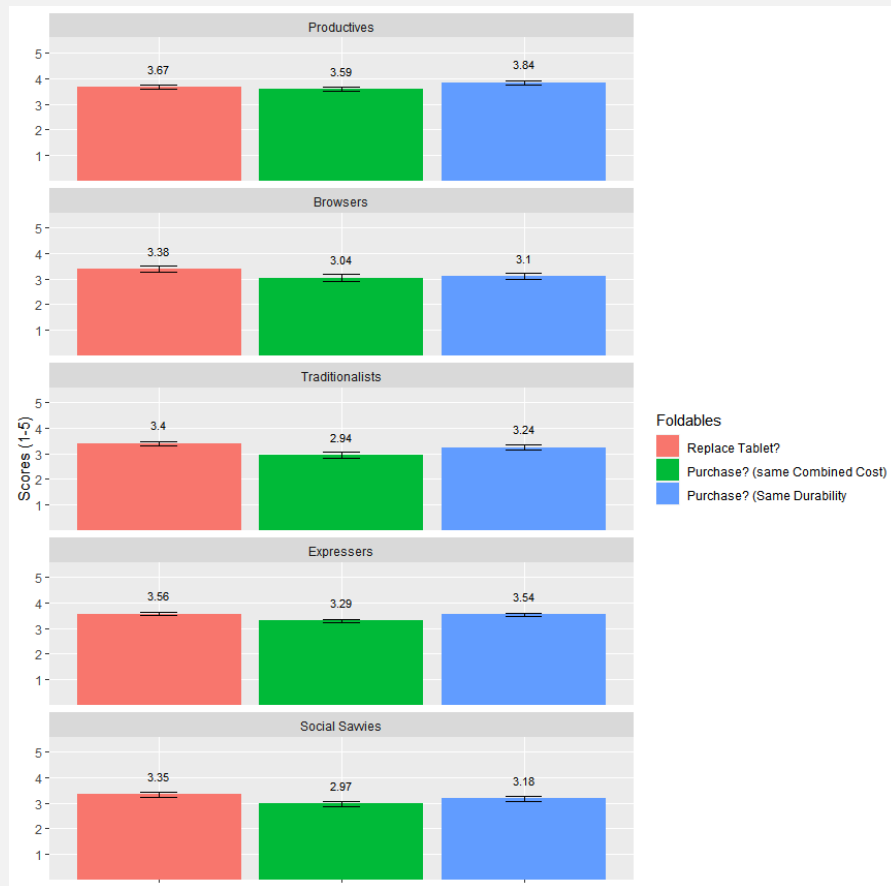
"Tablets are more convenient." **Male, iPhone XR, under 24**

"I like having dedicated devices and the ability to multitask with both at the same time for efficiency."
Male, iPhone 6, under 24

Likelihood of foldables purchase increases with tiers with Hyper-premium tier showing leading purchase intent.



Likelihood of foldables purchase is strongest with Productives followed by expressers.



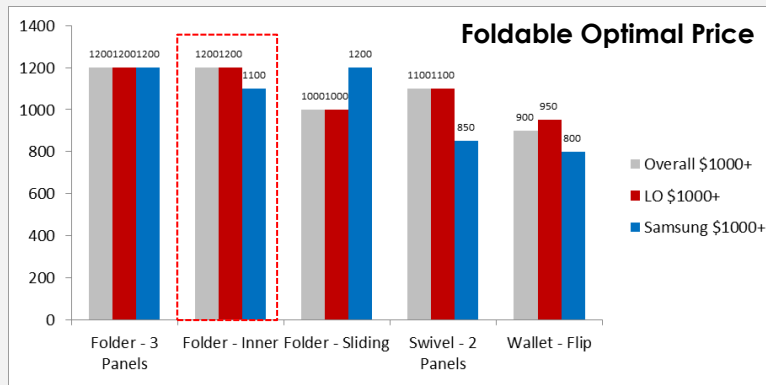
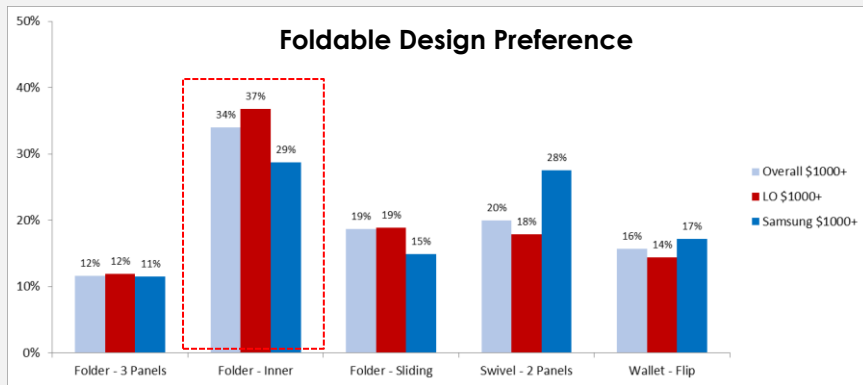
Foldable Designs and Pricing

Samsung should market book-format inner folder aggressively (similar to Galaxy Fold2 5G) to LO is Hi-premium and Hyper-premium tiers as it is popular with LO owners. They also are more inclined to pay more than Samsung owners for foldables with such design.

Book-format inner folder (similar to Galaxy Fold2 5G) is the most popular form factor among the different designs. Samsung high-premium and hyper-premium tier shows higher interest in Swivel form factor than LO while LO 1000+ tier shows more interest in book-format inner folder.

The optimum price of inner folder of LO 1000+ tier owners is higher than Samsung owners while Samsung owners may be willing to pay more for foldables with rolling/sliding display.

(Van Westendorf Pricing model)



Caution: small sample size with some of the less-preferred designs

Pricing of Foldables (all designs) – Overall Market

A base premium folder priced at around \$700 will likely have the best appeal to the general market.

Price Sensivity Meter	
Acceptable Range	\$600 to \$900
Optimum Price	\$700
Indifferent Price	\$700

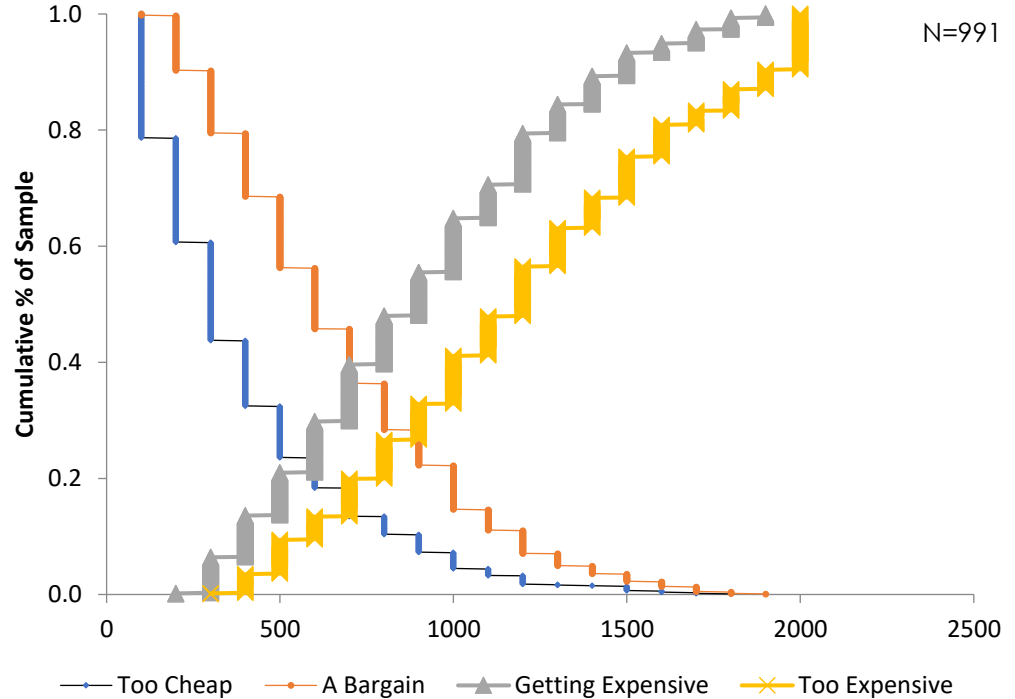
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



Inner Folder Pricing: High & Hyper Premium Tiers

If the price is over \$1,300, more than 15% 1000+ tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,300
Optimum Price	\$1,200
Indifference Price	\$1,100 - \$1,200

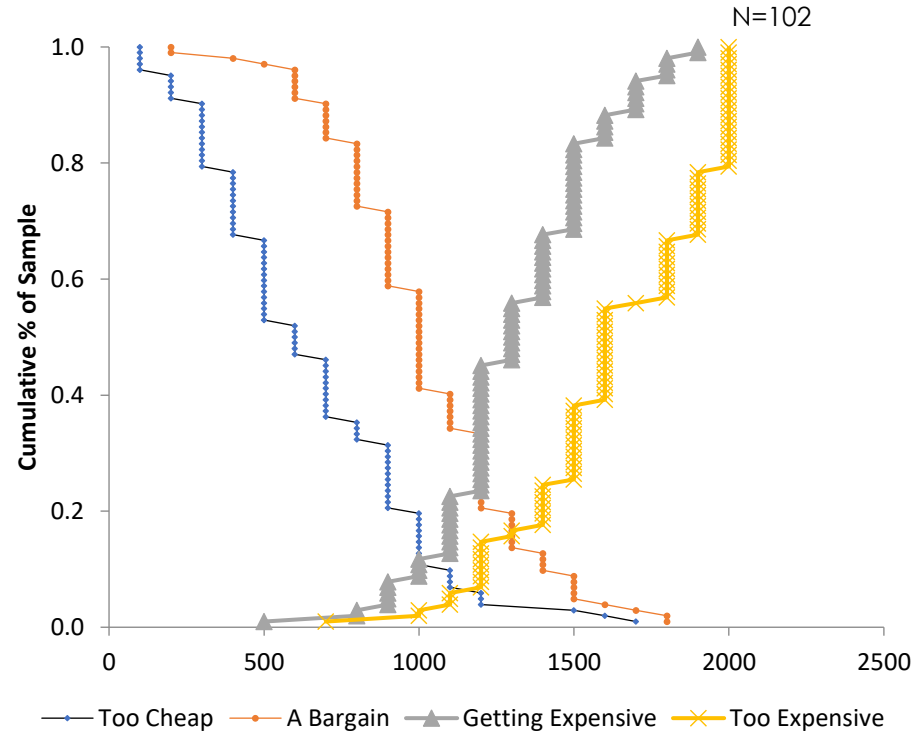
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



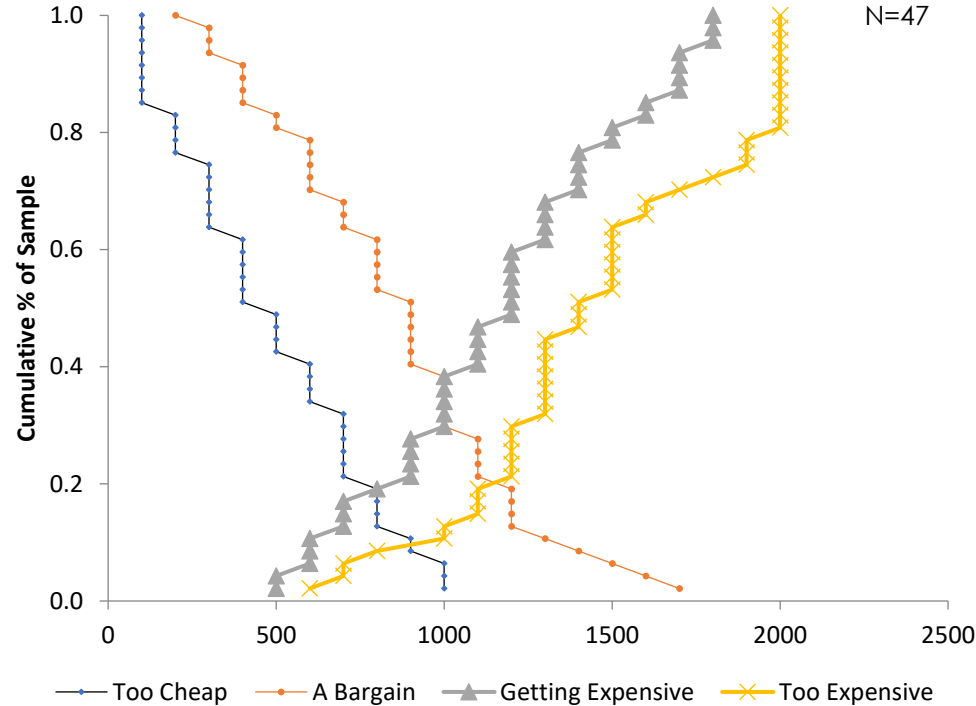
Wallet Foldable Pricing : High & Hyper Premium Tiers

If the price is over \$1,200, more than 20% 1000+ tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$800 to \$1,200
Optimum Price	\$900
Indifference Price	\$1,000

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price – "too cheap" and "too expensive" intersection
 Indifference Price – "bargain" and "getting expensive" intersection



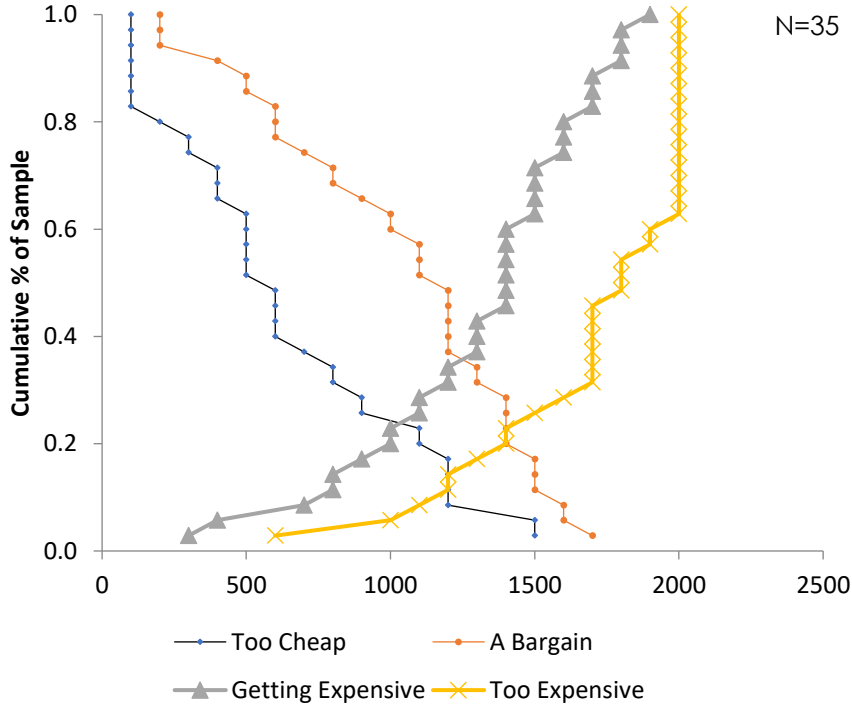
3-Panel Foldable Pricing : High & Hyper Premium

If the price is over \$1,400, more than 20% 1000+ tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,400
Optimum Price	\$1,200
Indifference Price	\$1,200

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



Rolling/Sliding Foldable Pricing : High & Hyper Premium

If the price is over \$1,200, more than 20% 1000+ tiers will find it too expensive. If the price is below \$900, they will start to doubt its quality, starting at around 25%.

Price Sensitivity Meter	
Acceptable Range	\$900 to \$1,200
Optimum Price	\$1,000
Indifference Price	\$1,000

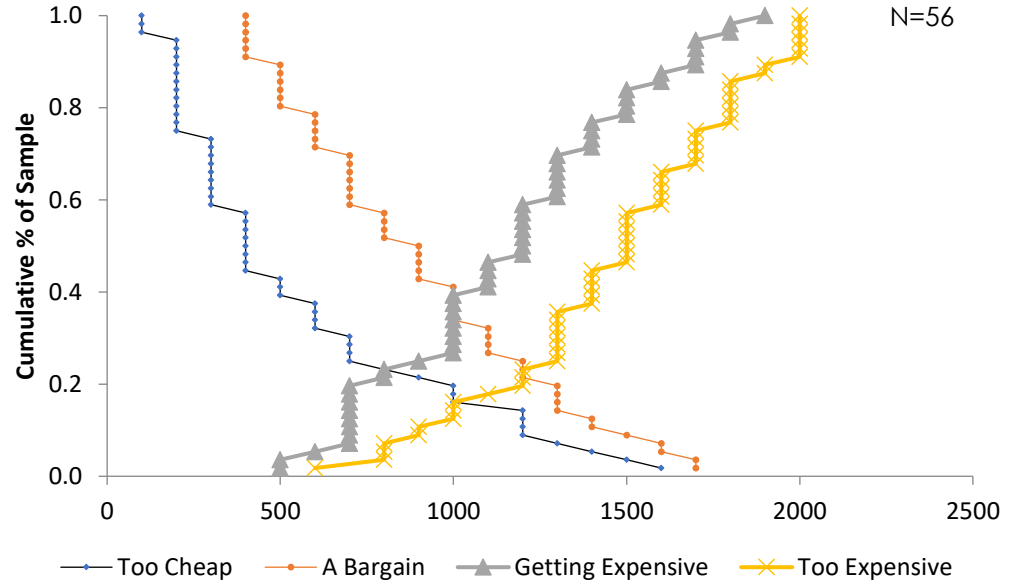
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



Swivel Foldable Pricing : High & Hyper Premium

If the price is over \$1,200, about 18% 1000+ tiers will find it too expensive. If the price is below \$800, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$800 to \$1,200
Optimum Price	\$1,100
Indifference Price	\$1,000

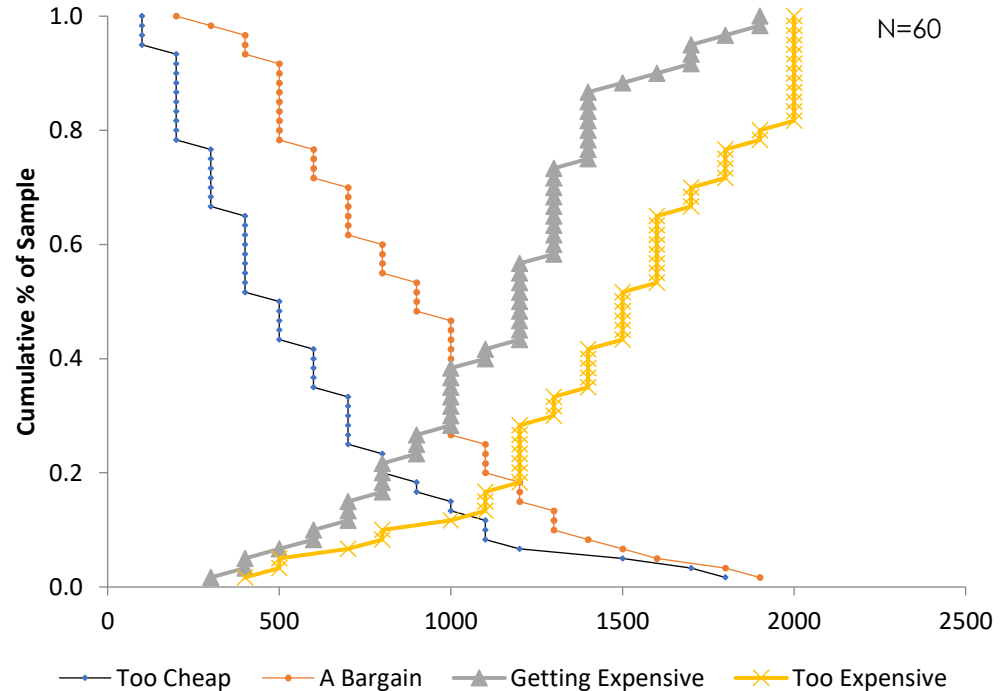
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



Inner Folder Pricing: LO High & Hyper Premium

If the price is over \$1,300, more than 20% LO 1000+ tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 10%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,300
Optimum Price	\$1,200
Indifference Price	\$1,200

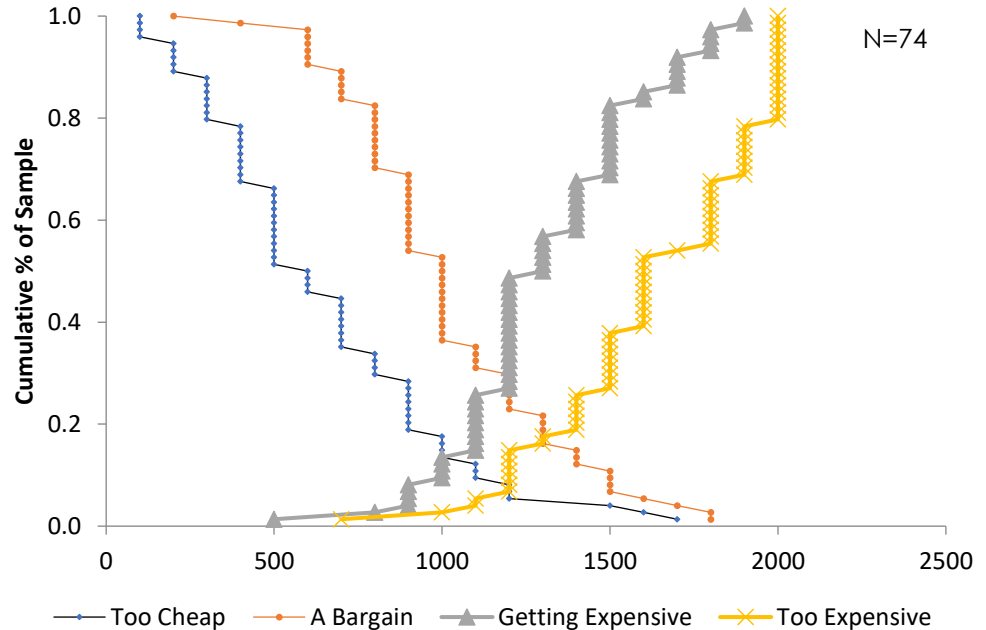
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



Wallet Foldable Pricing : LO High & Hyper Premium

If the price is over \$1,200, more than 15% LO 1000+ will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 15%.

Price Sensitivity Meter	
Acceptable Range	\$800 to \$1,200
Optimum Price	\$950
Indifference Price	\$1,000

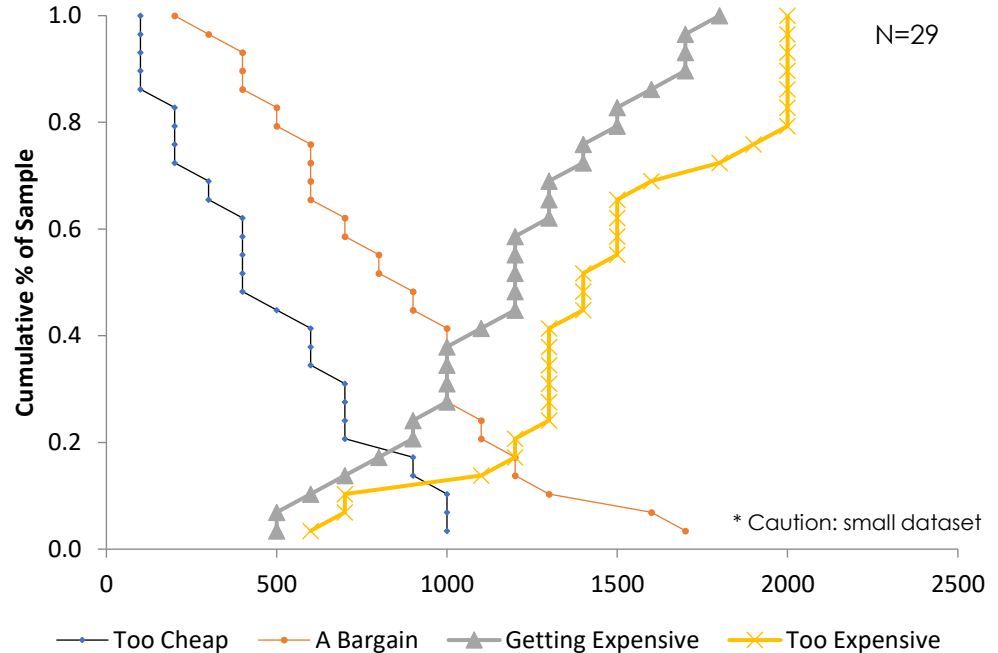
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



3-Panel Foldable Pricing : LO High & Hyper Premium

If the price is over \$1,400, more than about 20% LO 1000+ tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$800 to \$1,400
Optimum Price	\$1,200
Indifference Price	\$1,200

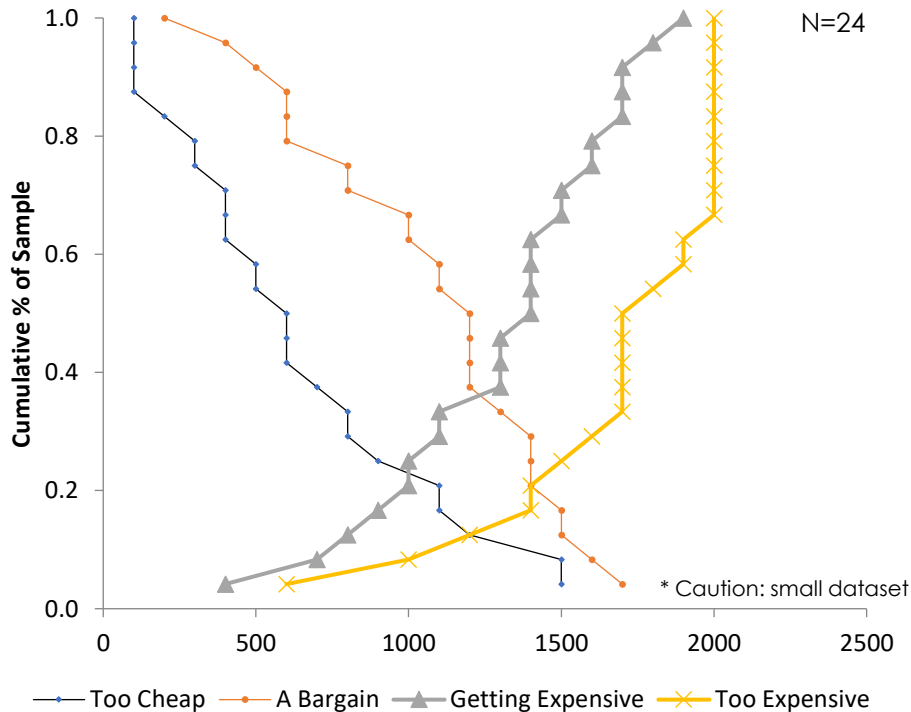
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



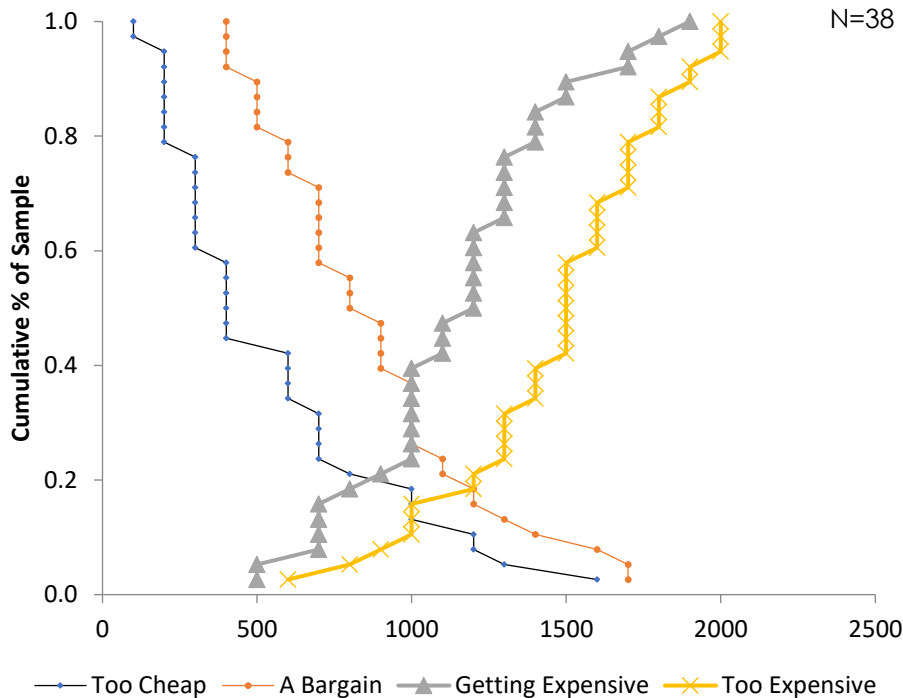
Rolling/Sliding Foldable Pricing : LO High & Hyper Premium

If the price is over \$1,200, about 18% LO 1000+ tiers will find it too expensive. If the price is below \$900, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$900 to \$1,200
Optimum Price	\$1,000
Indifference Price	\$1,000

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection

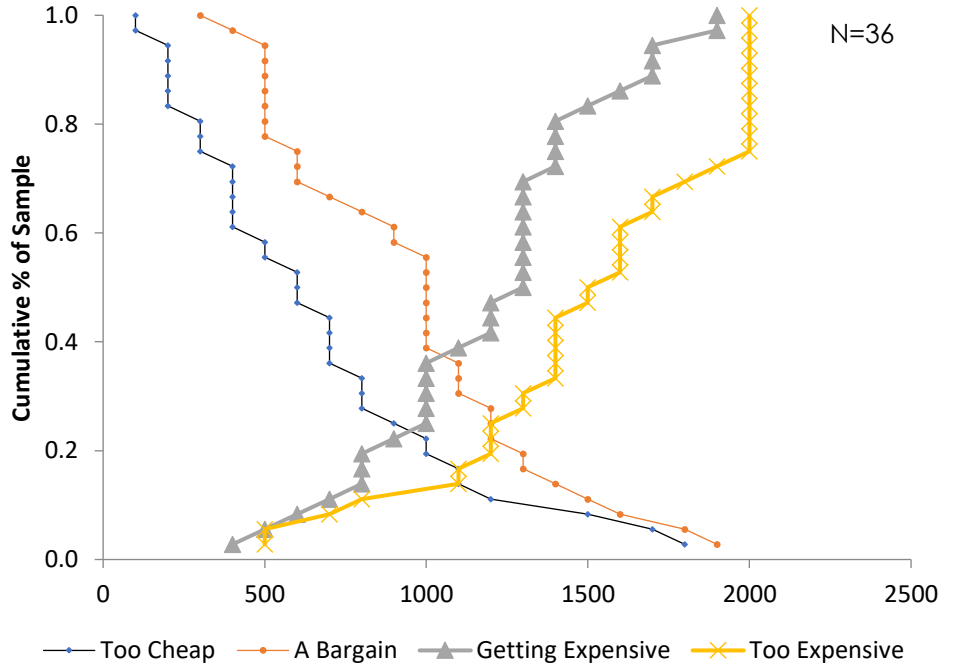


Swivel Foldable Pricing : LO High & Hyper Premium

If the price is over \$1,200, about 25% LO 1000+ tiers will find it too expensive. If the price is below \$900, they will start to doubt its quality, starting at around 20%.

Price Sensitivity Meter	
Acceptable Range	\$900 to \$1,200
Optimum Price	\$1,100
Indifference Price	\$1,100

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection
 Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



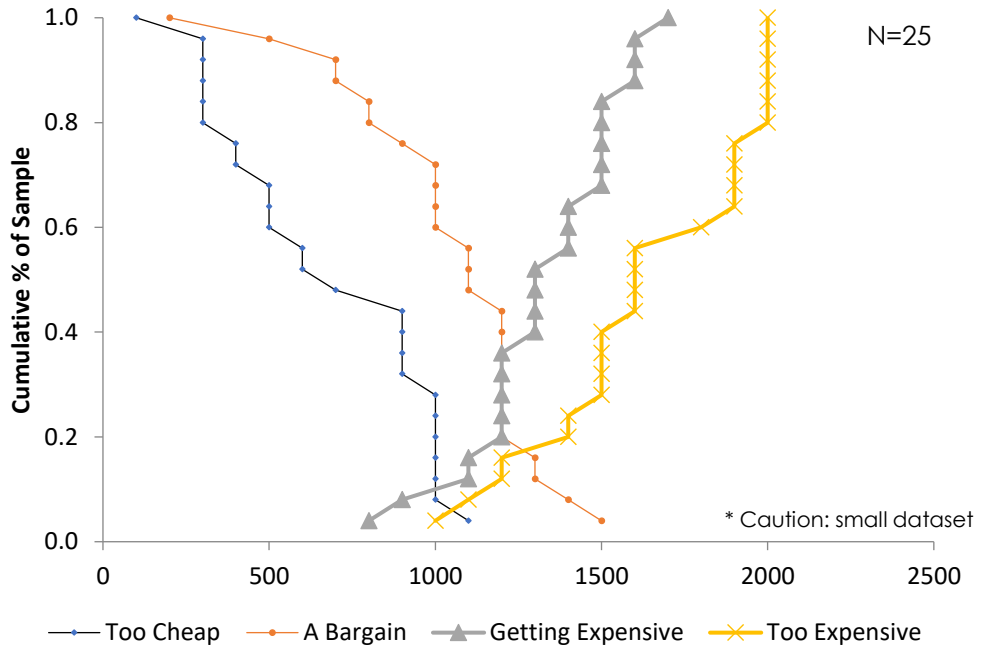
Inner Folder Pricing: Samsung High & Hyper Premium

If the price of foldable (with inner folder design) is over \$1,300, about 20% Samsung 1000+ Tier find it too expensive.
 If the price is below \$1,000, they will start to doubt its quality, starting at around 28%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,300
Optimum Price	\$1,100
Indifference Price	\$1,200

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection

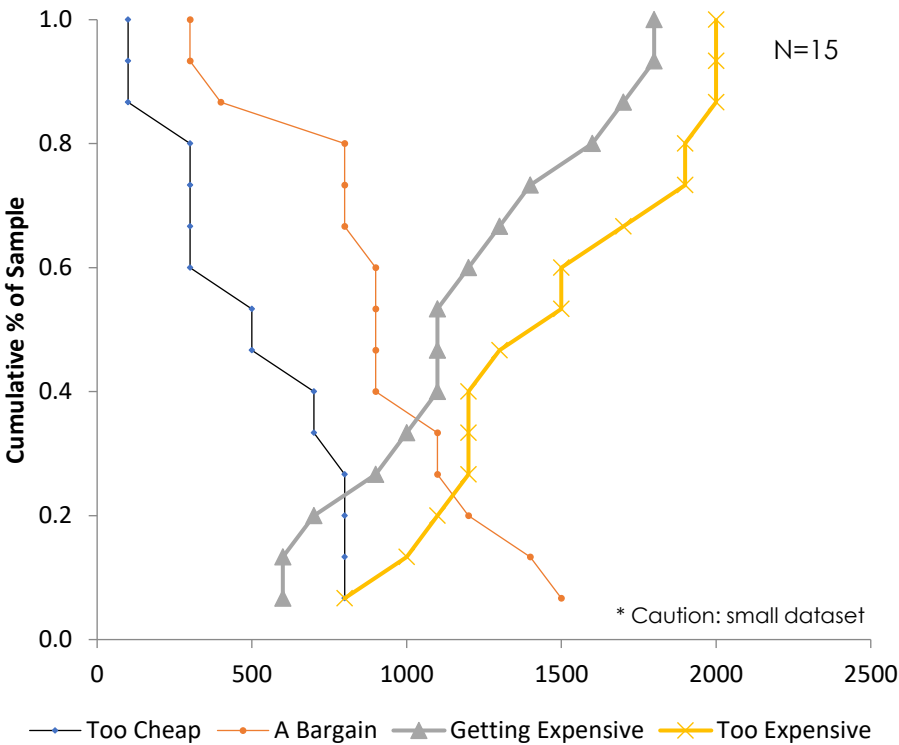


Wallet Foldable Pricing : Samsung High & Hyper Premium

If the price of foldable is over \$1,200, more than 25% Samsung 1000+ will find it too expensive. If the price is below \$800, they will start to doubt its quality, starting at around 25%.*

Price Sensitivity Meter	
Acceptable Range	\$800 to \$1,200
Optimum Price	\$800
Indifference Price	\$1,000

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection
 Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



3-Panel Foldable Pricing: Samsung High & Hyper Premium

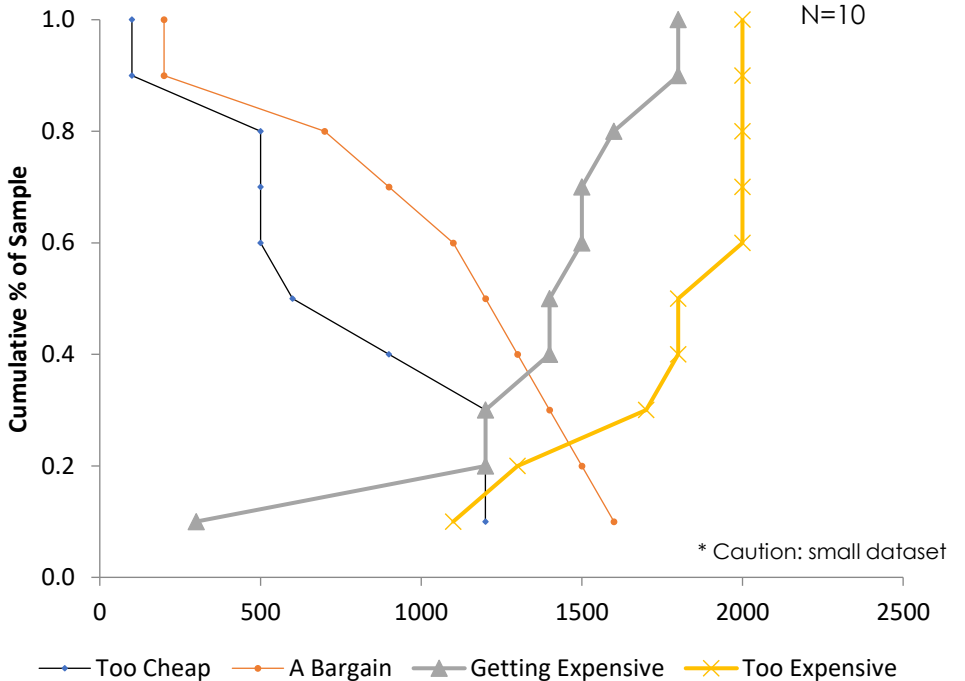
If the price is over \$1,200, more than 15% Samsung 1000+ tiers will find it too expensive. If the price is below \$1,200, they will start to doubt its quality, starting at around 25%.

Price Sensitivity Meter	
Acceptable Range	\$1,200 to \$1,400
Optimum Price	\$1,200
Indifference Price	\$1,350

Acceptable Range:
Lower bound - "too cheap" and "getting expensive" intersection
Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
Indifference Price - "bargain" and "getting expensive" intersection

N=10

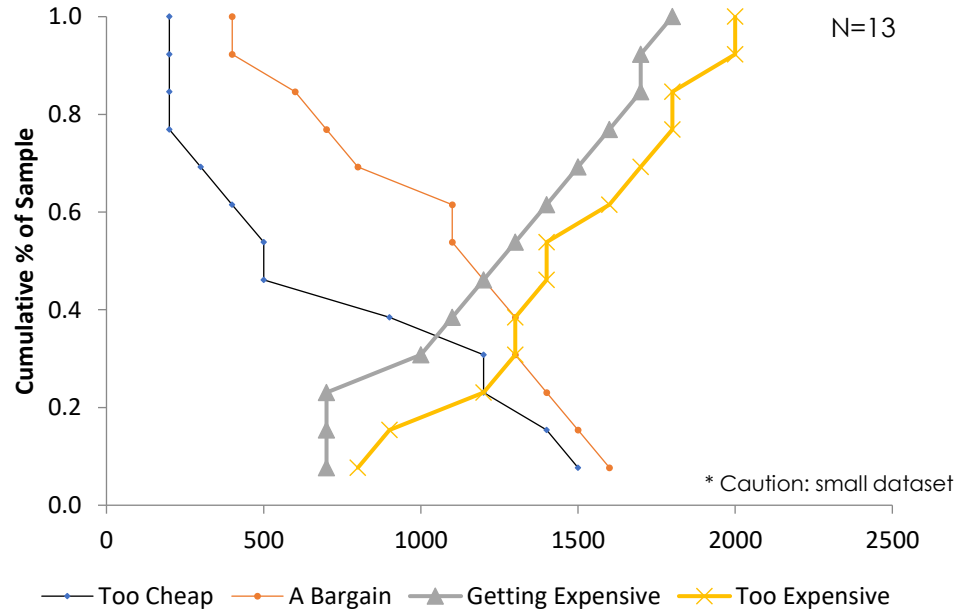


Rolling/Sliding Foldable Pricing : Samsung High & Hyper Premium

If the price of foldable (with inner folder design) is over \$1,300, about 30% LO high and hyper-premium tiers will find it too expensive. If the price is below \$1,000, they will start to doubt its quality, starting at around 35%.

Price Sensitivity Meter	
Acceptable Range	\$1,000 to \$1,300
Optimum Price	\$1,200
Indifference Price	\$1,200

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection
 Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



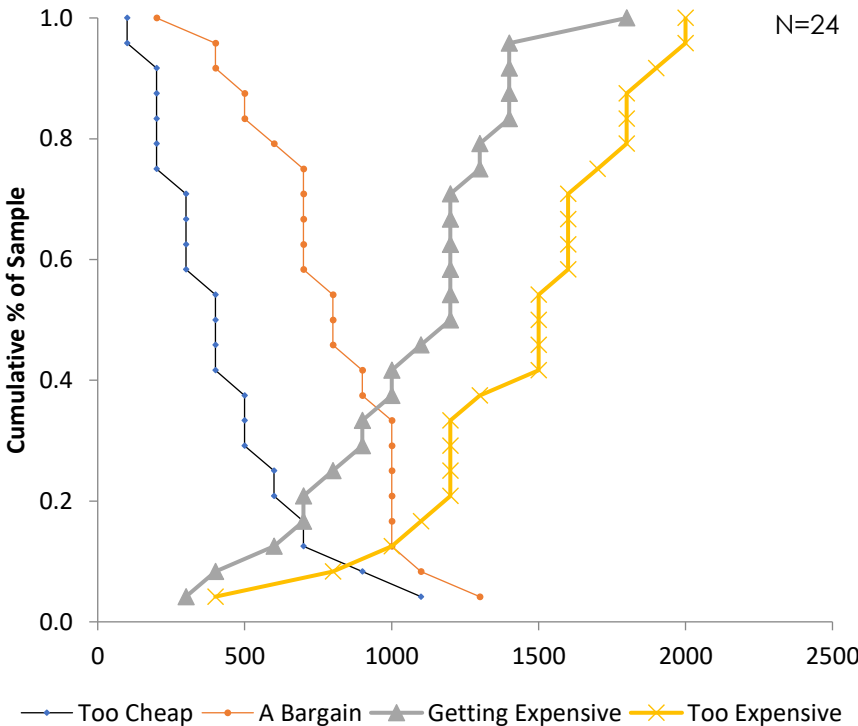
Swivel Foldable Pricing : Samsung High & Hyper Premium

If the price of foldable (with inner folder design) is over \$1,000, more than 12% LO high and hyper-premium tiers will find it too expensive. If the price is below \$700, they will start to doubt its quality, starting at around 12%.

Price Sensitivity Meter	
Acceptable Range	\$700 to \$1,000
Optimum Price	\$850
Indifference Price	\$950

Acceptable Range:
 Lower bound - "too cheap" and "getting expensive" intersection
 Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection
 Indifference Price - "bargain" and "getting expensive" intersection



Folder Pricing – Gen Z

Gen Z optimum price and indifference price are \$600 respectively, \$100 lower than the overall market. They also show lower purchase intent of foldables (similar trend for Baby Boomers)

Price Sensivity Meter	
Acceptable Range	\$500 to \$800
Optimum Price	\$600
Indifference Price	\$600

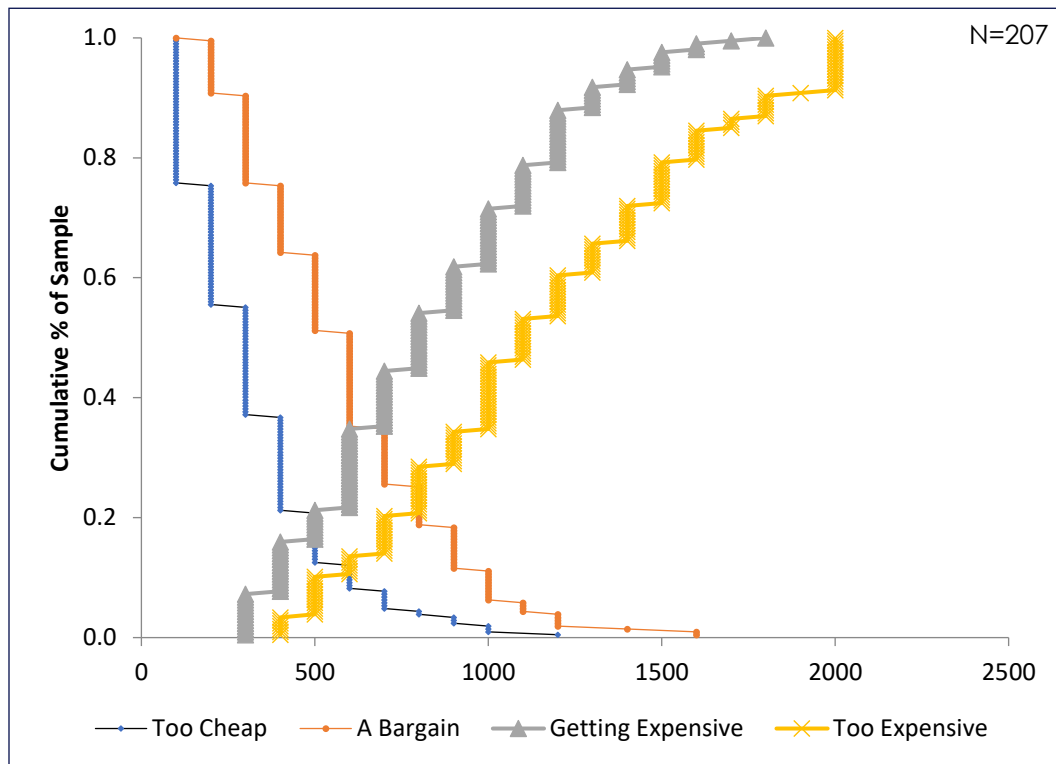
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price – "too cheap" and "too expensive" intersection

Indifference Price – "bargain" and "getting expensive" intersection



Folder Pricing – Millennials

A base premium folder priced at around \$700 will likely have the best appeal to Millennials.

Price Sensitivity Meter	
Acceptable Range	\$600 to \$900
Optimum Price	\$700
Indifference Price	\$800

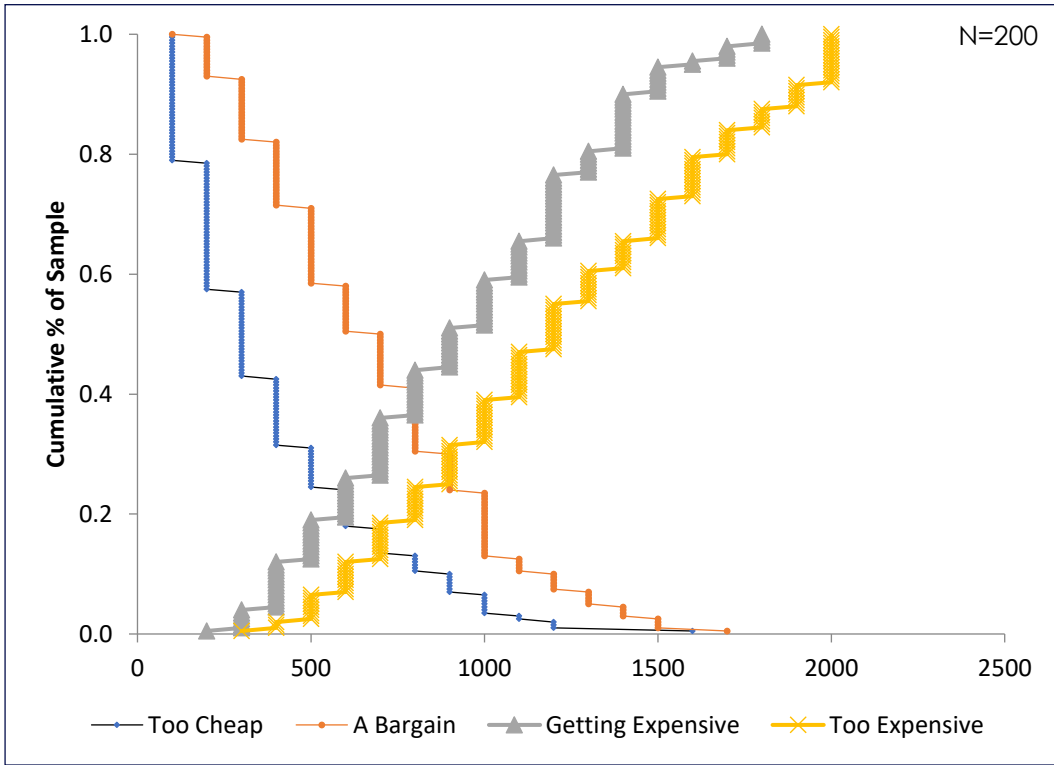
Acceptable Range:

Lower bound - "too cheap" and "getting expensive" intersection

Upper bound - "bargain" and "too expensive" intersection

Optimum Price - "too cheap" and "too expensive" intersection

Indifference Price - "bargain" and "getting expensive" intersection



Loyalty vs. Satisfaction

LO has highest loyalty at all satisfaction levels

- 1 LO loyalty is higher than Samsung and other Android owners at all satisfaction levels
- 2 Android (other than Samsung) owners have lower loyalty compared to both LO and Samsung owners at all satisfaction levels.

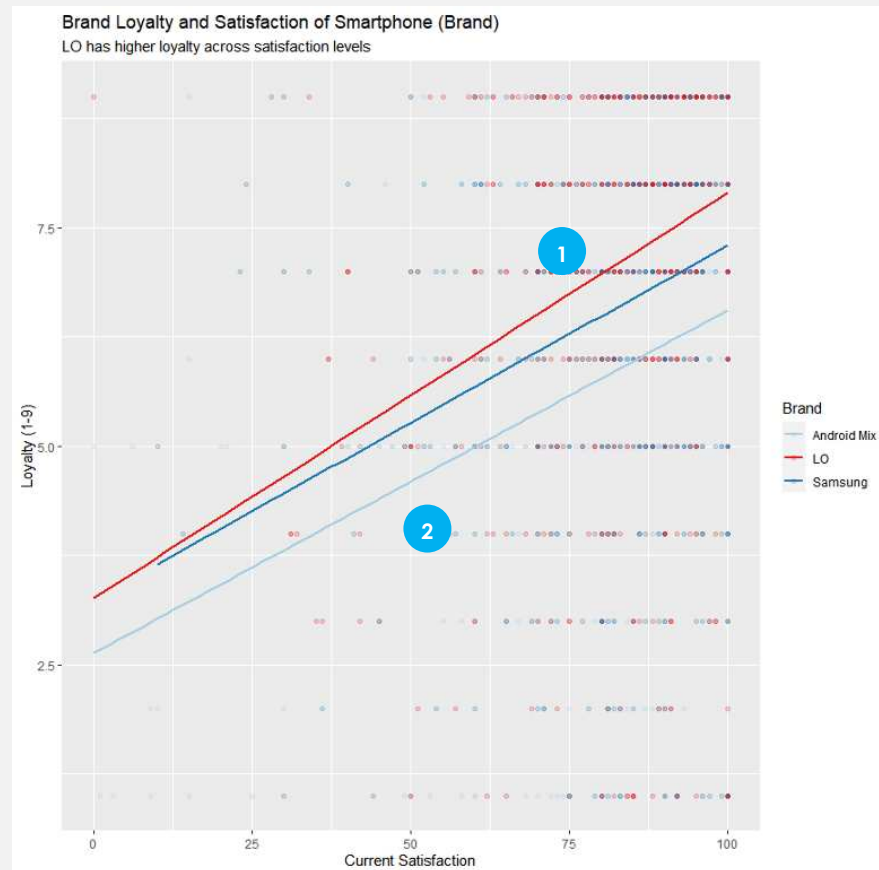
Source: Elements of Value Surveys (N=2096)

Loyalty - 1:9 Likert scale (1: Totally disagree, 9:Totally agree)

I love my current smartphone and will not consider other brands during my next upgrade.

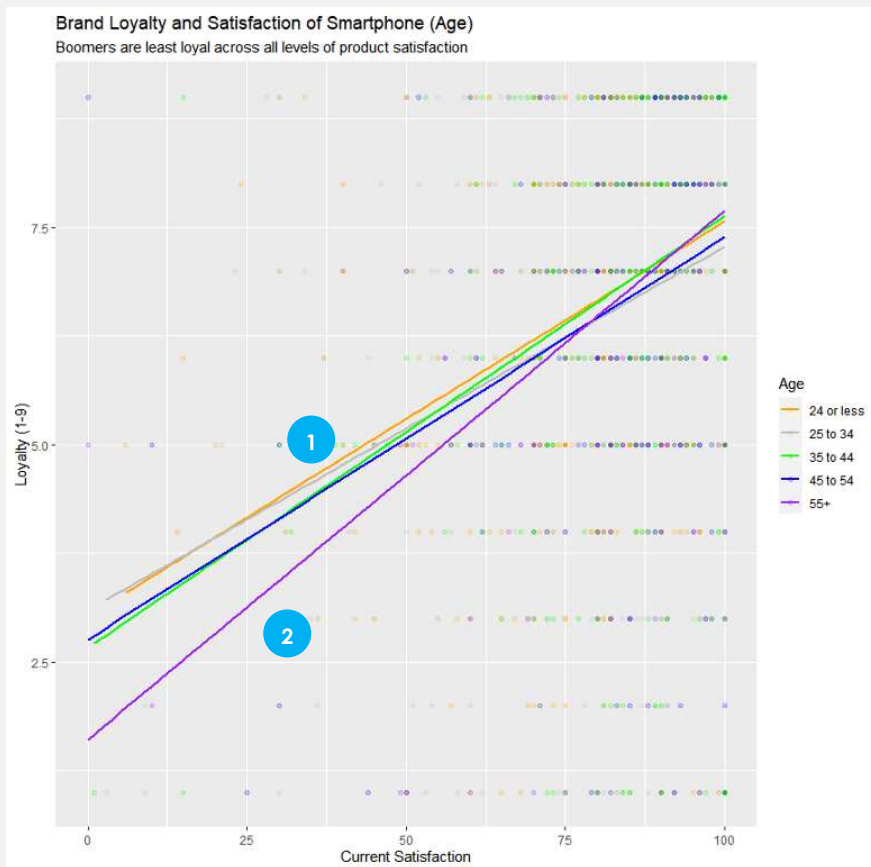
Satisfaction - On a scale of 1 to 100 points, please rate your overall satisfaction with your current smartphone (including physical design, user interface, various features, app and services you use supported by the device).

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Boomers have lower brand loyalty than other age groups across most satisfaction levels

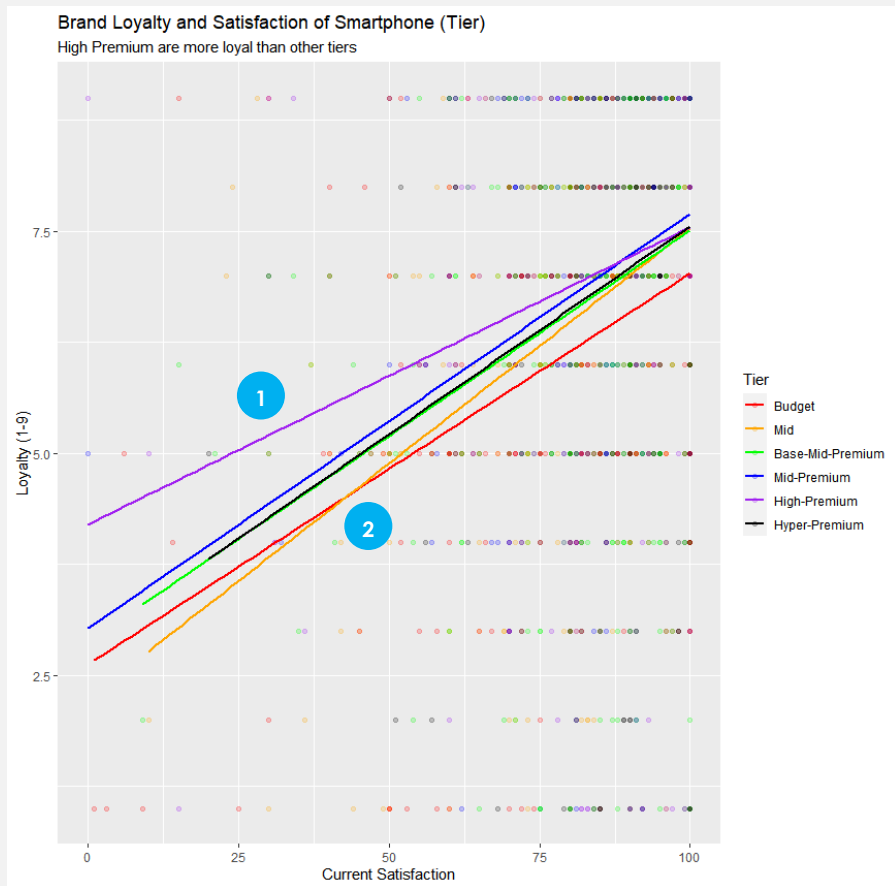
- 1 Brand loyalty for age groups under 55 have similar patterns with current satisfaction
- 2 Baby boomers are less loyal at almost all satisfaction levels.



High-Premium Tier maintains loyalty with decreasing product satisfaction

- 1 Loyalty of High Premium Tier declines more gradually than others despite low satisfaction scores
- 2 Loyalty of Value and Mid Tier are lower than other tiers across all satisfaction levels.

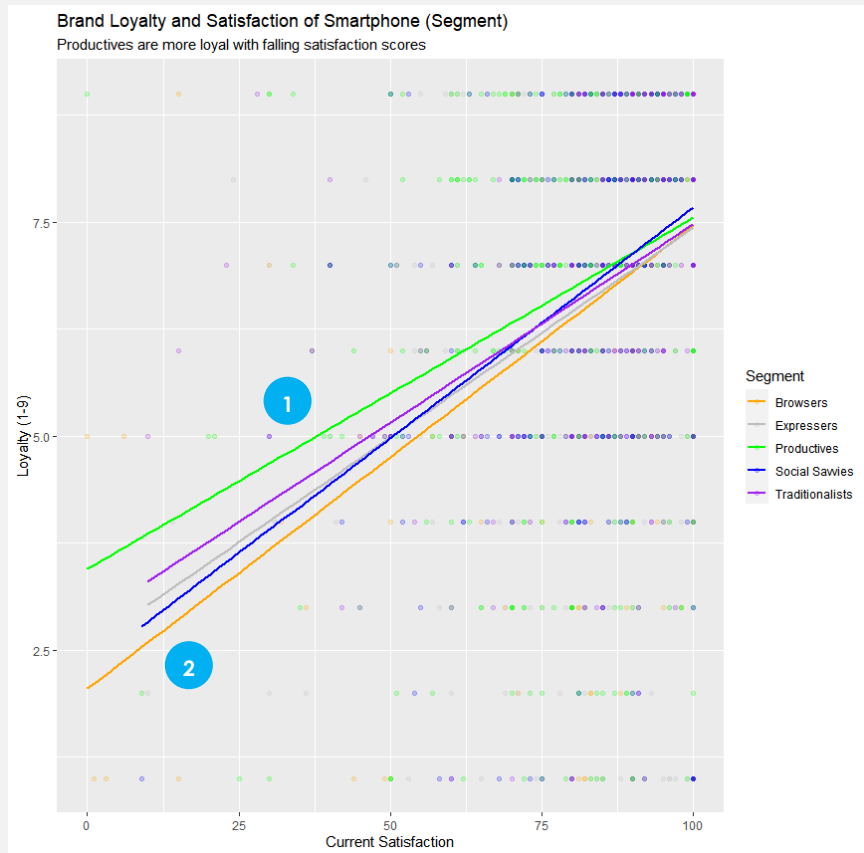
Source: Elements of Value Surveys (N=2096)



Browsers are least loyal with falling satisfaction scores

- 1 Productives have higher loyalty across most satisfaction levels
- 2 Browsers are least loyal with falling satisfaction scores, though the trends for other segments are quite similar.

Source: Elements of Value Surveys (N=2096)



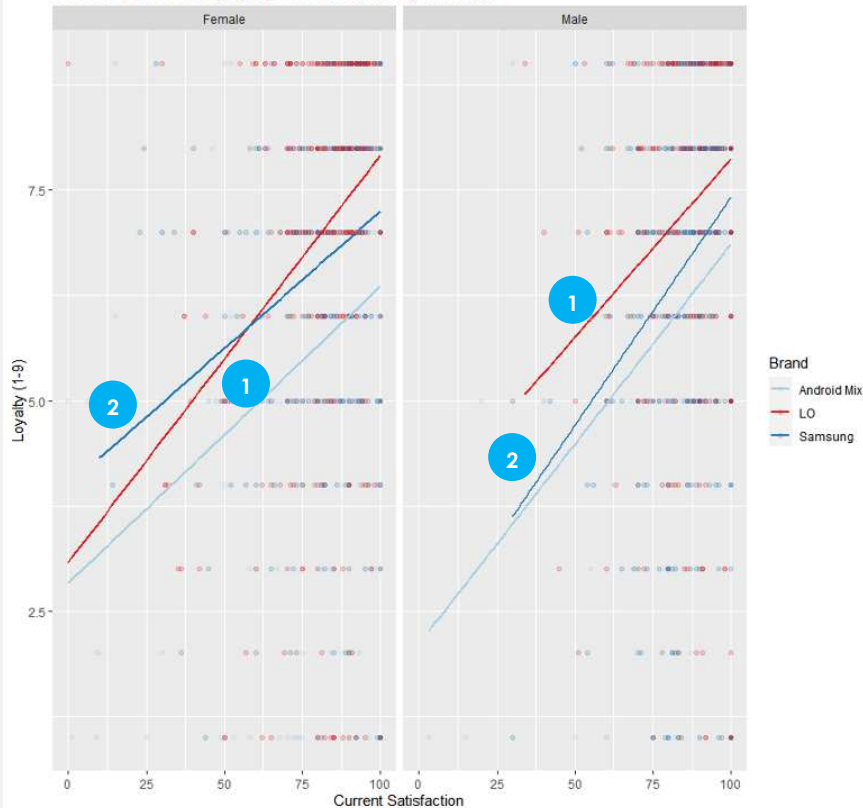
LO female loyalty drops steeply with decreasing satisfaction

- 1 LO female loyalty drops steeply with decreasing satisfaction while LO male loyalty remains higher than others with falling satisfaction.
- 2 Samsung Male loyalty drops more steeply than with less satisfaction.

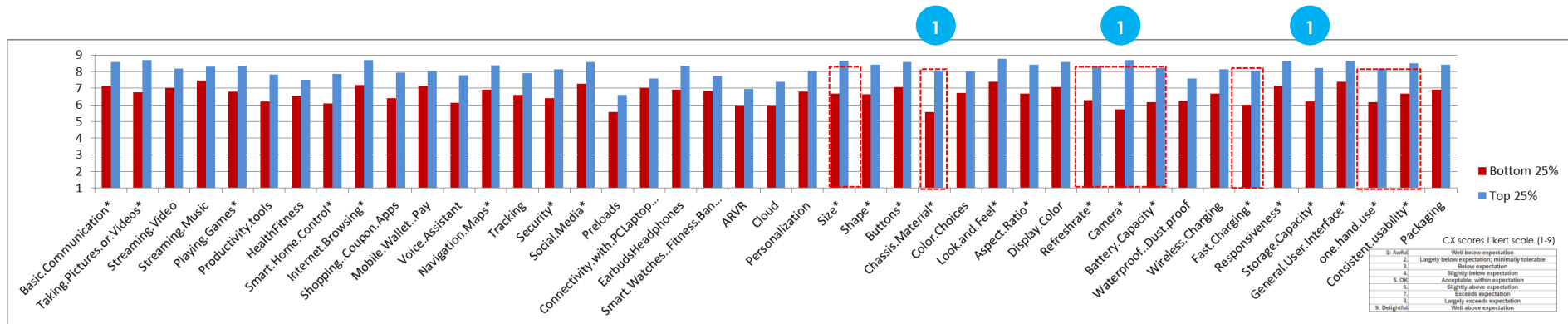
Source: Elements of Value Surveys (N=2096)

Brand Loyalty and Satisfaction of Smartphone (Brand & Gender)

LO Males are the most loyal group here with decreasing satisfaction



Switching LO Female with low loyalty to Samsung



Top loyal vs Bottom loyal LO female

The satisfaction scores of 23 out of 43 apps, design, hardware, software and UI features from LO female (N=51) with top 25% of loyalty and smartphone satisfaction scores are significantly higher than LO female (N=25) with bottom 25% loyalty and satisfaction scores.

t-test unequal samples with Bonferroni correction)

Top Dissatisfaction Drivers 1

The top 8 drivers* are identified as Camera, Chassis Material, Size, Refresh Rate, Consistent Usability, Battery Capacity, One-Hand Use and Fast Charging

*statistically significant dissatisfaction drivers.

Recommendation

With iPhone 12 improvement of camera and chassis material and availability of different sizes, Samsung may want focus on **refresh rate and battery capacity** as marketing message to LO female.

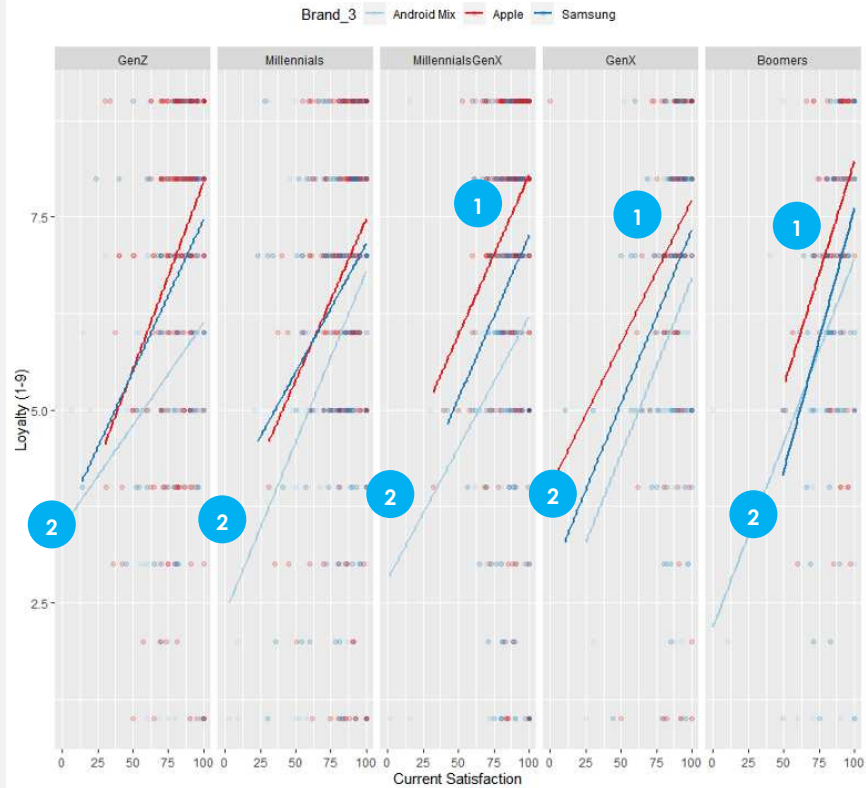
LO Loyalty is higher than others for age groups above 35

- 1 LO has higher loyalty 34 to 55+ age group at similar satisfaction levels compared to both Android and Samsung owners.
- 2 Other Android owners have consistently lower loyalty compared to both LO and Samsung owners across almost all age groups.

Source: Elements of Value Surveys (N=2096)

Loyalty and Satisfaction of Smartphone (Brand & Age)

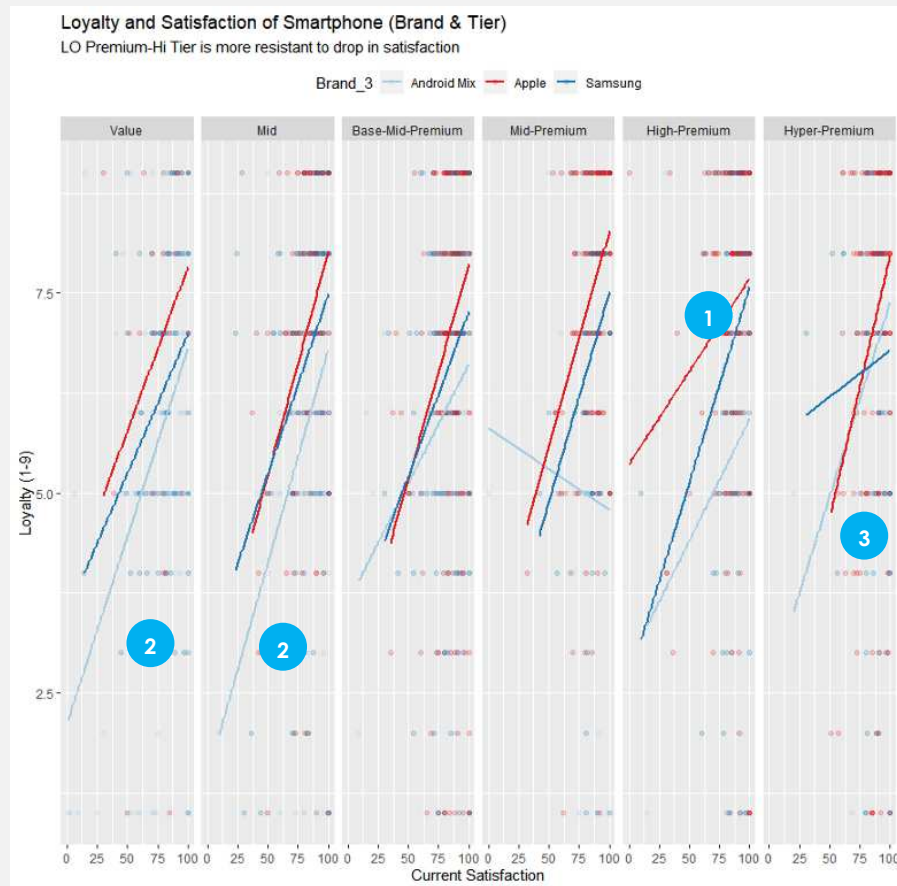
Loyalty and satisfaction of LO is higher than all Android owners for age groups 35 and above



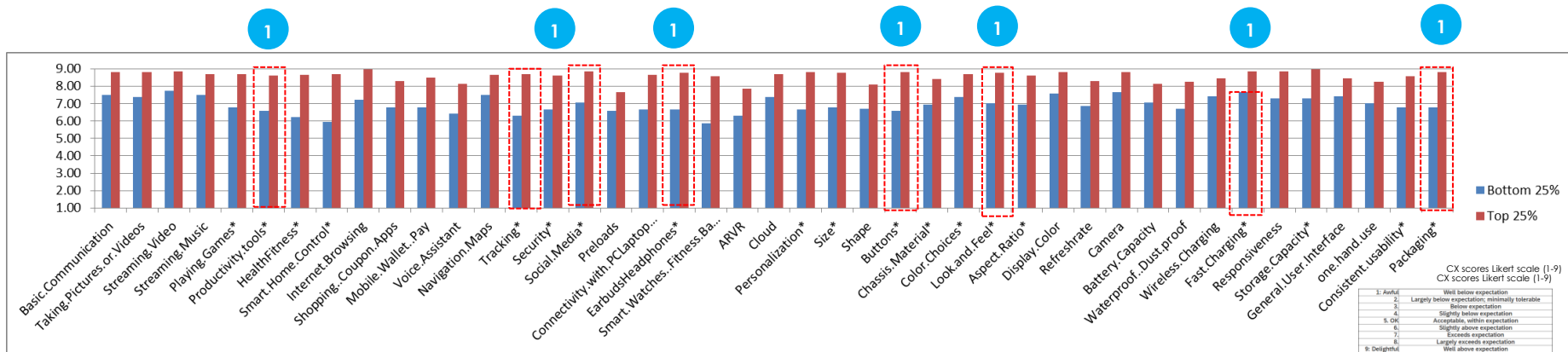
LO Premium loyalty is most resistant to drops in satisfaction

- 1 Loyalty drops steeply for almost all brands except LO Premium-Hi Tier which is more resistant to drops in satisfaction.
- 2 Other mid Android smartphones tier has low loyalty to begin with and drops very steeply with declining satisfaction
- 3 Loyalty of LO hyper-premium tier drops steeply with low satisfaction

Source: Elements of Value Surveys (N=2096)



Keep Samsung High and Hyper-Premium Tiers from Churning

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Top loyal vs Bottom loyal Samsung 1000+

The satisfaction scores of 21 out of 43 apps, design, hardware, software and UI features from Samsung 1000+ (N=20) with top 25% of loyalty and smartphone satisfaction scores are significantly higher than those with bottom 25% loyalty and satisfaction scores (N=14).

t-test unequal samples with Bonferroni correction)

Top Dissatisfaction Drivers

The top 8 drivers* are identified as Packaging, Look & Feel, Buttons, Tracking, Health/Fitness, Earbuds/Headphones, Social Media and Productivity Tools.

*statistically significant dissatisfaction drivers.

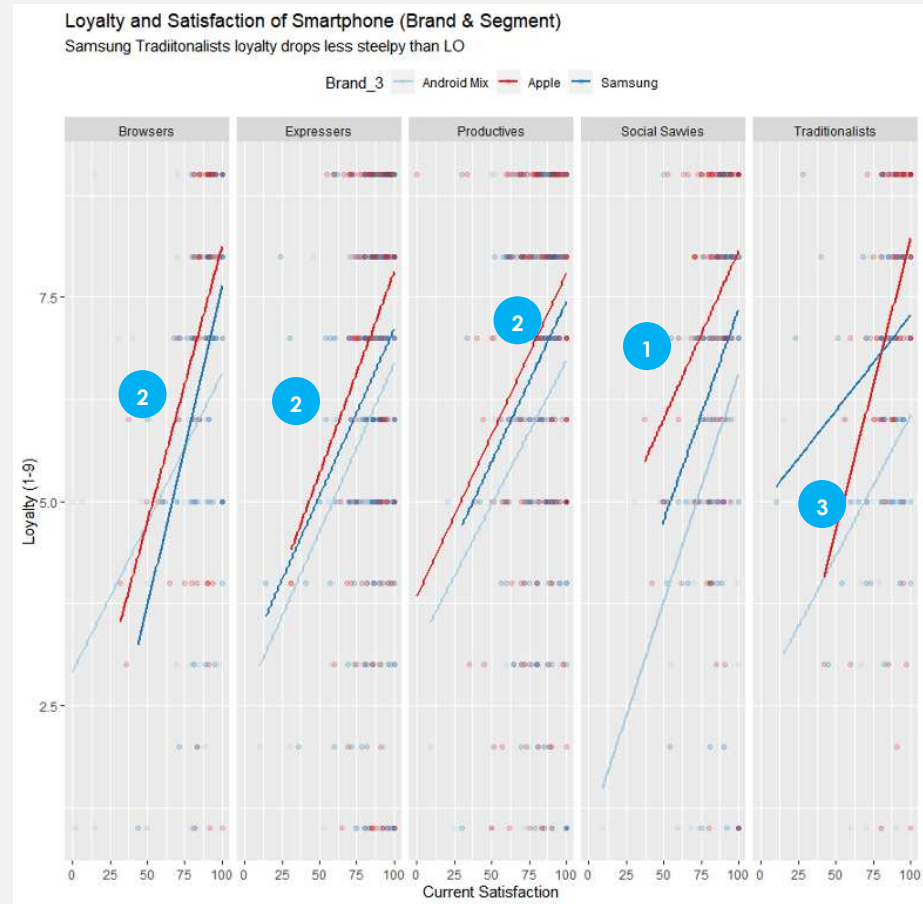
Recommendation

Packaging is also identified as top satisfaction driver for Productives. It seems to play a surprising role in driving satisfaction for high-tier users. Both Tracking and Earbuds have also been identified improvement areas for Gen Z and females and should receive priority in design improvement.

For Social Savvies loyalty of LO is higher across satisfaction levels

- 1 LO Social Savvies Loyalty is higher compared to all Android owners at different satisfaction levels.
- 2 Loyalty vs. satisfaction are similar among the brands for Browsers, Expressers and Productives
- 3 Loyalty of Samsung Traditionalists is more resistant to drop in satisfaction while LO drops steeply

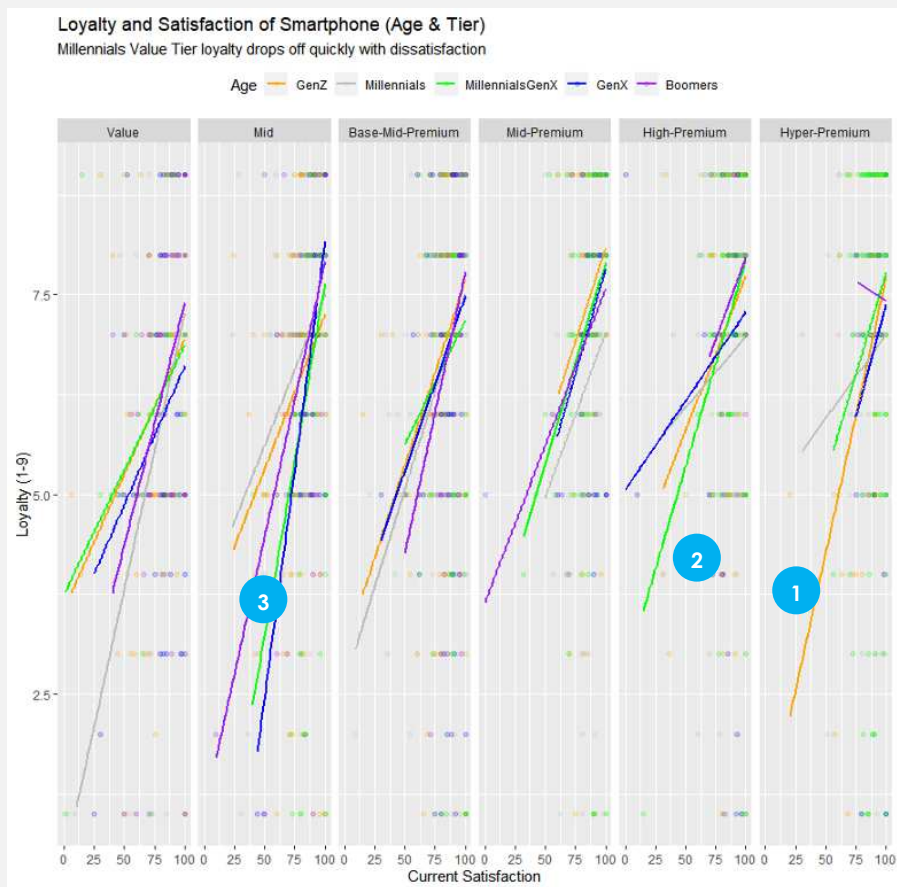
Source: Elements of Value Surveys (N=2096)



Gen Z Hyper-Premium Tier loyalty falls rapidly with dissatisfaction

- 1 Gen Z Hyper-premium loyalty falls rapidly with dissatisfaction
- 2 Loyalty of High-Premium Tier for age 35 to 44 drops rapidly with dissatisfaction.
- 3 Loyalty of Mid-Tier for ages 35 or above drops rapidly with dissatisfaction.

Source: Elements of Value Surveys (N=2096)



Satisfaction & Application Categories, Hardware, Design, Software, UI CX

CX of Apps, Design & Features and Current Satisfaction Overall

Overall	High Customer Experience Impact (Top 10)
Apps Experience	Basic Communication (Voice, video, Messaging Email), Taking Pictures and Videos, Security
Physical Design Experience	Look and Feel , Color Choices
Hardware Experience	Camera, Storage Capacity , Display Color, Responsiveness
UX and UI Design	Consistent Usability , General User Interface

Overall, Camera, Storage Capacity, Consistent Usability & Look and Feel have the highest impact (Top 10%) on Satisfaction.

1: Awful	Well below expectation
2	Largely below expectation; minimally tolerable
3	Below expectation
4	Slightly below expectation
5: OK	Acceptable, within expectation
6	Slightly above expectation
7	Exceeds expectation
8	Largely exceeds expectation
9: Delightful	Well above expectation

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables.

CX of Apps, Design & Features and Current Satisfaction

Gender and Brand

Gender	High Customer Experience Impact (Top 10%)
Male	Display Color, Look and Feel, Security, Storage Capacity
Female	Camera, Consistent Usability, Storage Capacity, Taking Pictures or Videos

Brand	High Customer Experience Impact (Top 10%)
Android Mix	Navigation/Maps, Social Media, Storage Capacity, Wireless Charging
LO	Camera, Consistent Usability, General UI, Shape
Samsung	Basic Communication, Camera, Consistent Usability, Storage Capacity

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

CX of Apps, Design & Features and Current Satisfaction

Overall & Gender

Relative impacts of each of 43 variables representing different apps categories, HW/SW Design and features on current satisfaction of smartphones shown*.

Notable Findings*

Top 2 Drivers of Smartphone Satisfaction

Females

- Camera
- Storage Capacity

Males

- Display
- Look and Feel

*Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

*Top 2 drivers in Notable Findings and Top 10% drivers are based on the numerical values of the coefficients without further statistical testing. Caution should be used for interpretation.

Features	Overall	Female	Male
ARVR	-0.97	-2.98	2.32
Aspect.Ratio	-0.17	-1.27	1.30
Basic.Communication	4.70	3.84	4.34
Battery.Capacity	2.15	3.73	-1.99
Buttons	1.27	0.30	1.77
Camera	7.25	7.82	3.24
Chassis.Material	1.88	1.74	1.55
Cloud	-0.01	1.53	-3.30
Color.Choices	3.90	3.40	2.67
Connectivity.with.PCLaptop..Tablet	-1.50	-2.66	0.89
Consistent.usability	6.67	6.38	4.45
Display.Color	4.61	2.40	6.24
EarbudsHeadphones	-0.65	-1.14	-0.06
Fast.Charging	1.90	4.07	-2.73
General.User.Interface	3.30	4.13	0.49
HealthFitness	-1.54	-2.42	0.24
Internet.Browsing	3.04	3.38	0.60
Look.and.Feel	4.94	2.52	6.94
Mobile.Wallet..Pay	-0.91	0.36	-2.81
Navigation.Maps	3.13	1.58	4.06
one.hand.use	-1.50	-0.82	-2.41
Packaging	2.92	1.06	4.27
Personalization	-0.01	-0.35	0.10
Playing.Games	1.18	1.06	1.03
Preloads	-0.67	1.16	-3.37
Productivity.tools	-0.57	-1.00	0.18
Refresh.rate	3.15	3.42	0.73
Responsiveness	4.23	4.13	2.73
Security	4.64	2.64	5.97
Shape	1.30	0.51	2.03
Shopping..Coupon.Apps	-0.93	1.07	-4.44
Size	3.25	2.36	3.25
Smart.Home.Control	-0.73	-0.27	-1.48
Smart.Watches..Fitness.Bands	-1.41	-2.43	0.53
Social.Media	3.15	2.81	1.95
Storage.Capacity	7.98	7.51	5.38
Streaming.Music	-1.66	-1.89	-1.18
Streaming.Video	0.79	0.65	0.53
Taking.Pictures.or.Videos	4.52	4.91	1.69
Tracking	-0.47	-0.59	-0.58
Voice.Assistant	-0.19	0.96	-2.47
Waterproof..Dust.proof	0.23	-0.29	0.75
Wireless.Charging	0.04	0.45	-0.96

CX of Apps, Design & Features and Current Satisfaction

Brand

Notable Findings

Top 2 Drivers of Smartphone Satisfaction

Android Mix users	- Social Media - Storage Capacity
LO users	- Camera - Consistent Usability
Samsung users	- Basic Communication - Storage Capacity

Featuers	Android	LO	Samsung
ARVR	-1.06	1.93	-3.44
Aspect.Ratio	0.04	2.57	-1.32
Basic.Communication	2.42	2.57	6.30
Battery.Capacity	3.11	2.19	-0.27
Buttons	2.23	2.44	0.67
Camera	2.38	2.97	5.42
Chassis.Material	-0.90	2.58	0.98
Cloud	-0.66	1.87	0.36
Color.Choices	2.06	2.41	3.00
Connectivity.with.PCLaptop..Tablet	-1.77	1.90	-0.83
Consistent.usability	2.69	2.93	6.12
Display.Color	2.11	2.69	4.67
EarbudsHeadphones	2.17	1.85	-2.26
Fast.Charging	4.07	2.27	-3.35
General.User.Interface	3.86	2.89	-0.56
HealthFitness	-0.48	1.89	-0.28
Internet.Browsing	3.10	2.50	3.03
Look.and.Feel	4.14	2.73	3.40
Mobile.Wallet..Pay	-3.87	2.06	0.30
Navigation.Maps	4.85	2.53	1.16
one.hand.use	0.24	2.26	-4.71
Packaging	1.35	2.72	1.60
Personalization	-0.37	2.43	-0.12
Playing.Games	-0.72	2.29	1.90
Preloads	-0.51	1.58	-1.21
Productivity.tools	-2.43	2.33	0.47
Refreshrate	3.44	2.69	1.25
Responsiveness	2.34	2.60	2.17
Security	3.62	2.63	4.11
Shape	2.25	2.73	-4.07
Shopping..Coupon.Apps	1.03	2.01	-2.03
Size	1.06	2.60	2.00
Smart.Home.Control	-2.03	2.06	-0.04
Smart.Watches..Fitness.Bands	-3.28	2.28	-1.51
Social.Media	6.70	2.22	1.81
Storage.Capacity	6.86	2.21	7.12
Streaming.Music	-0.89	1.83	0.30
Streaming.Video	1.65	2.35	0.92
Taking.Pictures.or.Videos	1.25	2.70	4.93
Tracking	-2.95	2.14	2.18
Voice.Assistant	-1.86	2.15	1.51
Waterproof..Dust.proof	0.41	1.41	1.26
Wireless.Charging	4.81	2.01	-5.05

CX of Apps, Design & Features and Current Satisfaction Age

Age	High Customer Experience Impact (Top 10%)
Gen Z (up to 24)	Consistent Usability, Display Color, Look and Feel, Packaging
Millennials (25 to 34)	Basic Communication, Camera, Consistent Usability, Storage Capacity
Millennials/Gen X (35 to 44)	Basic Communication, Security, Size, Taking Pictures and Videos
Gen X (45 to 54)	Color Choices, Internet Browsing, Look and Feel, Storage Capacity
Boomers (55+)	General UI, Refresh Rate, Responsiveness, Storage Capacity

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

CX of Apps, Design & Features and Current Satisfaction

Age

Notable Findings

Top 2 Drivers of Smartphone Satisfaction

Gen Z	- Consistent Usability - Display Color
Millennials	- Camera - Consistent Usability
Millennials/Gen X	- Security - Taking Pictures/Videos
Gen X	- Look and Feel - Color Choices
Boomers	- General UI - Storage Capacity

Features	Gen Z	Millennials	Millennials/Gen X	Gen X	Boomers
ARVR	1.60	1.70	1.97	-4.20	1.33
Aspect.Ratio	2.52	2.13	2.24	1.55	2.61
Basic.Communication	2.84	2.83	2.87	0.91	2.18
Battery.Capacity	2.32	1.86	2.47	1.82	2.56
Buttons	2.40	2.40	2.53	-0.07	2.78
Camera	2.81	2.94	2.52	3.13	2.77
Chassis.Material	2.26	2.15	2.35	2.17	2.90
Cloud	1.48	2.55	1.97	-0.33	1.86
Color.Choices	2.37	2.52	2.13	5.92	2.40
Connectivity.with.PCLaptop..Tablet	1.95	2.06	2.16	0.01	1.63
Consistent.usability	2.95	2.86	2.71	3.19	3.13
Display.Color	2.94	2.57	2.48	3.97	2.95
EarbudsHeadphones	2.22	2.13	2.07	-2.32	2.37
Fast.Charging	2.40	2.33	2.27	-1.64	2.59
General.User.Interface	2.83	2.44	2.43	0.71	3.45
HealthFitness	2.35	2.21	2.28	-6.21	1.15
Internet.Browsing	2.06	2.54	2.44	5.21	2.84
Look.and.Feel	2.93	2.40	2.40	7.38	3.33
Mobile.Wallet..Pay	1.90	1.95	2.03	0.68	1.62
Navigation.Maps	2.67	2.42	2.10	-0.36	2.42
one.hand.use	2.43	1.78	1.77	-2.03	2.35
Packaging	2.87	2.34	2.47	3.00	3.19
Personalization	2.35	2.42	2.09	-1.17	1.85
Playing.Games	2.02	2.50	2.50	1.95	1.66
Preloads	1.57	1.49	1.69	1.05	1.17
Productivity.tools	2.53	2.21	2.11	-3.82	2.32
Refreshrate	2.67	2.69	2.52	0.60	3.38
Responsiveness	2.55	2.48	2.77	3.21	3.34
Security	2.56	2.55	2.90	0.98	2.51
Shape	2.50	2.13	2.20	1.04	2.72
Shopping..Coupon.Apps	1.77	1.71	2.61	2.74	1.73
Size	2.52	2.00	3.00	2.86	3.07
Smart.Home.Control	1.72	2.51	1.92	-0.58	0.69
Smart.Watches..Fitness.Bands	2.21	2.47	1.90	-4.89	1.09
Social.Media	2.54	2.68	2.50	1.23	2.56
Storage.Capacity	2.41	2.80	2.58	4.96	3.47
Streaming.Music	2.21	2.37	2.15	-1.11	1.95
Streaming.Video	2.63	2.71	2.49	0.53	2.22
Taking.Pictures.or.Videos	2.16	2.65	2.96	1.14	2.96
Tracking	1.85	2.28	2.48	-0.56	1.98
Voice.Assistant	2.01	2.40	2.04	-4.14	1.66
Waterproof..Dust.proof	1.80	1.74	1.67	2.82	1.67
Wireless.Charging	2.34	2.05	2.25	-1.80	1.57

CX of Apps, Design & Features and Current Satisfaction Tier

Tier	High Customer Experience Impact (Top 10%)
Value (under \$300)	Display Color, General UI, Look and Feel, Social Media
Mid (\$300 to \$599)	Camera, Cloud, Color Choices, Consistent Usability, Storage Capacity
Premium-Base/Mid (\$600 to \$899)	Basic Communication, Camera, Security, Shape
Premium-Mid (\$900 to \$999)	Camera, Look and Feel, Size, Taking Pictures or Videos
High-premium (\$1,000 to \$1,249)	Display Color, General UI, Look and Feel, Responsiveness
Hyper-Premium (\$1,250+)	ARVR, Basic Communication Apps, Navigation/Maps, Taking Pictures or Videos

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

CX of Apps, Design & Features and Current Satisfaction Tier

Notable Findings

Top 2 Drivers of Smartphone Satisfaction	
Value Tier	- Social Media - Display Color
Mid Tier	- Storage Capacity - Consistent Usability
Premium-Base/Mid	- Basic Communication - Shape
Premium-Mid	- Look and Feel - Taking Pictures/Videos
Premium-Hi	- General UI - Look and Feel
Premium-Hyper	- AR/VR - Taking Pictures/Videos

Features	Value	Mid	Premium Base/Mid	Premium Mid	Premium Hi	Premium Hyper
ARVR	1.52	-0.99	1.04	1.92	1.42	6.05
Aspect.Ratio	2.24	-0.11	2.51	2.58	2.66	-2.29
Basic.Communication	2.76	-1.41	3.49	2.68	2.56	5.81
Battery.Capacity	2.36	2.18	1.72	2.68	2.23	-2.68
Buttons	1.93	2.96	2.99	2.29	2.87	-2.41
Camera	2.55	5.18	3.07	3.06	2.90	3.00
Chassis.Material	2.15	3.61	2.33	2.27	2.68	-1.35
Cloud	1.94	5.20	1.64	2.17	1.47	-3.19
Color.Choices	2.08	4.64	2.61	2.51	2.37	1.26
Connectivity.with.PCLaptop..Tablet	2.09	-1.11	1.84	2.25	1.70	-1.50
Consistent.usability	2.81	6.93	2.95	2.80	2.87	3.78
Display.Color	3.02	3.88	2.64	2.35	2.95	0.62
EarbudsHeadphones	2.35	-2.82	2.03	2.23	1.92	2.43
Fast.Charging	2.24	2.44	1.92	2.20	2.23	-1.41
General.User.Interface	2.94	1.32	2.60	2.35	3.43	1.23
HealthFitness	2.46	-0.24	1.27	2.09	1.77	-1.20
Internet.Browsing	2.34	2.61	3.01	2.65	1.64	2.61
Look.and.Feel	2.83	3.18	2.68	3.14	3.42	1.69
Mobile.Wallet..Pay	2.14	-3.85	1.86	1.82	2.33	2.17
Navigation.Maps	2.40	2.47	2.26	2.35	2.83	4.65
one.hand.use	2.16	-0.82	2.09	2.03	2.08	-2.63
Packaging	2.68	1.61	2.57	2.87	2.48	3.29
Personalization	2.37	0.19	2.53	1.65	1.81	2.87
Playing.Games	2.15	-0.59	2.06	2.76	2.61	0.94
Preloads	1.25	1.05	1.97	1.25	0.83	-2.00
Productivity.tools	2.00	-0.91	2.40	2.18	2.17	2.19
Refreshrate	2.78	4.45	2.69	2.86	2.69	-0.14
Responsiveness	2.78	3.01	3.02	2.45	2.90	1.40
Security	2.74	0.60	3.09	2.73	2.13	3.98
Shape	2.42	-3.18	3.15	2.72	2.83	-0.76
Shopping..Coupon.Apps	1.84	-2.58	2.16	2.33	1.58	0.20
Size	2.59	1.33	2.38	2.94	2.30	1.57
Smart.Home.Control	1.89	-1.10	2.33	1.08	2.62	0.70
Smart.Watches..Fitness.Bands	1.34	0.58	2.11	1.29	2.54	0.15
Social.Media	3.31	-0.22	2.37	2.34	2.60	-0.02
Storage.Capacity	2.56	10.31	2.42	2.42	2.02	3.25
Streaming.Music	2.30	-4.39	2.25	2.37	1.89	2.89
Streaming.Video	2.37	-0.71	2.27	2.69	2.65	3.21
Taking.Pictures.or.Videos	2.55	1.27	2.60	3.16	2.50	7.16
Tracking	2.10	0.87	2.03	2.09	2.15	0.49
Voice.Assistant	1.91	0.95	1.92	1.88	1.99	3.43
Waterproof..Dust.proof	2.13	0.45	1.16	2.16	1.68	-2.89
Wireless.Charging	2.61	-1.69	1.99	1.36	2.72	-2.46

CX of Apps, Design & Features and Current Satisfaction Segment

Segment	High Customer Experience Impact (Top 10%)
Productives	Consistent Usability, Display Color, Look and Feel, Packaging
Expressers	Camera, General UI, Responsiveness, Storage Capacity
Social Savvies	Consistent Usability, General UI, Internet Browsing, Taking Pictures and Videos
Traditionalists	Camera, Personalization, Storage Capacity, Taking Pictures or Videos
Browsers	Display Color, Look and Feel, Responsiveness, Security

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

CX of Apps, Design & Features and Current Satisfaction Segment

Notable Findings

Top 2 Drivers of Smartphone Satisfaction

Productives	- Packaging - Display Color
Expressers	- Storage Capacity - Camera
Social Savvies	- Consistent Usability - Internet Browsing
Traditionalists	- Storage Capacity - Tracking
Browsers	- Security - Look and Feel

* Results are based on Partial Linear Square Regression results conducted with current smartphone satisfaction as dependent variable and app categories, HW/UI as predictor variables for each level in Gender, Age, Brand, Tier and Segment

Features	Productives	Expressers	SocialSavvies	Traditionalists	Browsers
ARVR	2.11	1.56	1.80	-4.05	0.70
Aspect.Ratio	2.24	2.48	2.14	0.54	2.51
Basic.Communication	2.72	2.54	2.75	4.96	2.39
Battery.Capacity	1.98	2.75	1.82	1.26	2.23
Buttons	2.39	2.49	2.45	2.09	2.29
Camera	2.71	3.25	2.62	5.18	2.89
Chassis.Material	2.54	2.69	1.73	-0.06	2.37
Cloud	2.05	1.91	2.11	1.21	2.27
Color.Choices	2.34	2.76	1.74	3.20	2.99
Connectivity.with.PCLaptop..Tablet	2.08	1.89	1.91	-2.53	2.46
Consistent.usability	2.81	2.83	3.26	4.19	2.84
Display.Color	2.90	2.49	2.75	2.63	3.00
EarbudsHeadphones	2.31	2.13	1.43	-1.84	2.10
Fast.Charging	2.13	2.73	1.75	-0.81	2.80
General.User.Interface	2.36	2.91	3.05	-0.53	2.97
HealthFitness	2.12	1.60	1.97	0.39	1.69
Internet.Browsing	2.22	2.51	3.10	4.26	2.61
Look.and.Feel	2.81	2.76	2.76	2.41	3.22
Mobile.Wallet..Pay	2.09	1.06	2.59	-1.21	1.96
Navigation.Maps	2.68	2.23	2.71	2.90	2.06
one.hand.use	1.98	2.42	1.49	-2.36	2.55
Packaging	2.90	2.68	2.32	3.39	2.38
Personalization	2.17	2.11	2.38	5.19	1.96
Playing.Games	2.16	2.28	2.46	-2.61	2.53
Preloads	1.42	1.73	1.88	-0.86	1.14
Productivity.tools	2.38	1.83	2.61	0.51	1.65
Refreshrate	2.53	2.81	2.80	2.05	3.03
Responsiveness	2.33	2.87	2.80	1.17	3.32
Security	2.69	2.49	2.63	2.75	2.51
Shape	2.52	2.58	2.41	-3.66	2.57
Shopping..Coupon.Apps	1.92	1.49	2.69	-0.96	2.11
Size	2.57	2.72	2.06	0.14	3.00
Smart.Home.Control	2.04	1.92	2.01	0.73	1.39
Smart.Watches..Fitness.Bands	2.55	1.76	2.08	-4.29	1.26
Social.Media	2.30	2.66	2.68	2.16	2.72
Storage.Capacity	2.57	3.16	2.41	5.49	2.99
Streaming.Music	2.04	1.88	2.48	-0.57	1.90
Streaming.Video	2.34	2.48	2.45	-0.20	2.61
Taking.Pictures.or.Videos	2.46	2.47	2.87	5.80	2.53
Tracking	2.07	2.05	2.05	-1.43	2.36
Voice.Assistant	2.24	2.05	2.53	-2.79	1.66
Waterproof..Dust.proof	2.05	1.99	1.17	2.00	1.50
Wireless.Charging	2.16	2.00	2.28	-2.64	1.98

Elements of Value – Current Smartphones

Elements of Value Patterns - Overall

Top Functional:

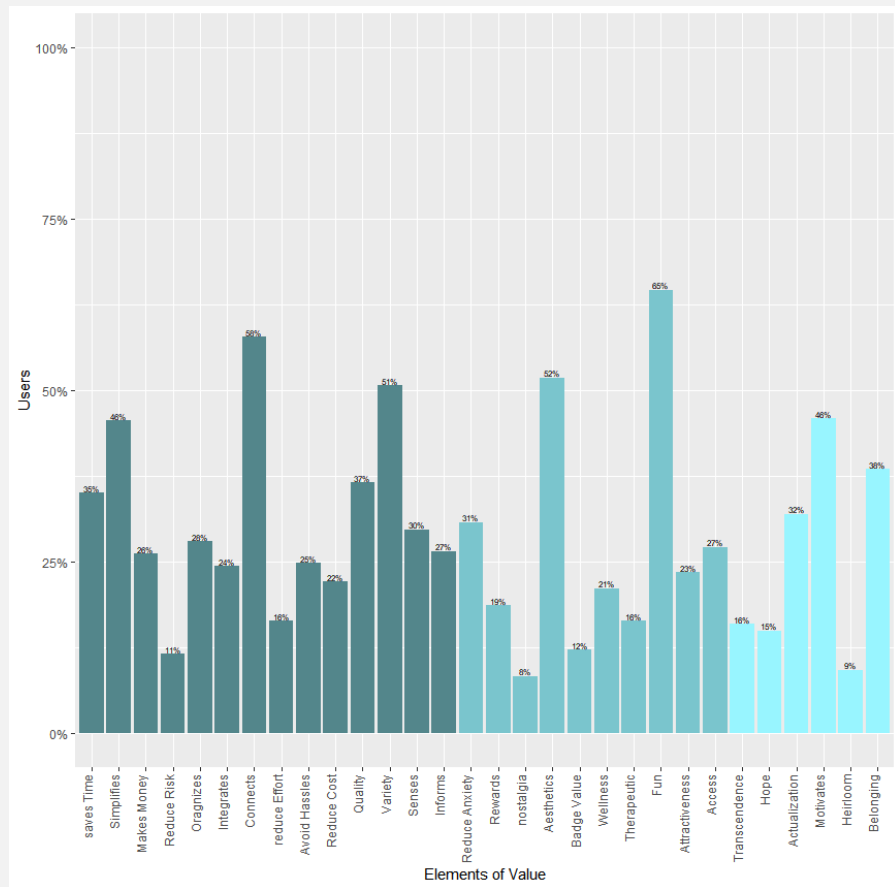
Connects	58%
Variety	51%
Simplifies	46%

Top Emotional:

Fun	65%
Aesthetics	52%
Reduces Anxiety	31%

Top Life Changing:

Motivation	46%
Belonging	38%
Actualization	32%

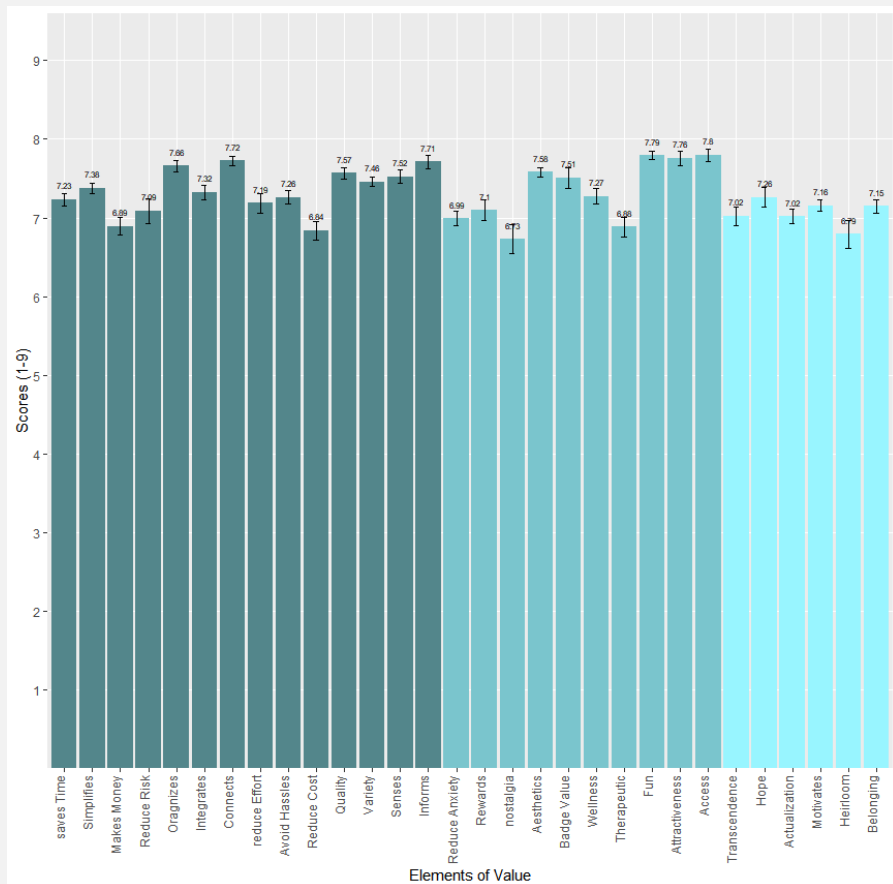


Elements of Value Scores - Overall

Using rating 7 as a rough baseline, the average of the following EOVs are below 7 :*

- Makes Money
- Reduce Cost
- Reduce Anxiety
- Nostalgia
- Therapeutic
- Heirloom

- Visual inspection only with no statistical testing performed
- statistical modeling will be conducted in subsequent phase



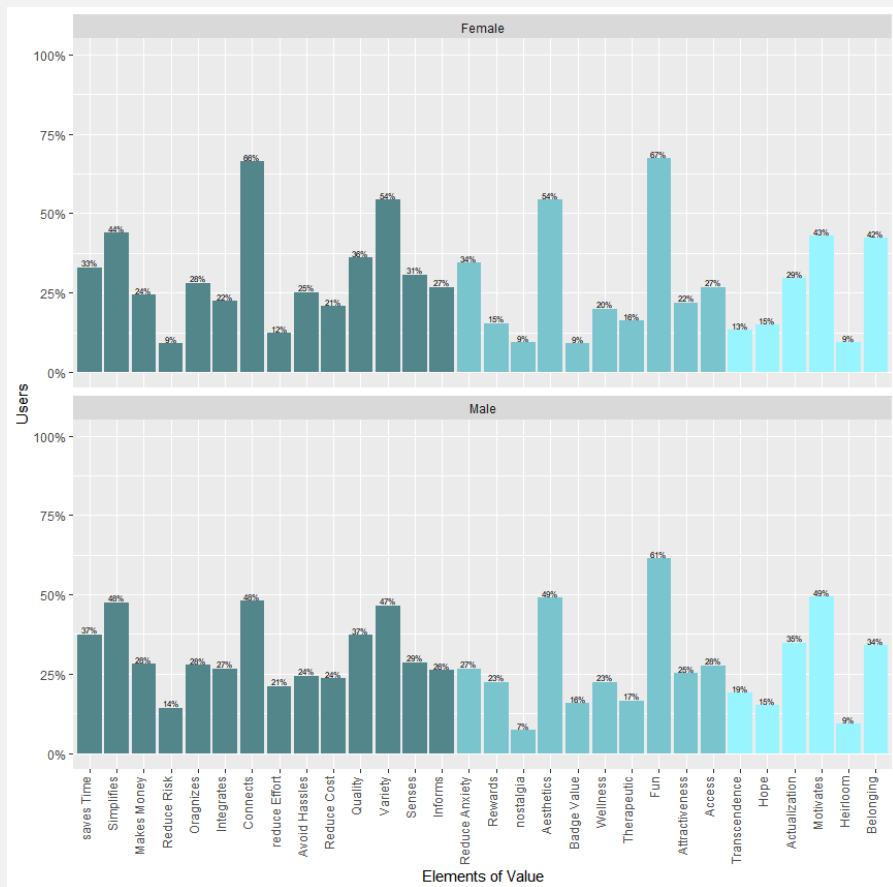
Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Elements of Value Patterns - Gender

Females generally select “Connects” more often as top functional value than males across different app categories

Males are more likely to select Rewards and Badge Value.

Females generally select “Belonging” more often as top life changing value than males while males select “Actualization” more often.



Elements of Value Score - Gender

Scores below 7:

Female

- Makes Money

- Reduce Risk

- Reduce Cost

- Rewards

- Nostalgia

- Therapeutic

- Transcendence

- Actualization

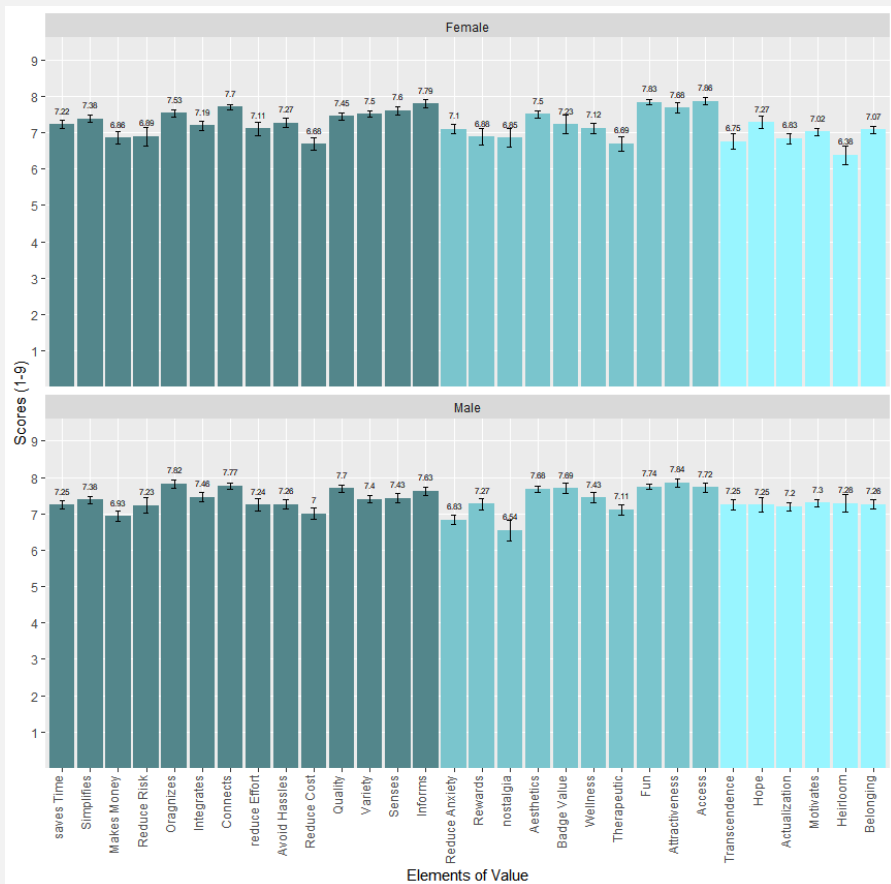
- Heirloom

Male

- Makes Money

- Reduce Anxiety

- Nostalgia

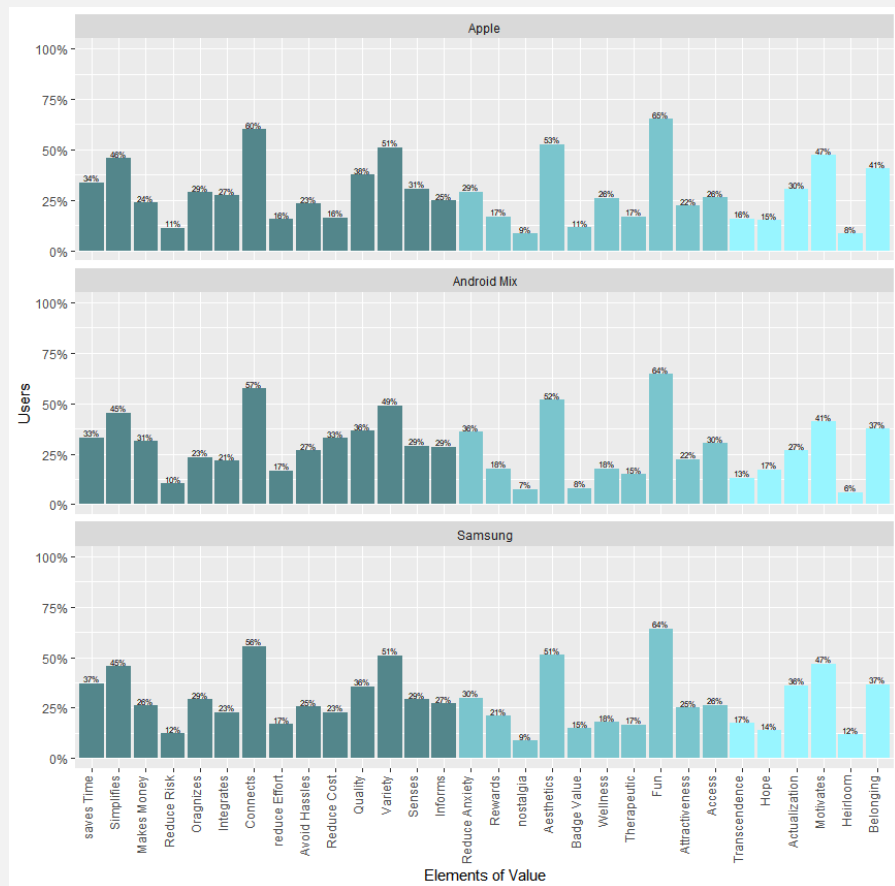


Elements of Value Patterns - Brand

“Reduce Cost” is selected less often by LO owners.

“Wellness” stands out a bit more with LO owners.

“Actualization” is strongest with Samsung owners .



Elements of Value Score - Brand

Scores below 7:

Android Mix

- Saves time
- Simplifies
- Makes Money
- Reduce Risk
- Reduce Effort
- Avoid Hassles
- Reduce Cost
- Quality

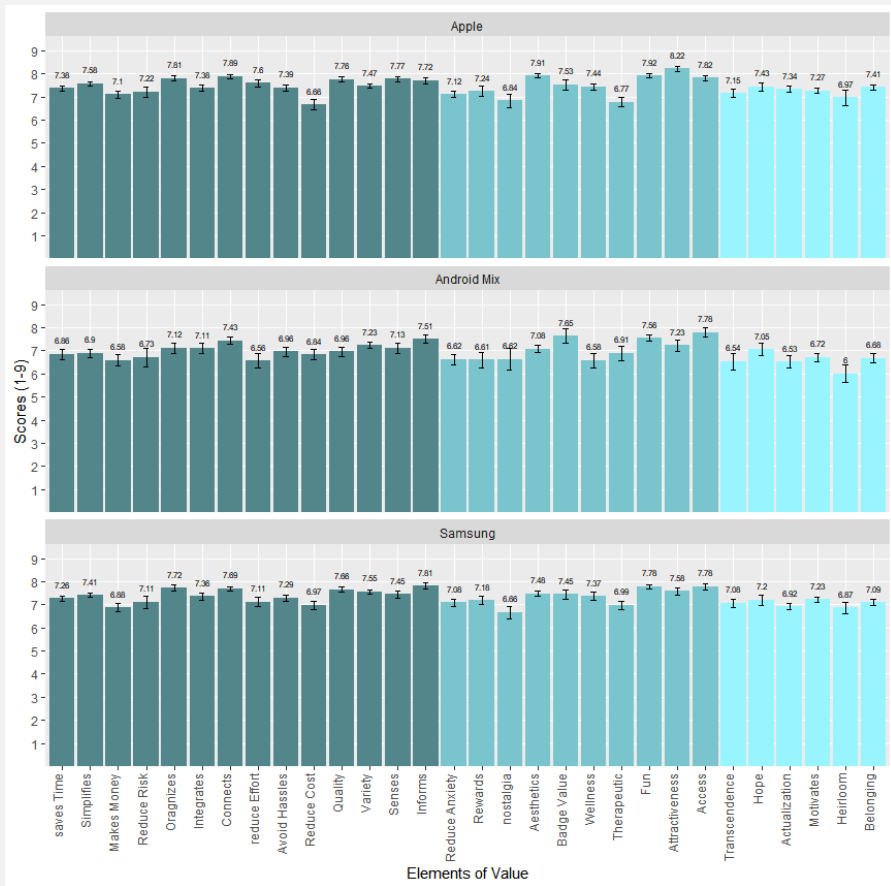
Apple

- Reduce Cost
- Nostalgia
- Therapeutic
- Heirloom

Samsung

- Makes Money
- Reduce Cost
- Nostalgia
- Therapeutic
- Actualization
- Heirloom

- Reduce Anxiety
- Rewards
- Nostalgia
- Wellness
- Therapeutic
- Transcendence
- Actualization
- Heirloom
- Transcendence
- Motivates
- Heirloom
- Belonging

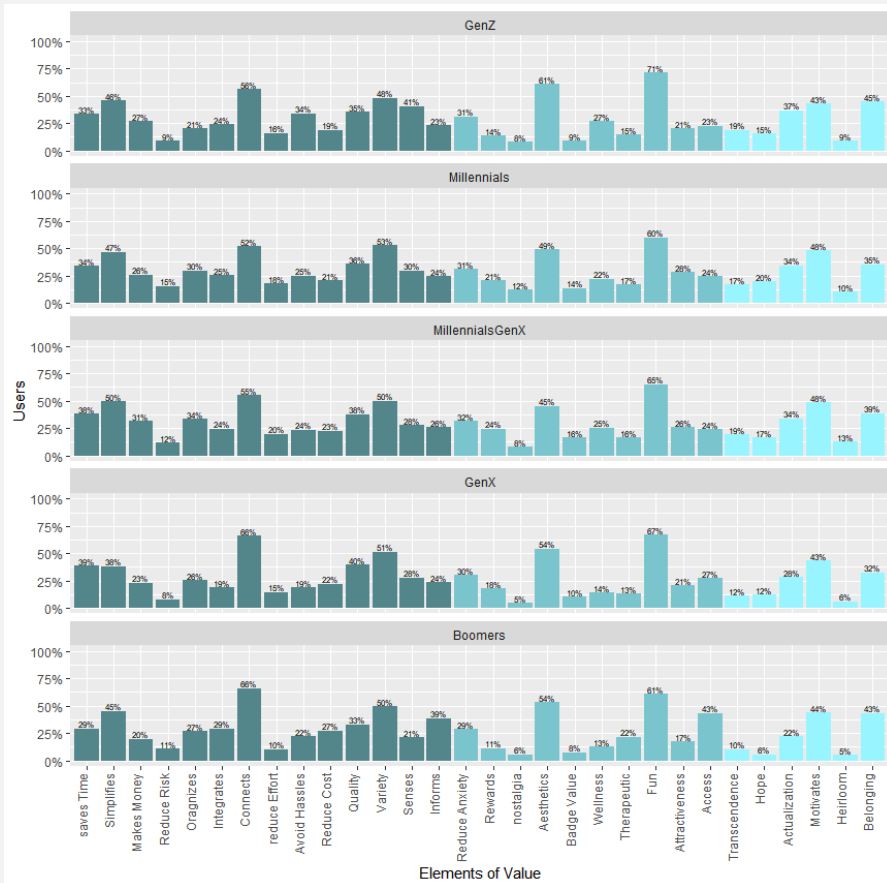


Elements of Value Patterns - Age

Boomers value “Informs” more often than other generations.

“Aesthetics” is strongest with Gen Z and declines with millennials and young Gen X but rises and levels off with Gen X and Boomers

“Hope” is less frequently selected with Boomers while “Belonging” is more frequently selected by Gen Z and Boomers



Elements of Value Scores -Age

Scores below 7:

Gen Z

- Saves time
- Makes Money
- Reduce Risk
- Integrates
- Reduce Anxiety
- Rewards
- Nostalgia
- Wellness
- Therapeutic
- Transcendence
- Actualization
- Motivates
- Heirloom

Millennials/Gen X

- Avoid Hassles
- Reduce Cost
- Rewards
- Reduce Anxiety
- Nostalgia
- Heirloom

Gen X

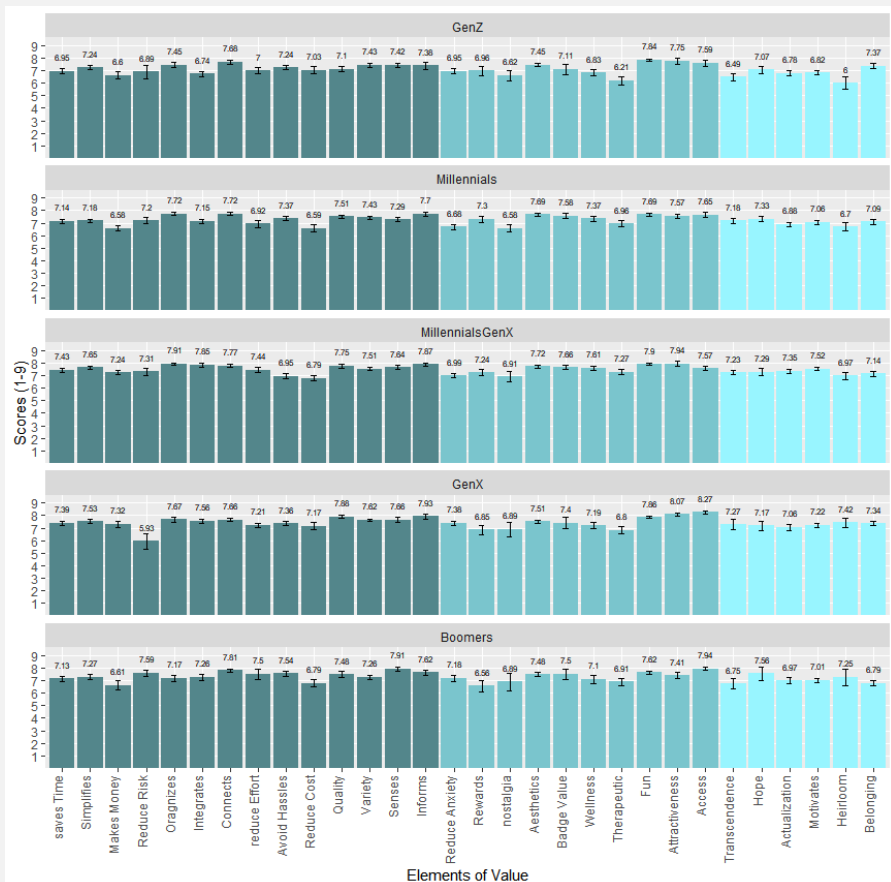
- Reduce Risk
- Rewards
- Nostalgia
- Therapeutic

Boomers

- Makes Money
- Reduce Cost
- Rewards
- Nostalgia
- Transcendence
- Belonging

Millennials

- Makes Money
- Reduce Cost
- Reduce Effort
- Reduce Anxiety
- Wellness
- Nostalgia
- Actualization
- Heirloom

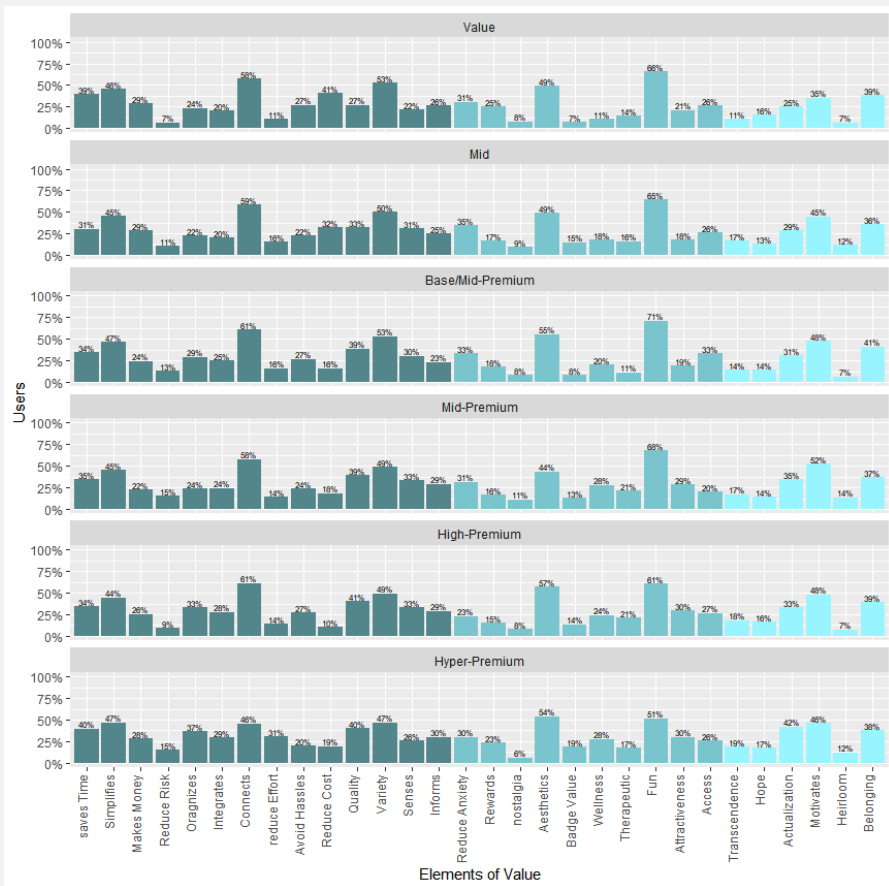


Elements of Value Patterns - Tier

“Reduce Effort” is higher with Hyper-Premium than other tiers.

“Fun” is lower with Hyper-Premium owners while “Badge Value” is lower with Value and Base/Mid- Premium.

“Actualization” is strongest with Hyper-Premium Tier .



Elements of Value Score - Tier

Scores below 7:
Value

- Saves time
- Simplifies
- Makes Money
- Reduce Risk
- Integrates
- Rewards
- Nostalgia
- Aesthetics
- Wellness
- Therapeutic
- Attractiveness
- Actualization
- Motivates
- Heirloom
- Transcendence
- Belonging

Mid

- Saves time
- Makes Money
- Reduce Risk
- Reduce Effort
- Reduce Cost
- Reduce Anxiety
- Rewards
- Nostalgia
- Badge Value
- Wellness
- Therapeutic
- Transcendence
- Hope
- Actualization
- Motivates
- Heirloom

Base/Mid-Premium

- Makes Money
- Reduce Cost
- Reduce Anxiety
- Nostalgia
- Therapeutic
- Transcendence

Mid-Premium

- Makes Money
- Reduce Cost
- Reduce Anxiety
- Nostalgia
- Therapeutic
- Actualization
- Heirloom
- Belonging

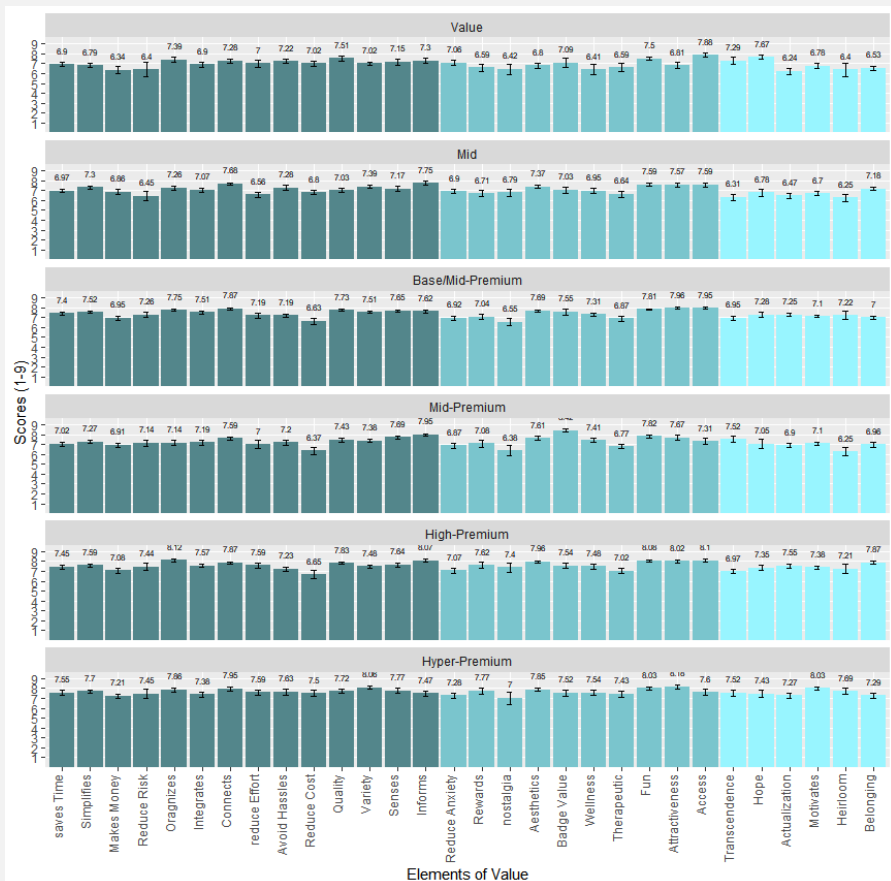
High-Premium

- Reduce Cost
- Transcendence
- Nostalgia
- Therapeutic
- Heirloom

Hyper-Premium

None

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

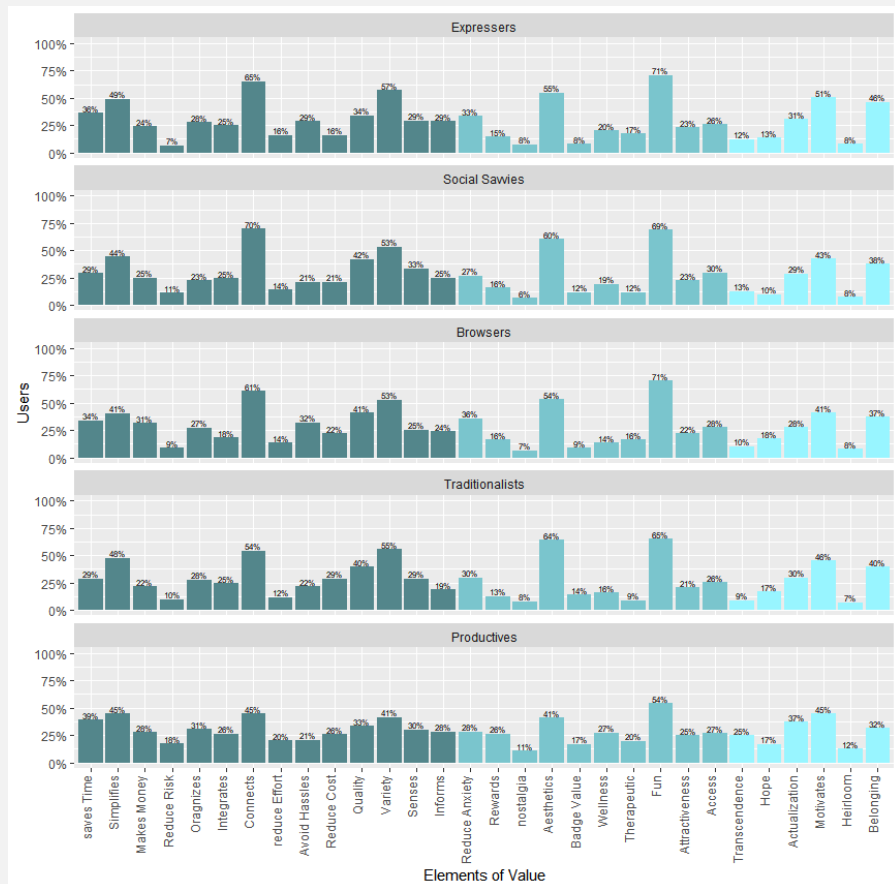


Elements of Value Score - Segment

“Variety” and “Connects are lower with Productives.

“Fun” and Aesthetics are lower with “Productives”.

“Belonging” is lowest with “Productives” while “Transcendence” is highest..



Elements of Value Patterns - Segment

Scores below 7:

Expressers

- Makes Money
- Reduce Cost
- Nostalgia
- Therapeutic
- Transcendence
- Hope

Social Savvies

- Makes Money
- Avoid Hassles
- Reduce Cost
- Reduce Anxiety
- Rewards
- Wellness
- Therapeutic
- Actualization
- Motivates
- Heirloom

Productives

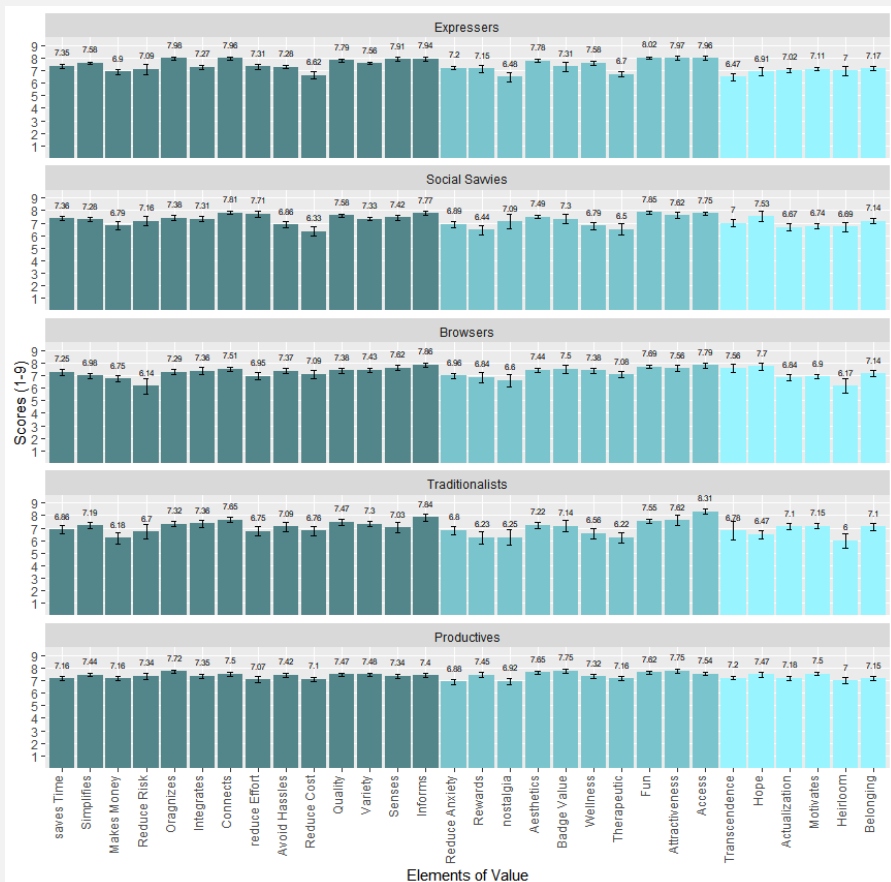
- Reduce Anxiety
- Nostalgia

Traditionalists

- Saves Time
- Makes Money
- Reduce Risk
- Reduce Effort
- Reduce Cost
- Reduce Anxiety
- Rewards
- Nostalgia
- Wellness
- Therapeutic
- Transcendence
- Hope
- Heirloom

Browsers

- Simplifies
- Makes Money
- Reduce Risk
- Reduce Effort
- Reduce Anxiety
- Rewards
- Nostalgia
- Actualization
- Motivates
- Heirloom

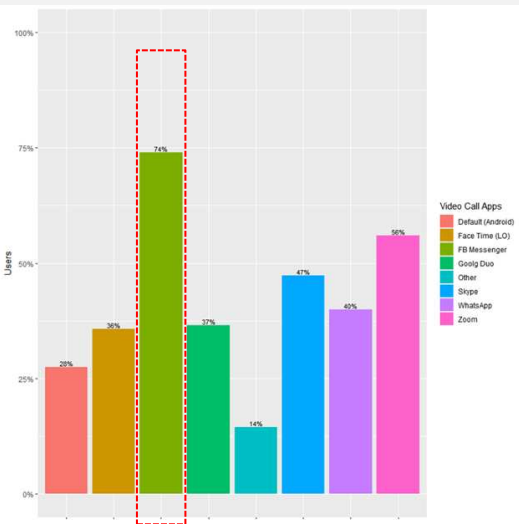


Basic Apps

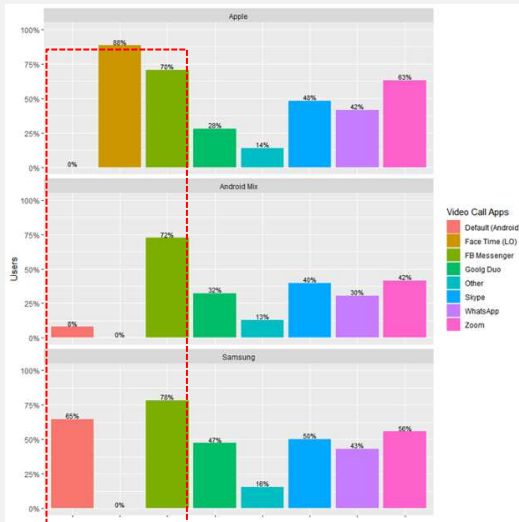
Video Call Apps

Facebook is the most widely used video chat app by gender, age, brand, product tier and segments. As Gen Z rate MS Skype and Google Duo lowest in CX Samsung should consider working with FB for video chat app or working with Google or MS to improve UX and market penetration of Duo or Skype.

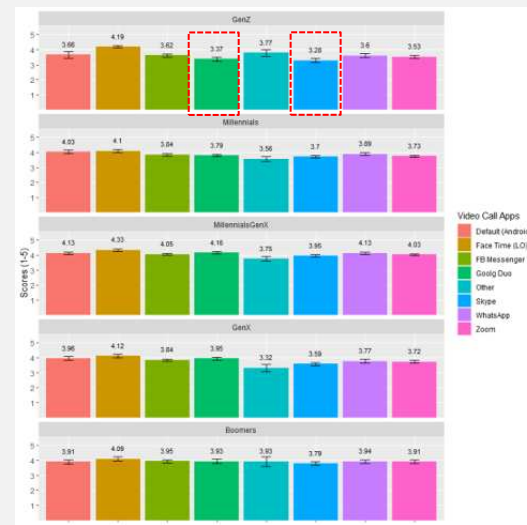
Facebook is the most widely used video chat app (74% market penetration)



Facebook is popular across brands, age, product tier and segment. Samsung & LO users use their default video call apps heavily



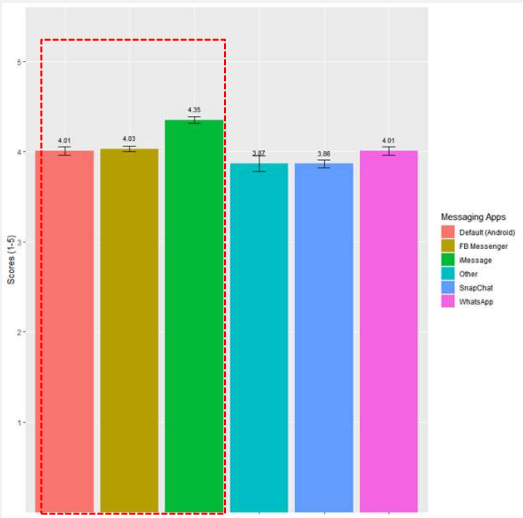
Gen Z rate CX of Google Duo and MS Skype second last and last



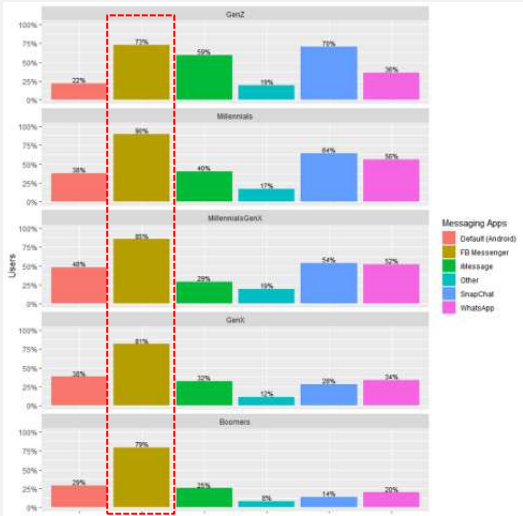
Messaging Apps

Facebook Messenger is the most widely used messaging app (overall 83% market penetration) by gender, age, brand, product tier and segments. With the overall CX score of 4.03, it may be worthwhile to integrate more directly into the UX of Samsung mobile devices to reduce friction in messaging and improve UX across different platforms and demographics.

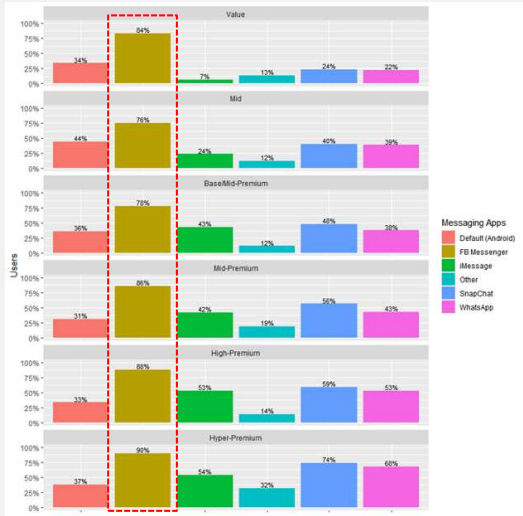
Facebook Messenger CX score as second best in messaging CX(4.03 vs. iMessage score of 4.34)



Facebook Messenger is also the most popular across brands, age, product tier and segment.



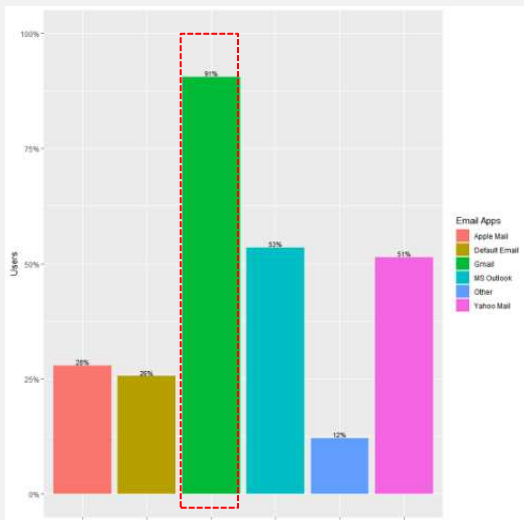
Facebook Messenger has a high and stable share across all product tiers



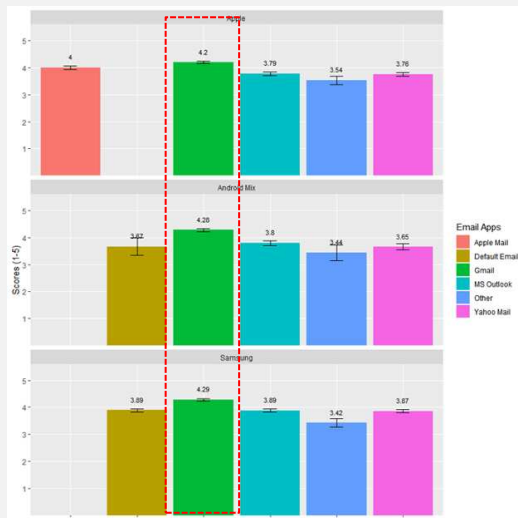
Email Apps

Gmail is the de facto email app for the market with top-rated CX. Samsung can consider de-prioritizing Email app as default email app in mobile devices.

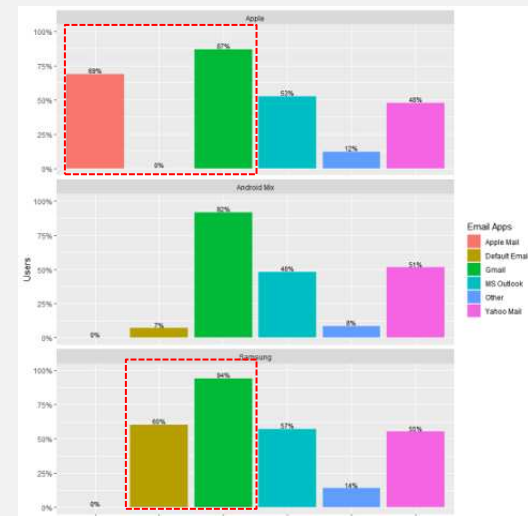
Gmail market penetration is the highest (91%) followed by MS Outlook (53%) and Yahoo Mail as distant second and third (51%).



Gmail also scores best in CX compared to other email apps compared.



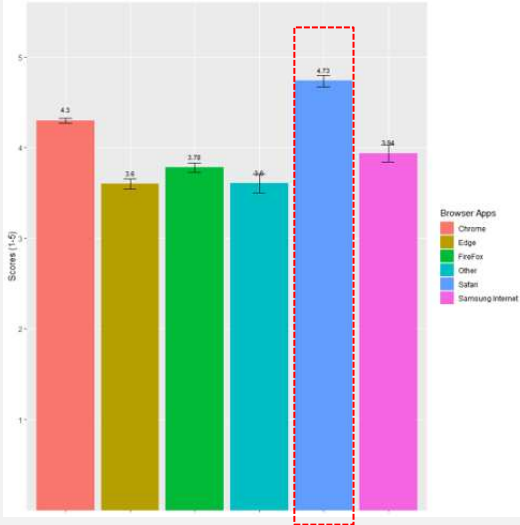
Gmail market penetration by brand surpasses both LO and Samsung Email App



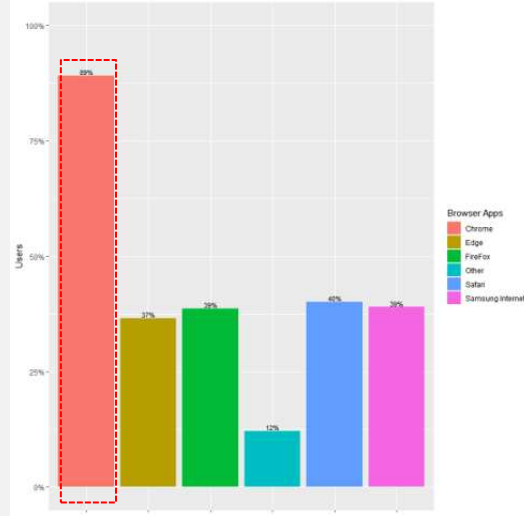
Browser Apps

Given that Google Chrome has the highest market adoption (89%) and its CX score is substantially better than other browsers (except LO Safari), Samsung can consider deprioritizing Samsung Internet for future mobile devices, especially for any models targeting Gen Z.

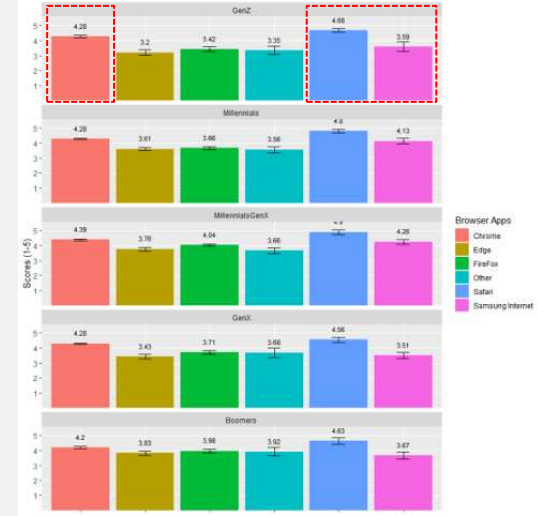
Safari (rated by LO users) has the best CX as mobile browser (compared to Samsung Internet as third place (rated by Samsung users)



Google Chrome has the highest market penetration of all mobile browsers.



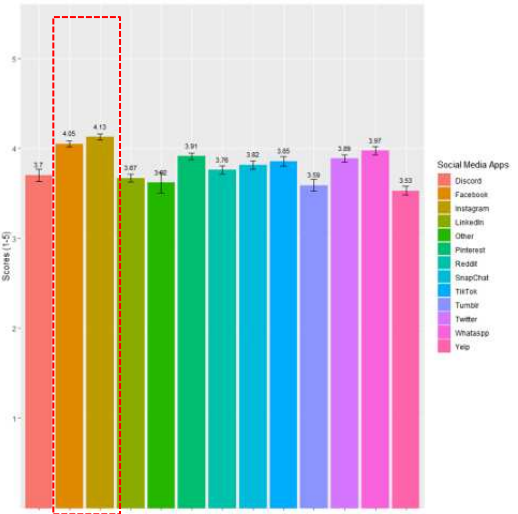
Gen Z rate CX of Samsung Internet significantly lower than Safari and Chrome



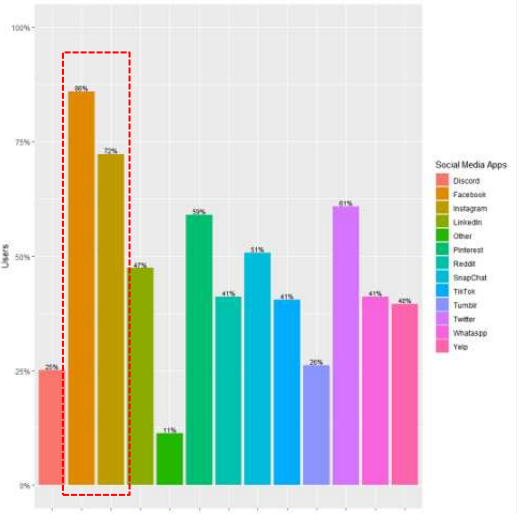
Social Media Apps

Given that Facebook and Instagram have the highest market adoption and their CX scores are substantially better than other social media apps, Samsung can consider more strategic relationship with Facebook to integrate their social media, messaging and video call experience to provide seamless UX.

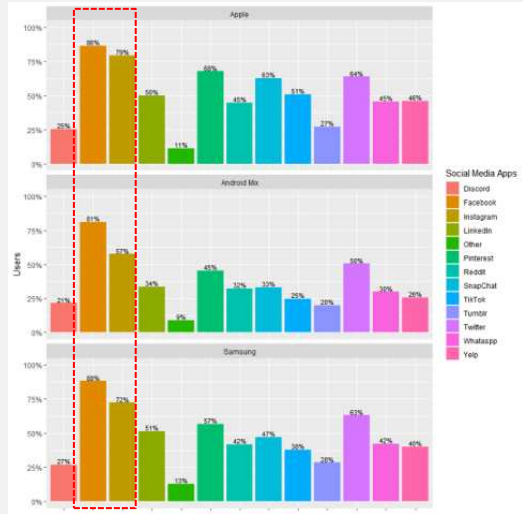
Facebook and Instagram have the highest CX scores among all social media apps



Facebook and Instagram are the most popular social media apps with 86% and 72% usage.



Facebook adoption is highest across all age groups. Instagram CX is highly rated by Gen Z.

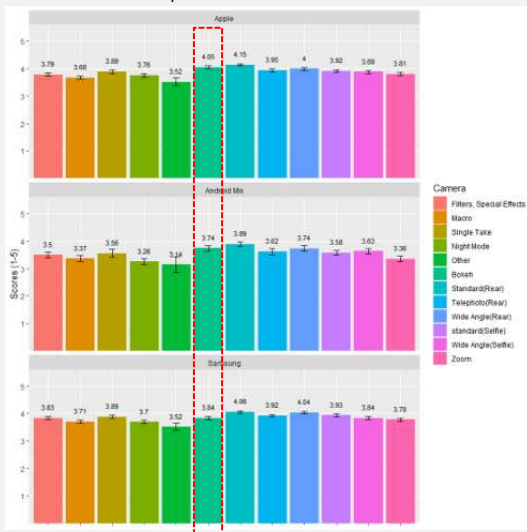


Photography & Videography

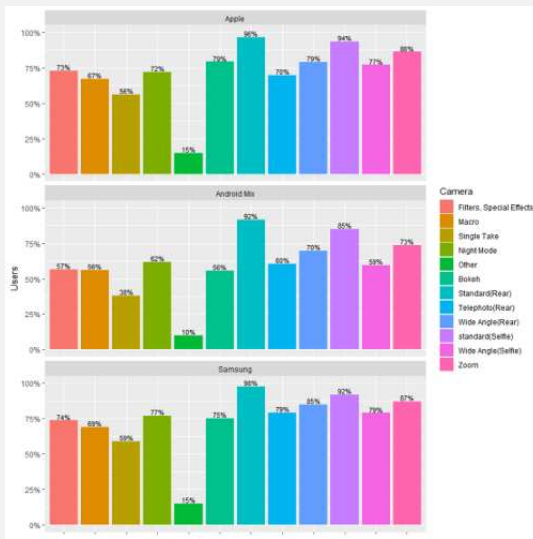
Camera

The lowest overall CX ratings of camera features* are “Night Mode”, “Macro” and “Zoom” (especially for Value and Mid Tier). These areas can form the focal area where improvements can be made for next generation products.

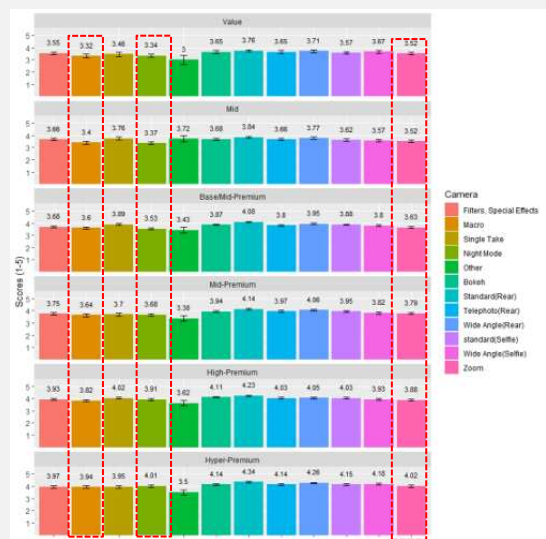
LO and Samsung Camera CX ratings (by features) are similar and outperform other Android smartphones in the US market



Camera features usage are also consistent across brands .



CX ratings of Macro, Night Mode and Zoom are especially low for Budget and Mid-Tier



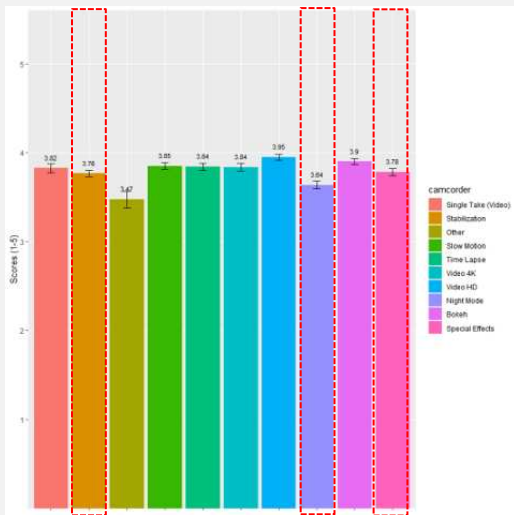
*excluding “Other” category which is a “catch all” bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

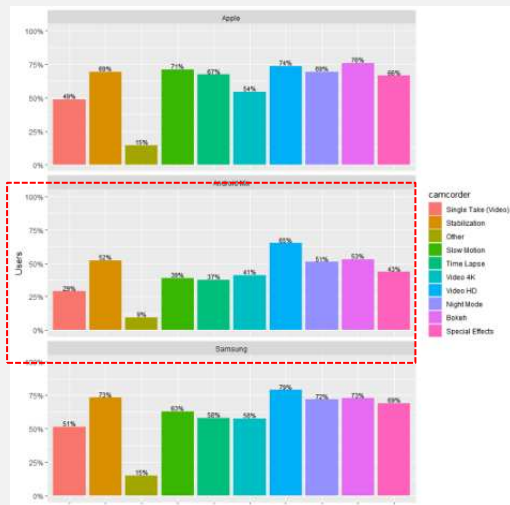
Camcorder

The lowest overall CX ratings of camcorder features* are “Stabilization”, “Night Mode” and “Special Effects” (especially for females). These areas can form the focal area where improvements can be made for next generation products.

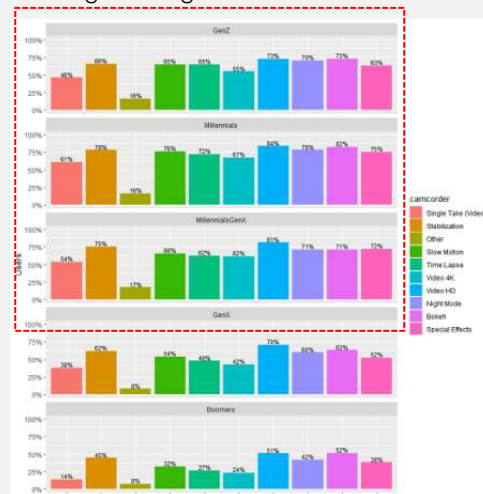
“Stabilization”, “Night Mode” and “Special Effects” are areas where female ratings of CX impact on the overall CX scores of those features



Camcorder features usage are also consistent across brands with less adoption of other Android users.



Gen Z, Millennials and younger Gen X use many camcorder features heavily. Usage declines with age starting at older Gen X.



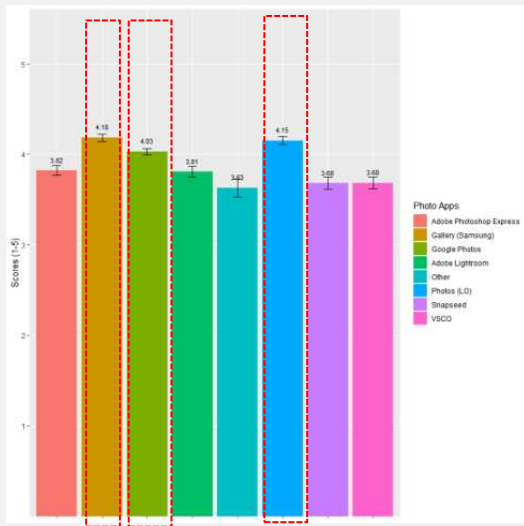
*excluding “Other” category which is a “catch all” bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

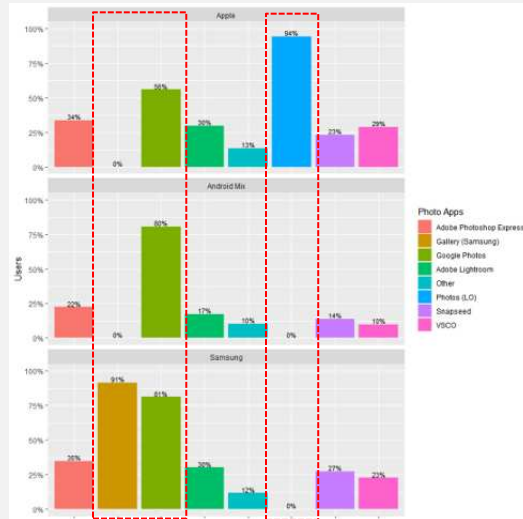
Photo Apps

Samsung Gallery is top-rated by Samsung users and tie with LO Photos by LO users. To maintain competitiveness, especially for Premium and Super premium devices, Samsung should consider building features supported by 3rd party apps such as Snapseed (through collaboration with Google?) or VSCO into the Gallery app.

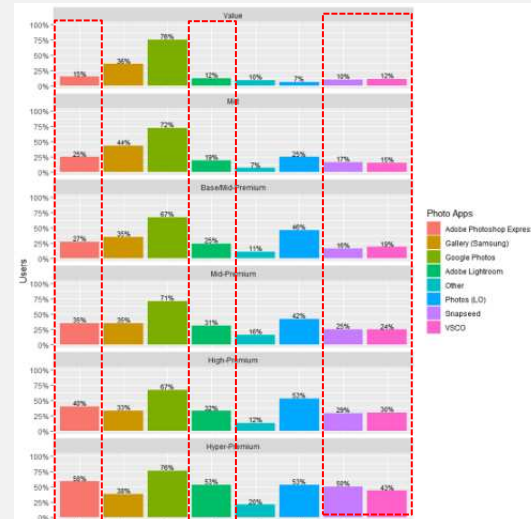
Both Samsung Gallery and LO Photos are rated highly by their respective owners with Google Photos following closely.



Samsung Gallery adoption rate is comparable to LO Photos (over 90%), while Google Photos has a strong presence across platforms



Usage of third-party photo apps increase rapidly with higher tiers reaching 50% or more for Snapseed, Adobe Photo Express & Lightroom



*excluding "Other" category which is a "catch all" bucket with many different responses other than the options offered.

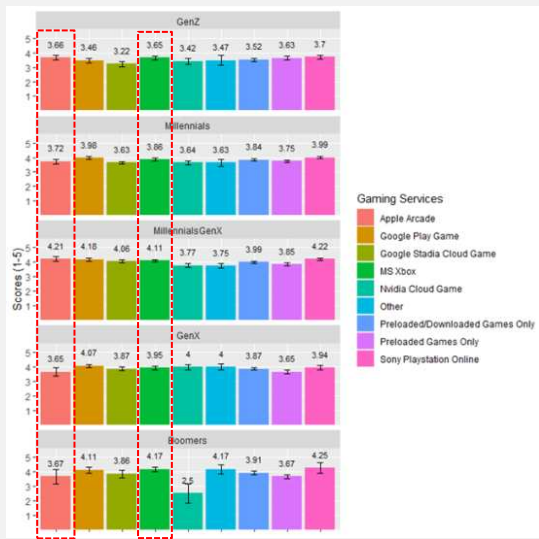
Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Entertainment

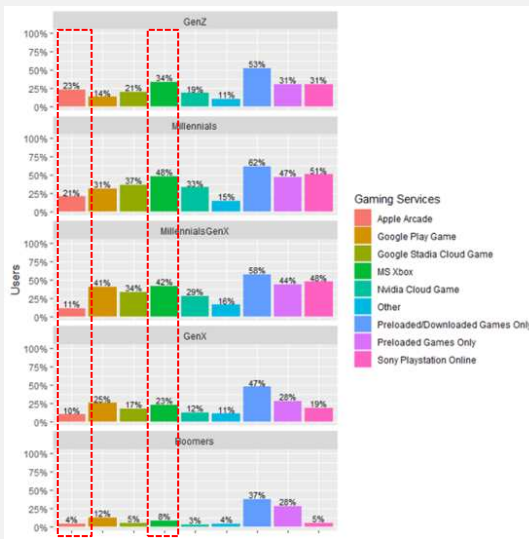
Gaming

Unlike LO Arcade whose presence declines rapidly with age, MS Xbox has a strong presence from Gen Z to Gen X. Coupled with the competitive CX scores with LO Arcade, Samsung and MS Xbox partnership have a distinct strategic advantage over LO. Samsung should also aim to bring MS Xbox experience to future premium and “premium lite” devices to further capture the market.

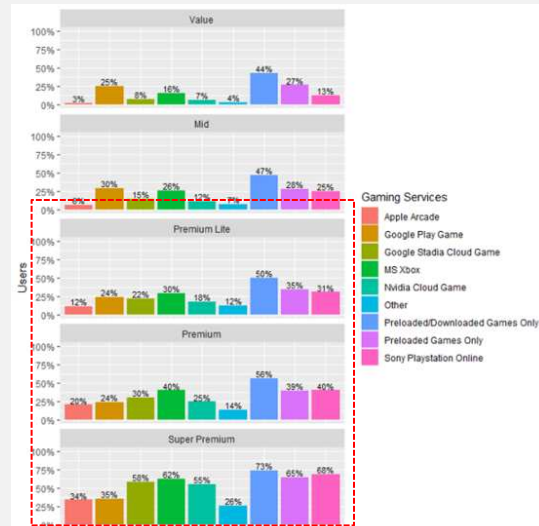
MS Xbox CX scores are competitive with LO Arcade scores for all age groups



Market penetration of MS Xbox exceeds LO Arcade for all age groups.



Super Premium Tier has the highest adoption for games and Value Tier has the lowest but good potential exists for premium and premium lite tiers



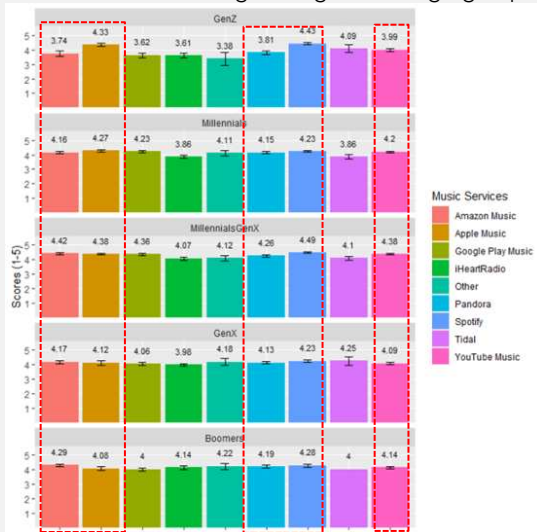
*excluding “Other” category which is a “catch all” bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

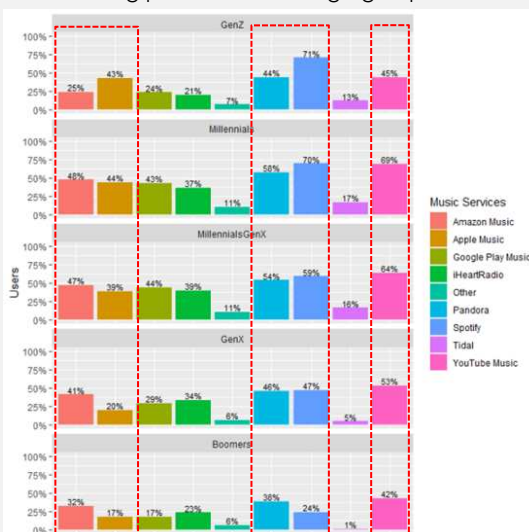
Streaming Music Services

With the strong market penetration of multiple music services beyond Spotify with the consideration of popularity of competitive music services with different age groups (or different products targeted to different age groups).

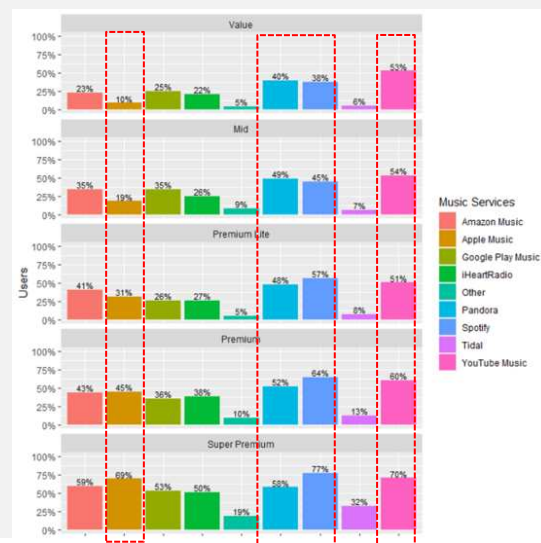
While Spotify has the highest CX ratings followed by LO, Amazon Music, YouTube Music and Pandora also have high ratings across age groups



Spotify has the highest adoption for younger age groups while YouTube Music and Pandora have strong presence for all age groups.



LO Music has a low market penetration into Gen X and Boomers as well as mid and value tier devices.



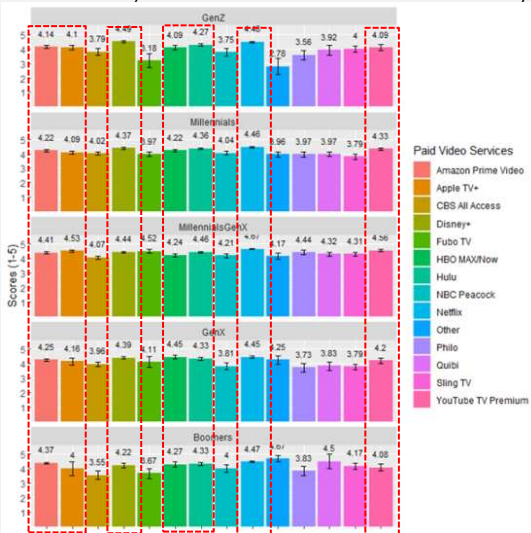
*excluding "Other" category which is a "catch all" bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

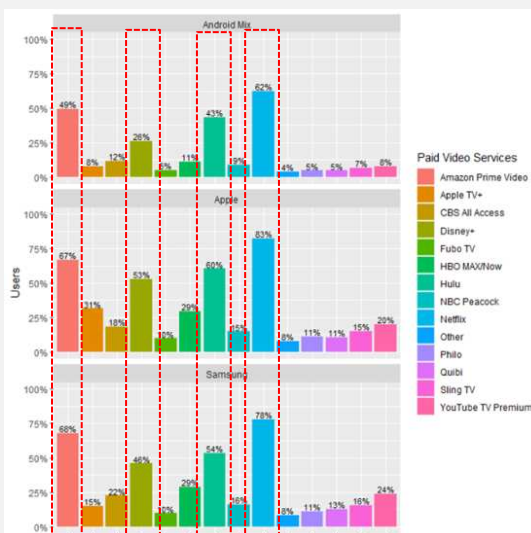
Paid Video Streaming Services

Collaboration with Netflix has given both Samsung and Netflix strategic advantage. If the partnership does not work out in the future, there are quite a few options in the market to consider based on the CX and the market penetration of the services. There are unique opportunities for bundling less known but attractive services to super premium tier to micro target prospective users.

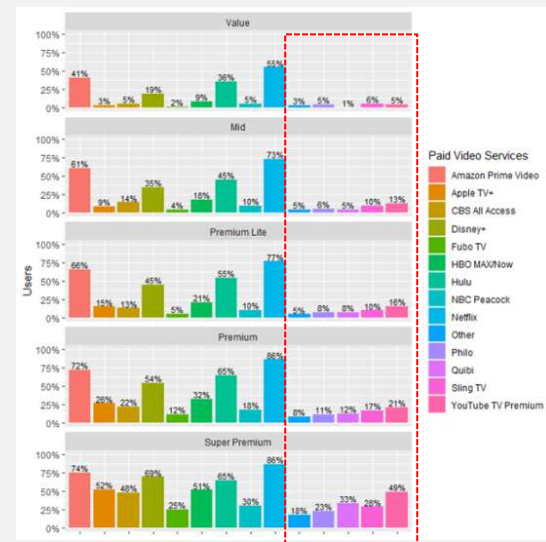
Netflix has the top CX scores for all age groups but popular services are also highly rated (Amazon, LO TV+, Disney+, HBO Max, Hulu, YouTube Premium)



Netflix, Amazon Prime Video, Hulu and Disney+ are the top 4 streaming video providers



Subscriptions of less "popular" video services increase with higher tiers.



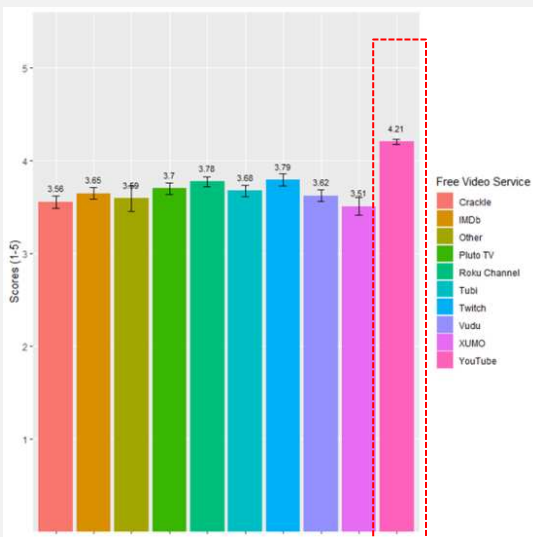
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Source: Elements of Value Phase 1 & 2 Survey (N=1105)

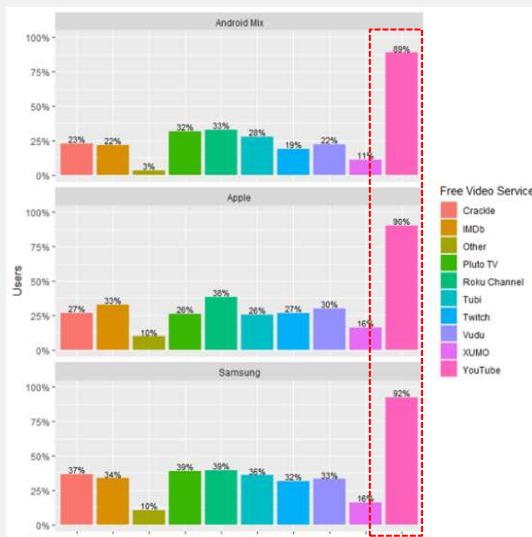
Free Video Streaming Services

As the need for free video streaming services cut across all tiers, Samsung has the strategic advantage of bringing the TV Plus service on Smart TV to mobile device as a value-added service to all device tiers and providing an additional cross sales opportunity with MDE integration.

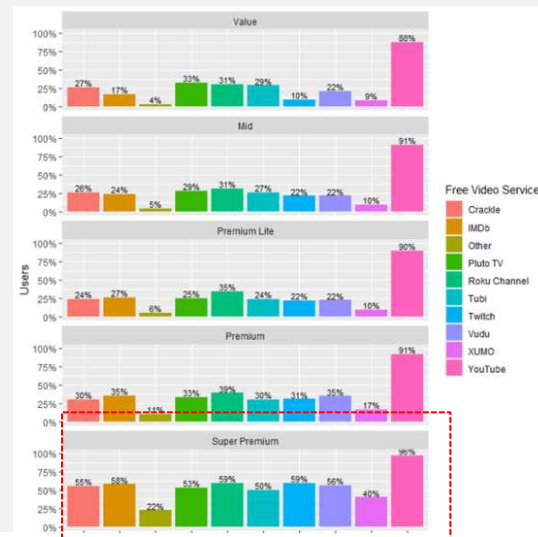
YouTube is by far the top free video streaming service substantially higher CX score.



YouTube has the largest market penetration across all age groups, brands, gender and tiers



Usage of different video services are relatively stable from value to premium tiers but increases sharply with super premium tiers



*excluding "Other" category which is a "catch all" bucket with many different responses other than the options offered.

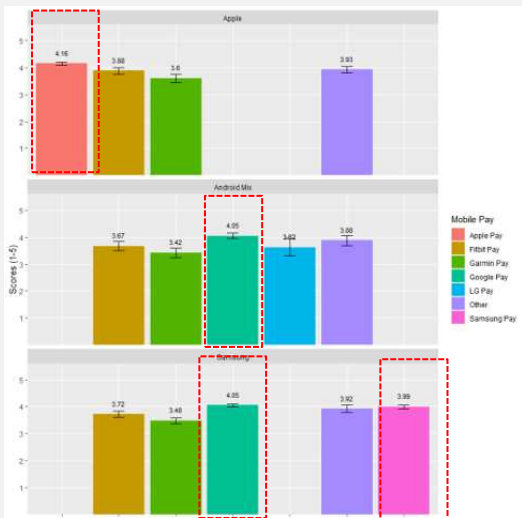
Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Practical Apps

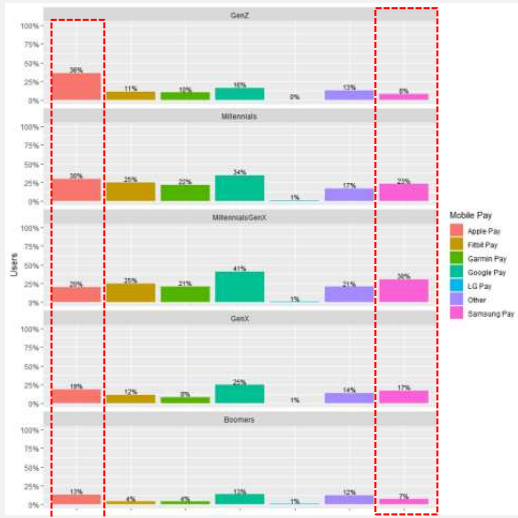
Mobile Pay

The use of Samsung Pay is under-indexed with Gen Z and Boomers . To expand the market reach of mobile pay Samsung Pay, it needs to be supported throughout the entire product tiers.

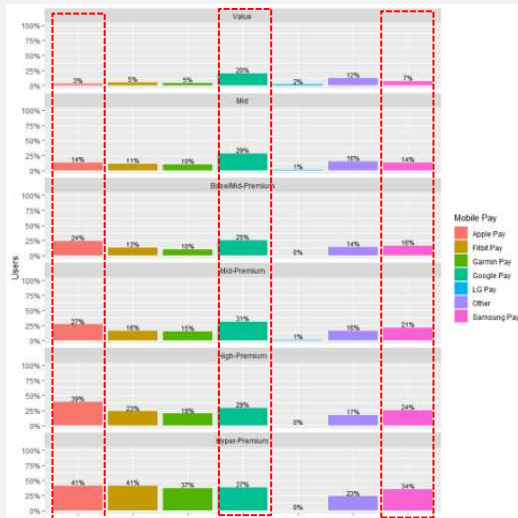
LO Pay has the highest CX score, followed by Google Pay and Samsung Pay.



LO Pay has the highest adoption from Gen Z and Millennials while Samsung Pay is more widely adopted by Millennials & younger Gen X



The adoption of Google Pay is consistent across all tiers compared to LO and Samsung (usage increases with higher-tiered devices)



*excluding "Other" category which is a "catch all" bucket with many different responses other than the options offered.

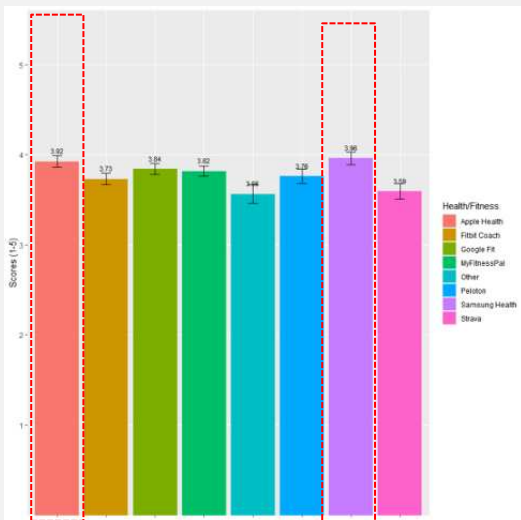
Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Health & Fitness App

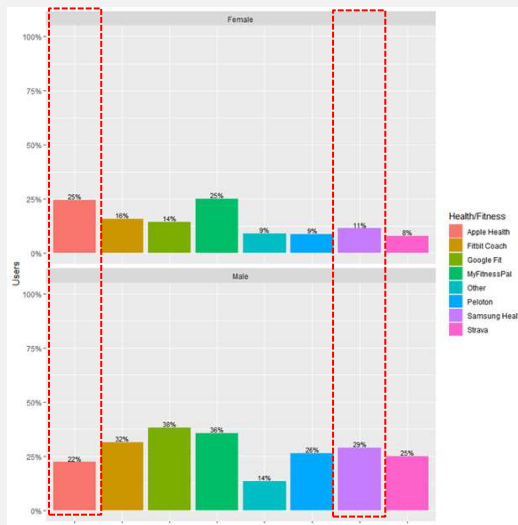
LO Health has a more balanced ratio of male and female users while Samsung Health (and most other health apps) have a ratio skewed towards male . Recent addition of female health features is a great stride towards diversity design of Samsung Health.

Data collected suggest blood pressure measurement to Samsung Health is a timely step.

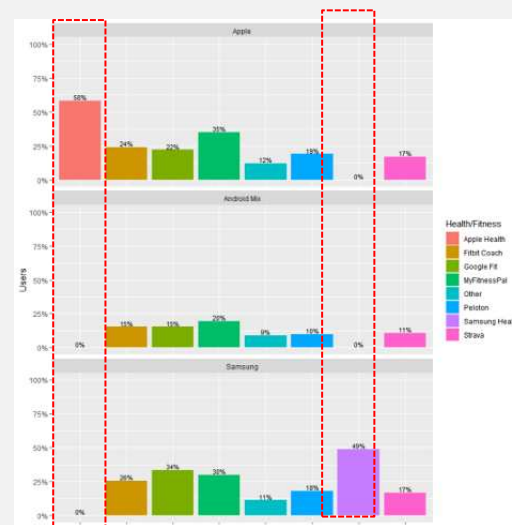
LO and Samsung Health lead the pack in CX scores, followed by FitBit Coach, MyFitnessPal and Google Fit.



LO Health has a much more even adoption ratio between genders than Samsung Health



LO Health has an adoption rate of 57% (Samsung at 48%)



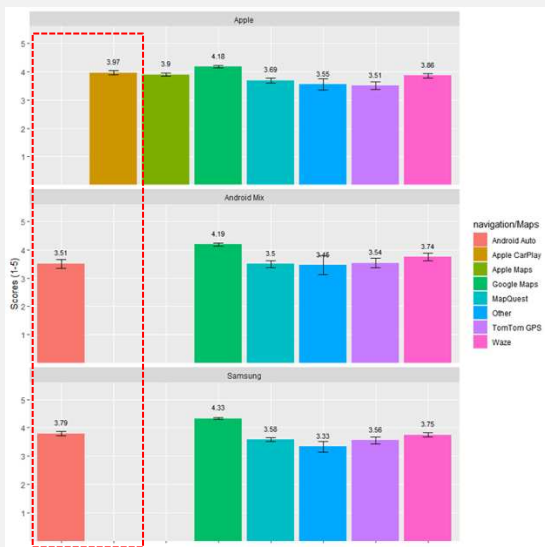
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Source: Elements of Value Phase 1 & 2 Survey (N=1105)

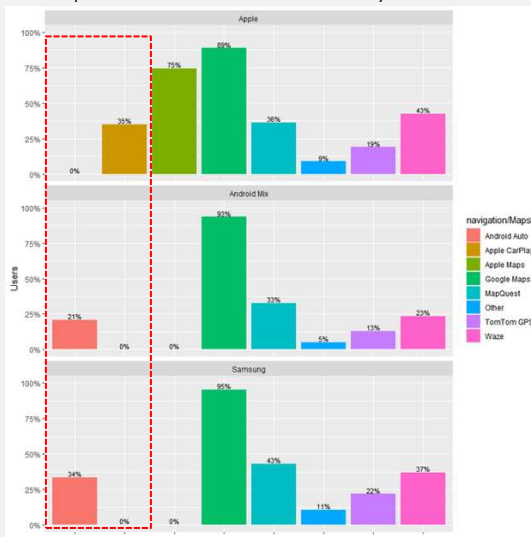
Navigation Apps

Google Maps has substantially higher CX scores than all other navigation maps. Samsung users are also found to rate CX score of Android Auto is substantially better than other Android users, suggesting that efforts making Android Auto UX better with Samsung devices paid off.

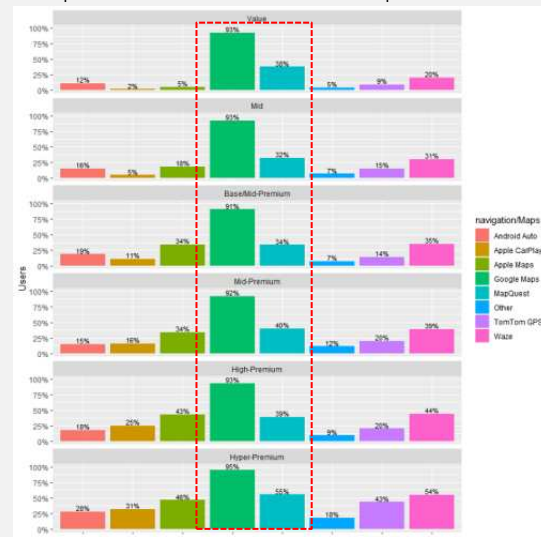
Android Auto CX score is lower than LO CarPlay.



Adoption of Android Auto on Samsung smartphones is similar to LO CarPlay.



Google Maps and Android Auto have higher adoption across all Android-based product tiers.



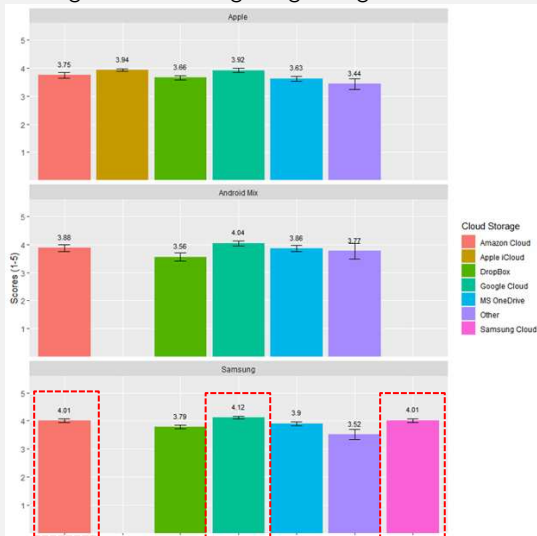
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Source: Elements of Value Phase 1 & 2 Survey (N=1105)

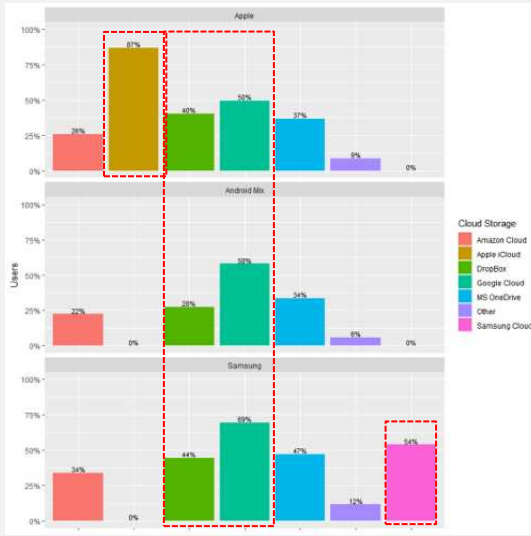
Cloud Storage Apps

LO iCloud is one of the most successful strategy to keep LO users locked in their ecosystem. The best chance to break into this wall-garden is to provide a cross-platform cloud service that can rival LO's CX with Samsung Smart Switch as a pry bar. Samsung will need to further collaborate with MS and/or Google to improve UX/CX of those cloud services.

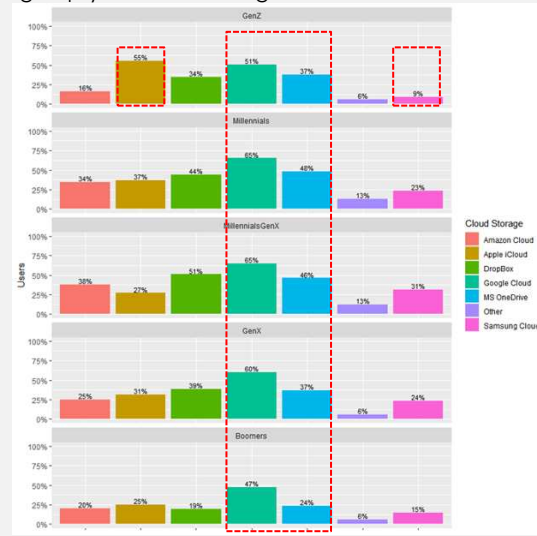
Both Google Cloud, Amazon Cloud and Samsung Cloud CX scores are comparable with Google cloud having a slight edge.



LO iCloud (87%) has a very high adoption compared to Samsung Cloud at 54%.



Both MS OneCloud and Google Cloud are more widely used by Gen Z (and other age groups), unlike Samsung Cloud.



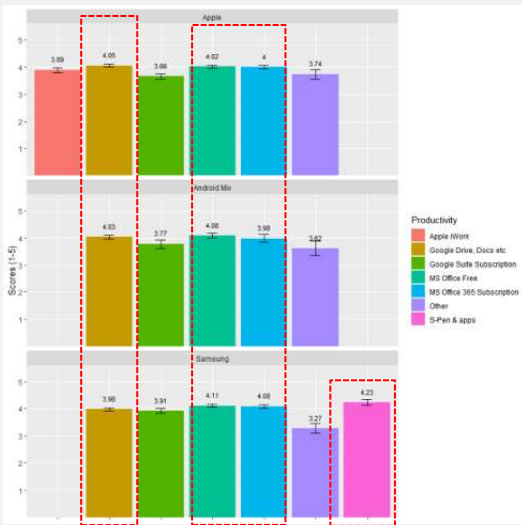
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Source: Elements of Value Phase 1 & 2 Survey (N=1105)

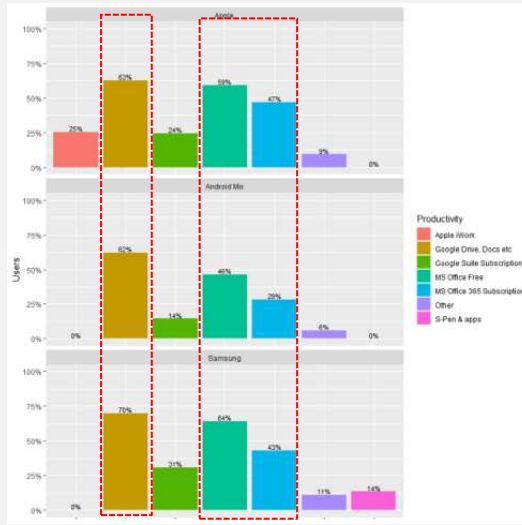
Productivity Apps

Productivity apps are one area that LO lags behind with iWorks. This can be another crack of the wall garden that Samsung, Google and MS can capitalize on. Getting productivity content to either Google Drive or MS OneDrive are important strategic steps to take. Samsung and MS can enable more integration of S Pen and apps to bring Note apps into mainstream.

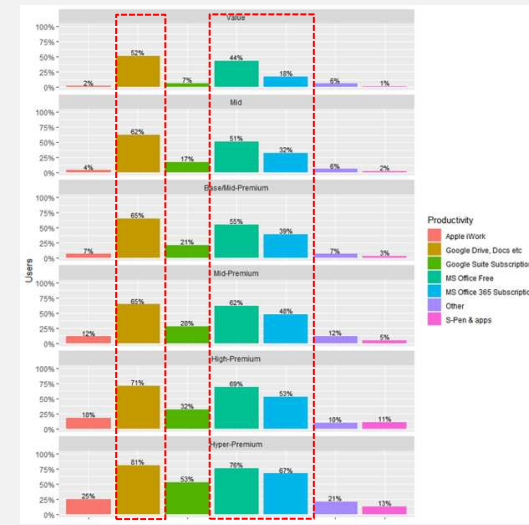
MS Office and Google Drive and Docs have the high CX scores. As productivity tool, Samsung S Pen & apps have highest CX rating



Google Drive and Docs are widely adopted by all brands (60%+), MS Office and 365 Subscription adoption are also relatively high across all brands



Common productivity tools should be expanded to lower tier devices as the adoption data below suggest.



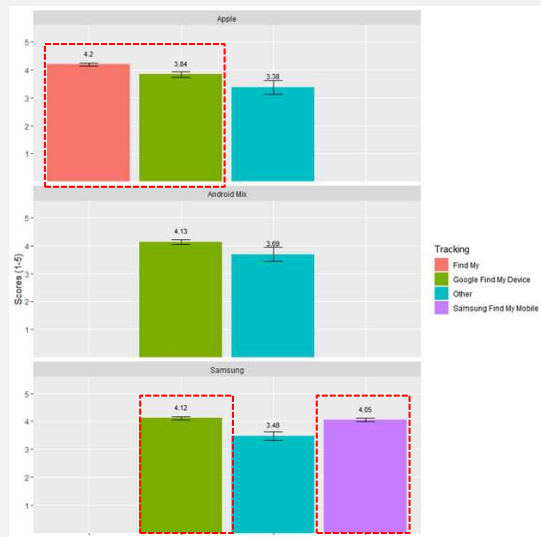
*excluding "Other" category which is a "catch all" bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

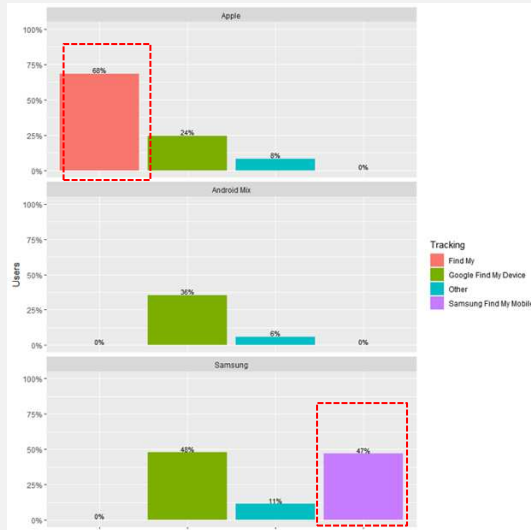
Tracking Apps

The relative high adoption of LO Find My is its integration of “Find My Friend” with “Find My Device”, making it especially appealing for Gen Z. In order to capture more Gen Z users, Samsung and partners like Google or Microsoft will need to take a holistic approach to app/product design by taking the emotional and social needs (aka elements of value) of Gen Z (and other age groups).

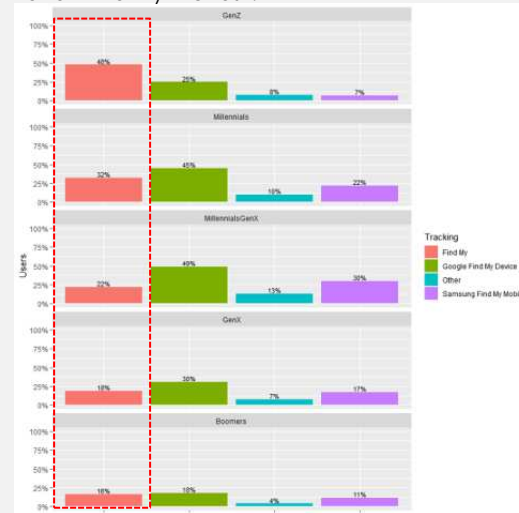
Both LO, Google and Samsung tracking apps have comparable CX scores



Adoption of device tracking apps is highest for LO, followed by Samsung.



The most likely reason for LO's high adoption for Gen Z is the combination of “Find My Phone” and “Find My Friends”.



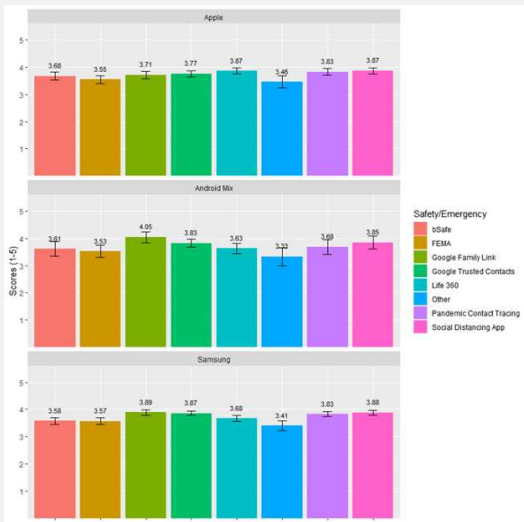
*excluding “Other” category which is a “catch all” bucket with many different responses other than the options offered.

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

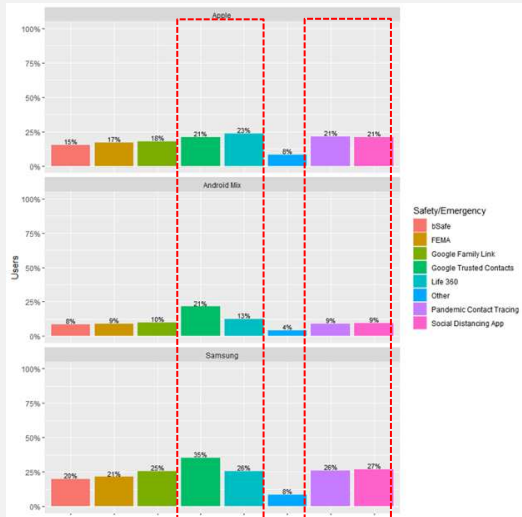
Personal/Family/Community Safety Apps

This class of apps generally have substantially lower CX ratings compared with other app categories. With the pandemic, social unrest and more frequent occurrence of natural disasters, safety apps may become more important in meeting users' emergent needs (e.g., to satisfy the need to “reduce anxiety” in the elements of value)

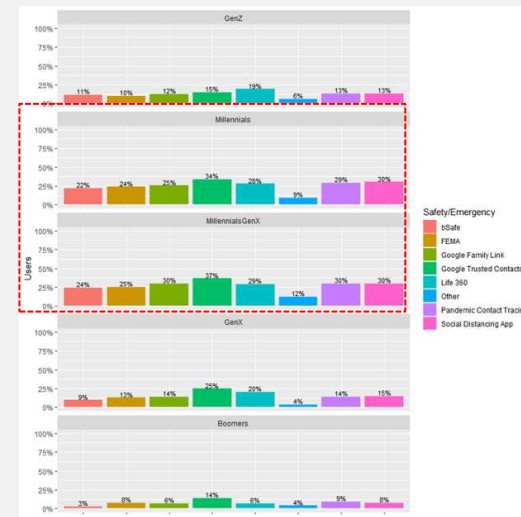
Most safety apps considered here have average CX scores lower than other categories.



Google Trusted Contacts and Life 360 are more widely used across brands. Apps related to pandemic are also used more widely.



Personal/Family/Community Safety Apps are more popular with Millennials and younger Gen X, possibly because of young children



*excluding “Other” category which is a “catch all” bucket with many different responses other than the options offered.

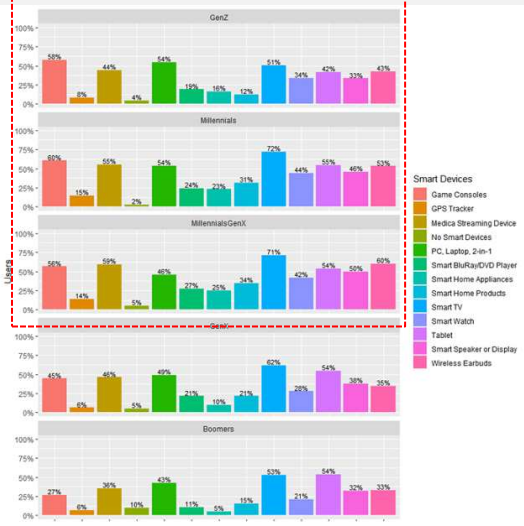
Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Ecosystem

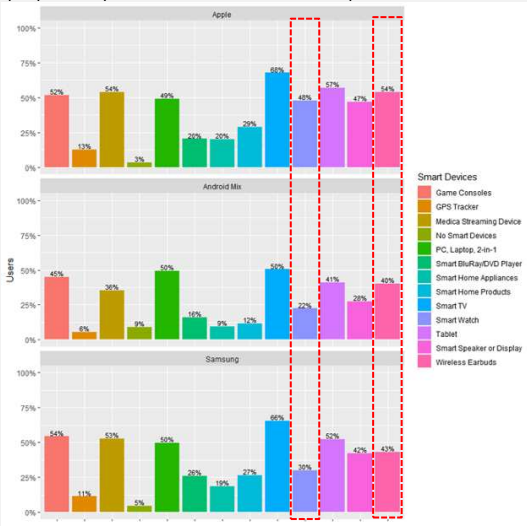
Smart Device Ownership

Smart devices are prevalent across all age groups starting with Gen Z, peaking with Millennials and younger Gen X, and declining lightly of most smart types from Gen X to Boomers. Smart TV and tablets are popular with any age group, with Boomer ownership of more than 50%.

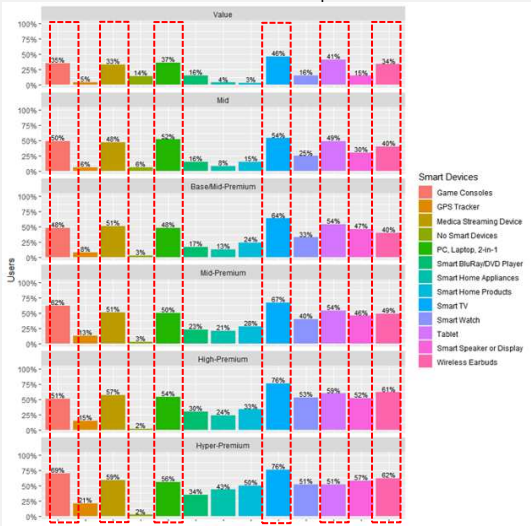
Gen Z, Millennials and younger Gen X have higher ownership of different types of smart device.



LO has higher ownership of wireless earbuds (54%) and smart watches (48%), reflecting the popularity of LO watches and airpods.



Game consoles, streaming media devices, PC/Laptop/2-in-1, smart TV, tablet and wireless earbuds are at 30%+ ownership with Value Tier



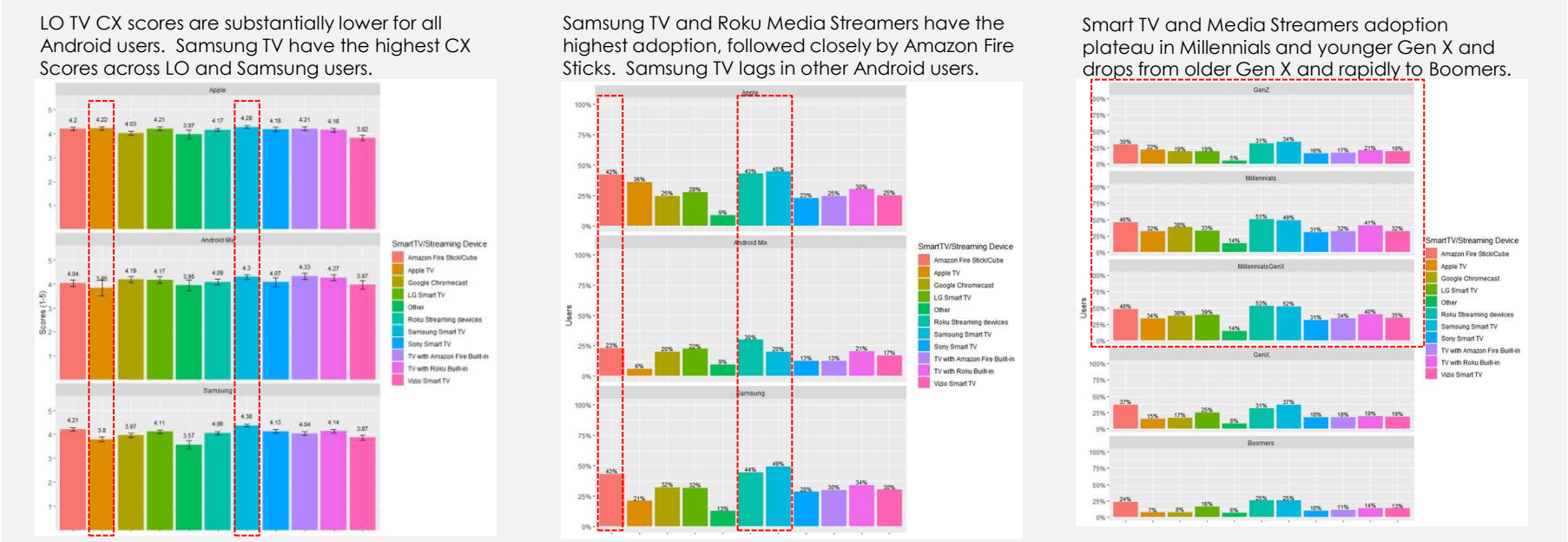
Smart TV and Media Streaming Devices

Samsung TV has a strong market penetration and highest CX ratings across all brands, all age groups, gender, tiers and segments. This strategic advantage can be leveraged to expand the reach and brand value of Samsung products. Roku is the major competitor to watch out with the value and quality of its TV (e.g., from TCL). Amazon is also a competitor in value smart TV market.

LO TV CX scores are substantially lower for all Android users. Samsung TV have the highest CX Scores across LO and Samsung users.

Samsung TV and Roku Media Streamers have the highest adoption, followed closely by Amazon Fire Sticks. Samsung TV lags in other Android users.

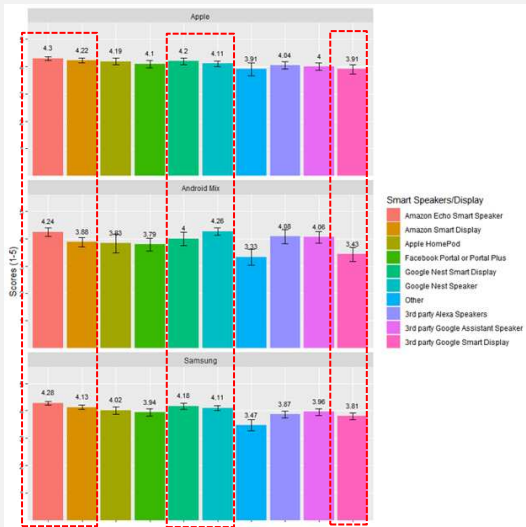
Smart TV and Media Streamers adoption plateau in Millennials and younger Gen X and drops from older Gen X and rapidly to Boomers.



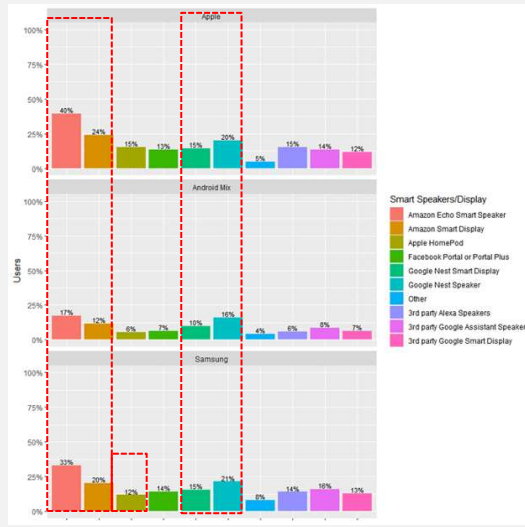
Smart Speakers & Display

Amazon Echo products dominate the market across all demographic and segmentation breakouts. Ownership of LO HomePod by Samsung users (12% vs. 15% by LO owners) suggests that there are quite a few dual ecosystem users. Gen Z is also found to be less satisfied with HomePod, which may represent a crack in the strong grip of LO ecosystem in Gen Z.

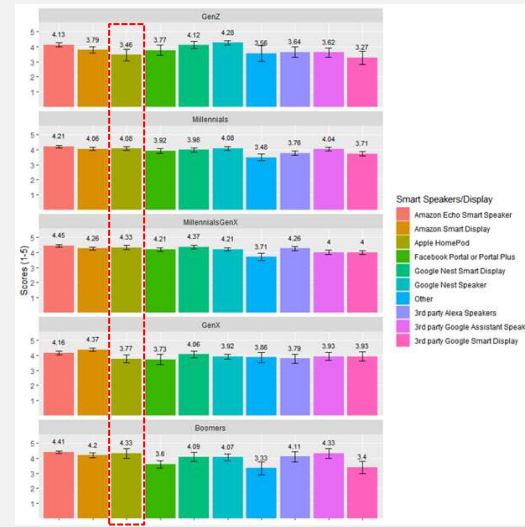
Both Amazon and Google smart speakers have high CX scores. Third-party Google smart displays do not score well.



Amazon echo products have the highest adoption, followed by Google Nest products. 12% Samsung owners have LO HomePod.



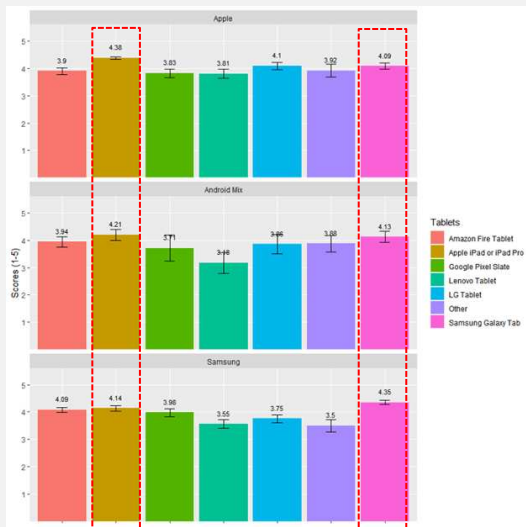
Gen Z rated LO HomePod lower than other age group, which is contrary to their typical ratings of LO products. Gen X also rate it lower.



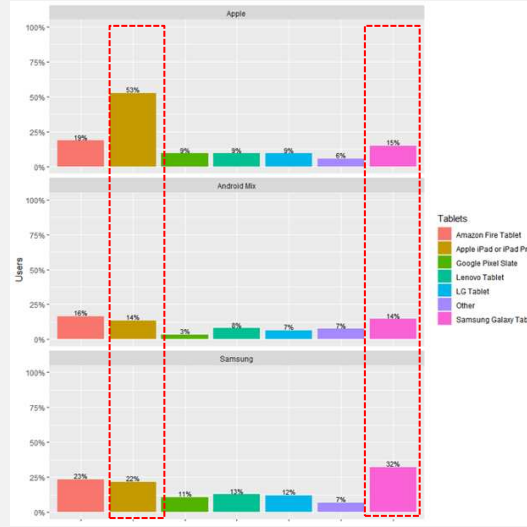
Tablets

Tablets ownership stretches across all demographic, brand and product tier breakouts. One common theme identified is that Gen Z has a substantially low CX ratings for non-LO products. If this trend continues, Samsung and Android will lose this generation.

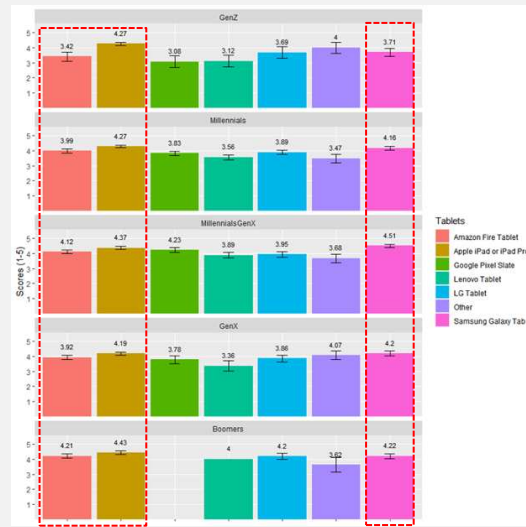
Both LO and Samsung tablets receive high scores across brands, with LO tablets highest scores by LO and other Android users



53% LO users own LO tablets. 32% Samsung users own Samsung tablets. Amazon, LO and Samsung have more than 10% cross brand penetration.



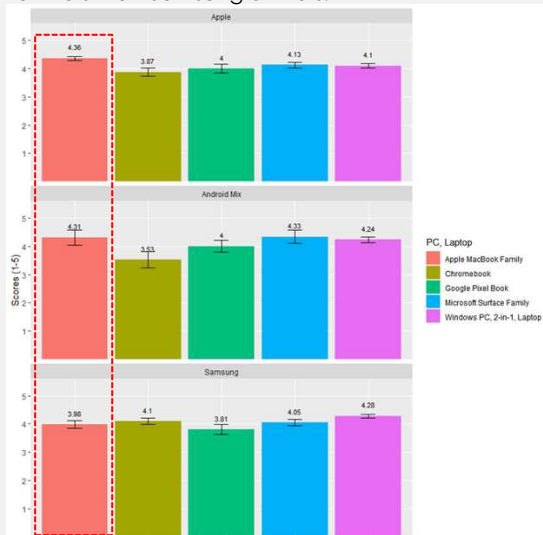
Gen Z rated both Amazon Fire Tablet and Samsung Tablet substantially lower than LO tablets.



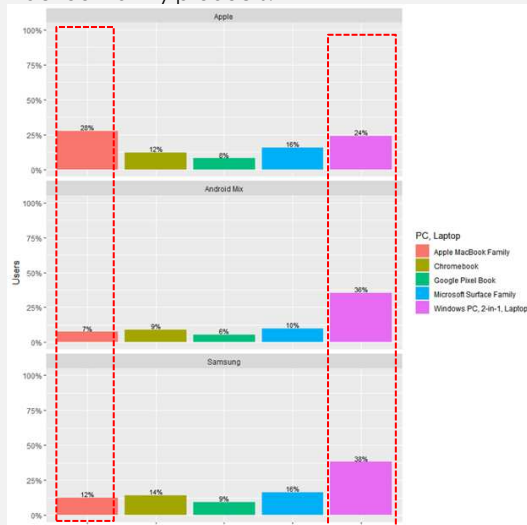
PC, Laptop, 2-in-1

Windows PC and Laptop have the largest user base across all tiers, age groups and segments. A smooth and seamless integration of mobile experience is important across all tiers for long term success.

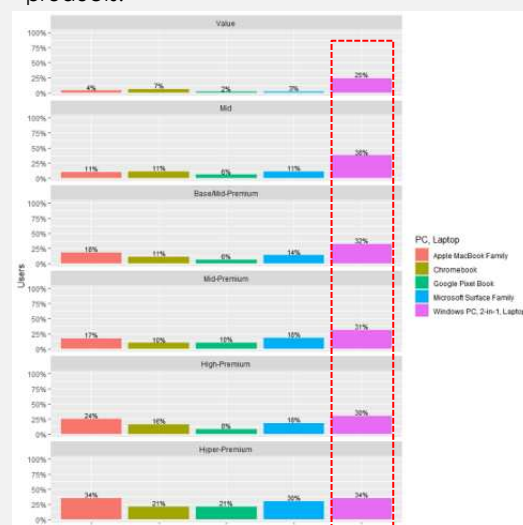
MacBook family products CX scores are substantially higher with LO and other Android owners than Samsung owners.



Windows PC with Surface Family have highest market share across brands. 28% LO users own MacBook family products.



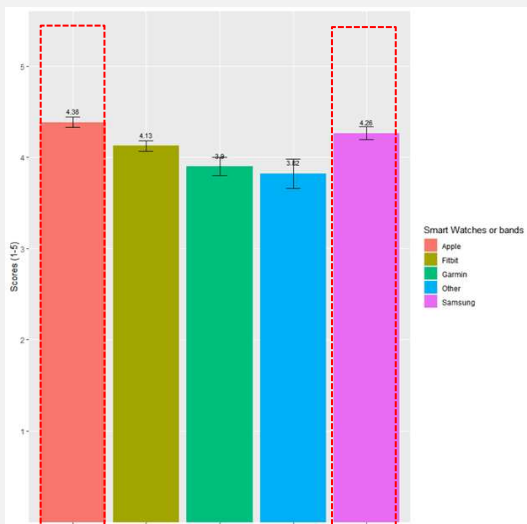
Windows PC and laptop have sizable share across all product tiers compared to other products.



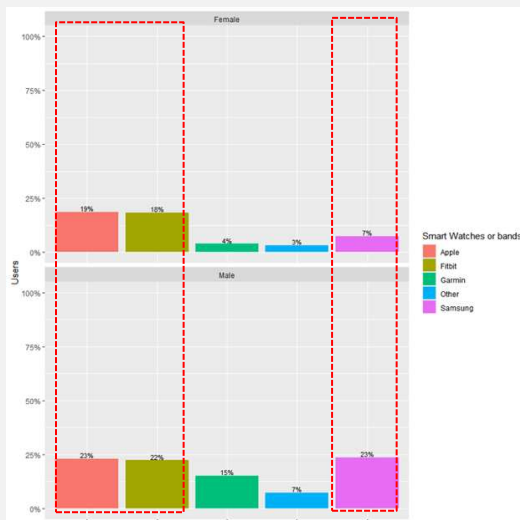
Smart Watches

Although LO dominates the smart watch market, Fitbit maintains a competitive presence among LO, Samsung and other Android brand smartphones. Smart watches from LO is more widely adopted by Gen Z and Boomers compared to Samsung Smart Watches. More females use LO and Fitbit smart watches than Samsung (and Garmin) smartwatches.

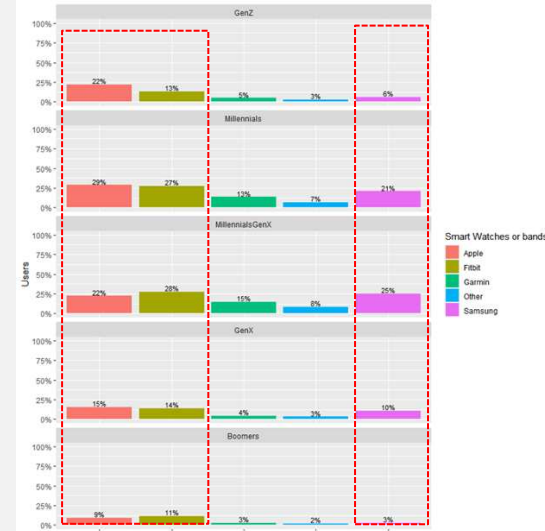
LO and Samsung smart watches have high CX scores across brands.



Both LO and Fitbit have more balanced gender ratio while Samsung ownership is skewed to male.



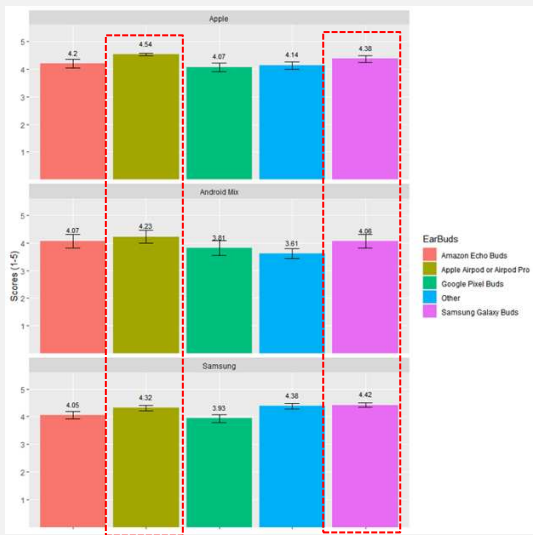
Samsung Smartwatch has a substantially lower adoption for Gen Z users.



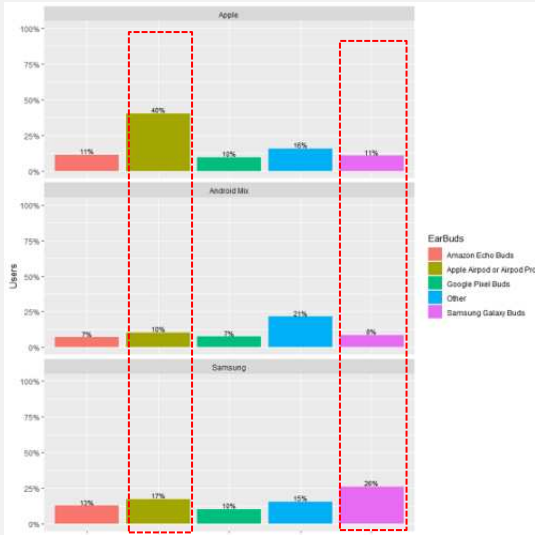
Wireless Earbuds

There is a sizable third-party wireless Earbuds market with quality or value products. Design for Gen Z may be an important area that Samsung need to invest in with a more holistic approach spanning across both hardware and software.

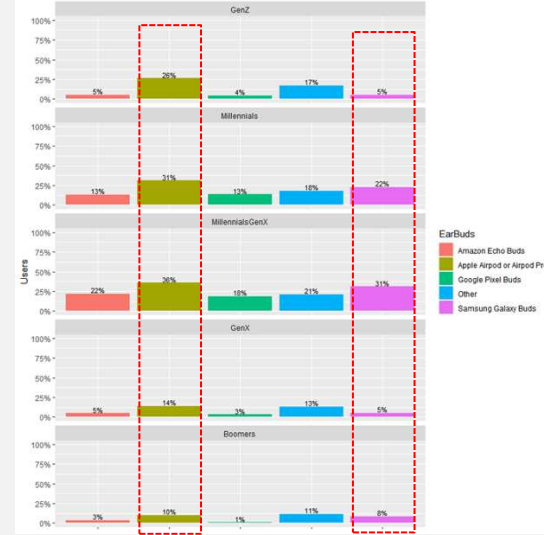
LO Airpod and Samsung Galaxy Buds receive higher CX scores.



LO has higher ownership of wireless earbuds but 10% of LO users use Samsung Galaxy Buds



Gen Z has a substantially low ownership of Galaxy EarBuds compared to LO.

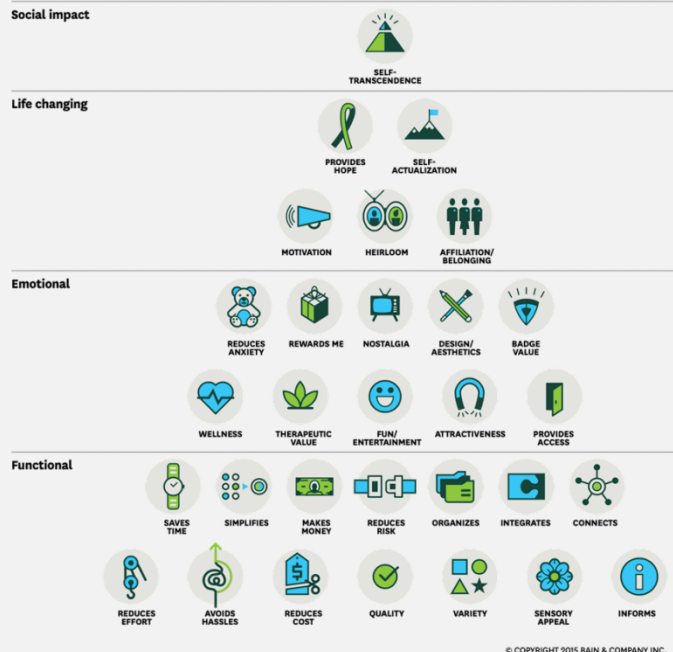


Basic Demographics

Background Information (Phase 1 & 2)

Data reported in this document collected in July & August 2020 for use in Elements of Value Phase 1 & 2 Exploratory Analysis. The online survey has 109 questions and the median time for completion of this survey was about 31 minutes with 1105 were used in this analysis after cleaning up the data set (999 in Phase 1 and 106 in Phase 2)

The purpose of the survey is to develop a comprehensive understanding of the role of elements in the context of smartphones and their relationship to brand loyalty, product/service satisfaction and the customer experience of important apps categories, HW/SW/UI features. The study also provides CX benchmark for selected apps and product categories.



Brand	LO	Android Mix	Samsung
F	286	122	186
M	161	92	258

Background Information

The survey also was originally intended to use user segmentation from HQ Segmentation Model ("Productives", "Expressers", "Social Savvies", "Traditionalists" and "Browsers") as basis of analysis. Due to changes of HQ direction after the survey and the report was completed, the data set was re-analyzed with basic demographic breakouts, product tier (for next upgrade) and reported in this series of data library reports.

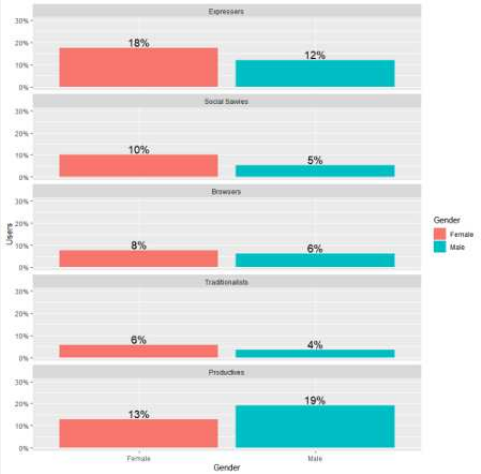
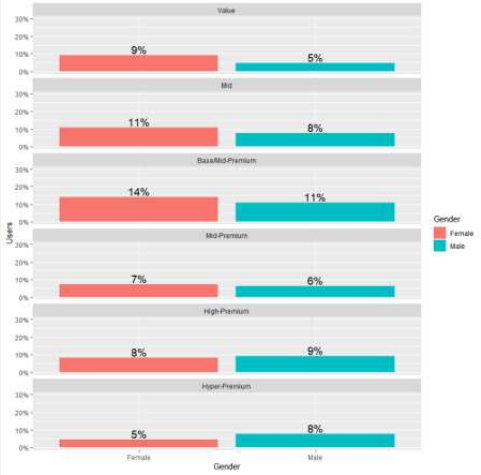
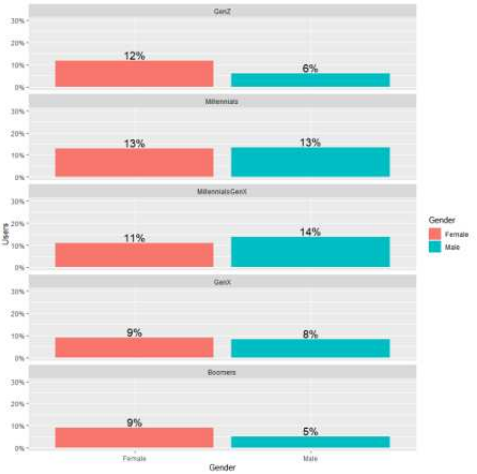
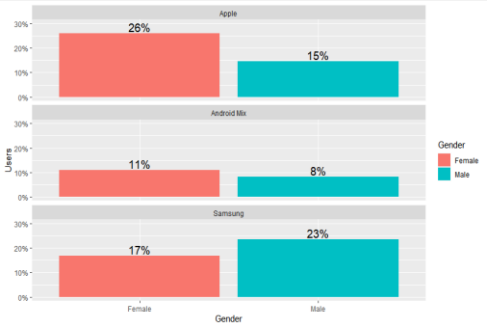
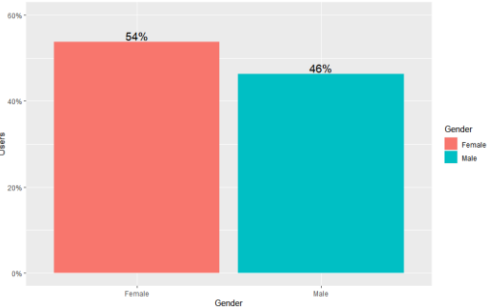
Age	Gen Z (up to 24)	Millennials (25 – 34)	Millennials/GenX (35 – 44)	GenX (45 – 55)	Boomers (55+)
F	130	143	120	100	101
M	65	148	151	91	56

Tier	Value (300-)	Mid (300 -599)	Base/Mid Premium (699-899)	Mid- Premium (900 – 999)	High Premium (1000 - 1249)	Hyper- Premium (1250+)
F	101	121	154	78	90	50
M	52	85	119	69	102	84

Seg	Expressers	Social Savvies	Browsers	Traditionalists	Productives
F	194	112	85	62	141
M	133	59	68	39	212

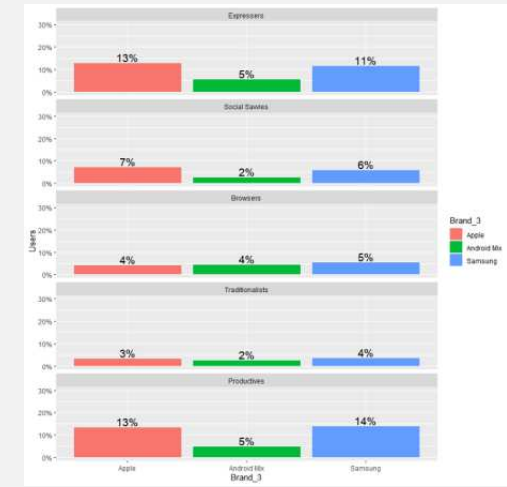
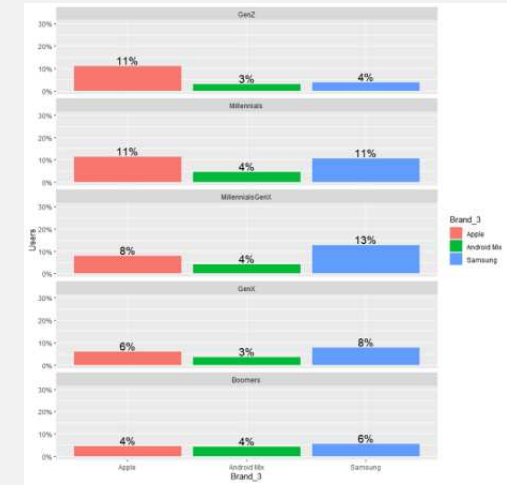
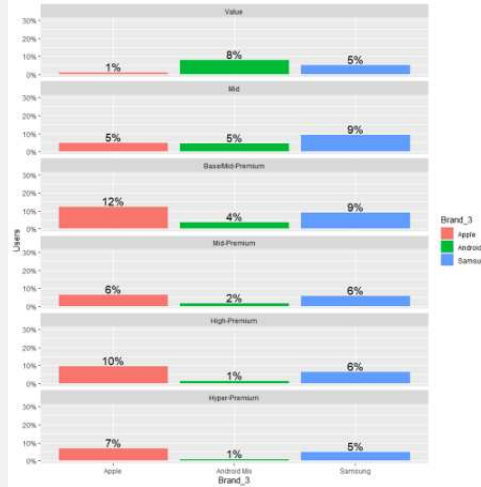
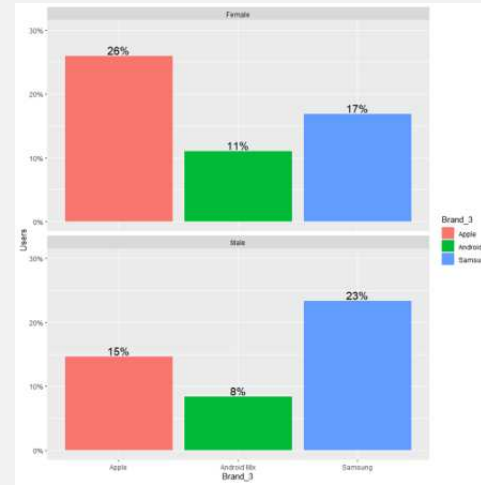
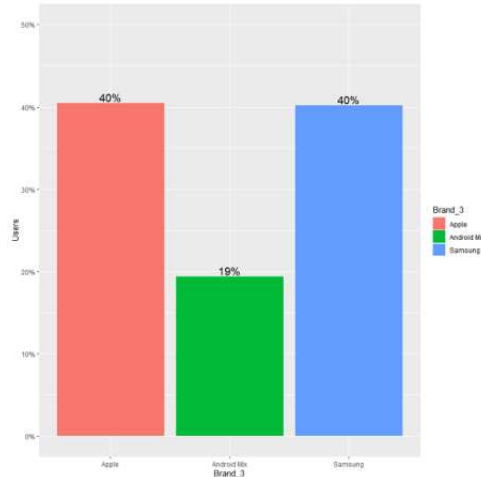
Gender

- Male
- Female



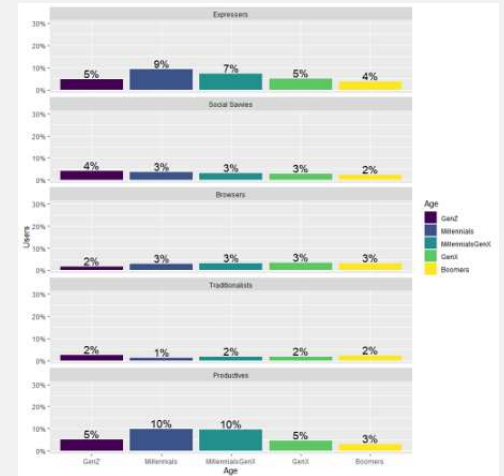
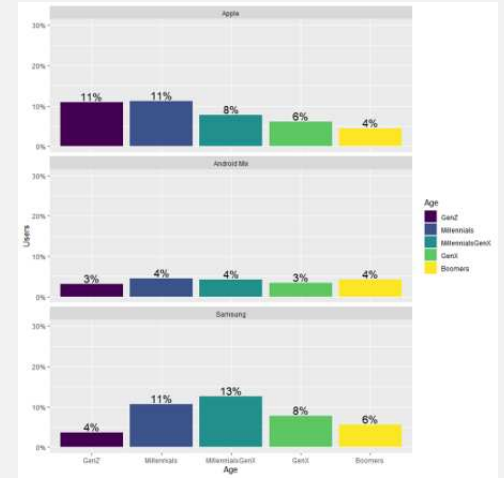
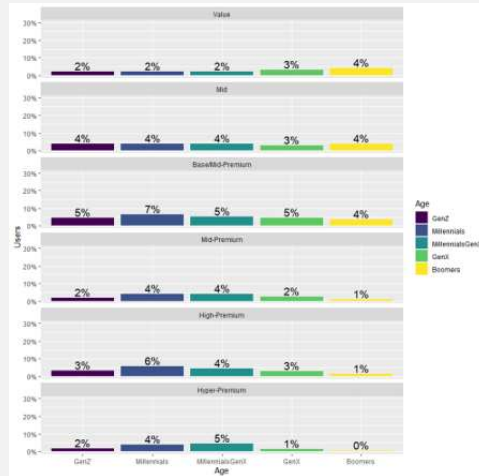
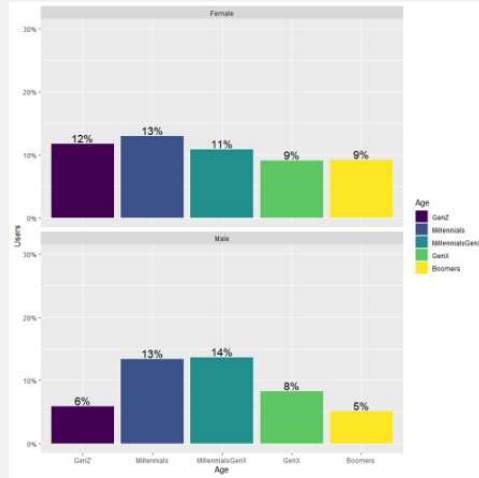
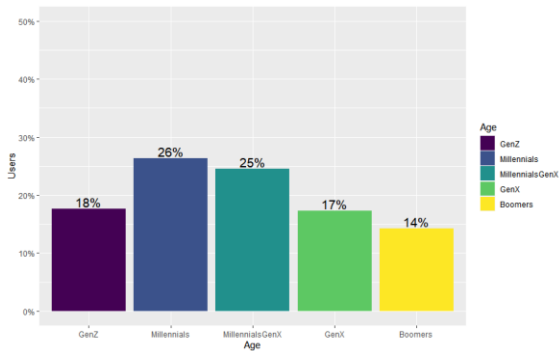
SmartPhone Brand

- Android Mix
- Apple
- Samsung



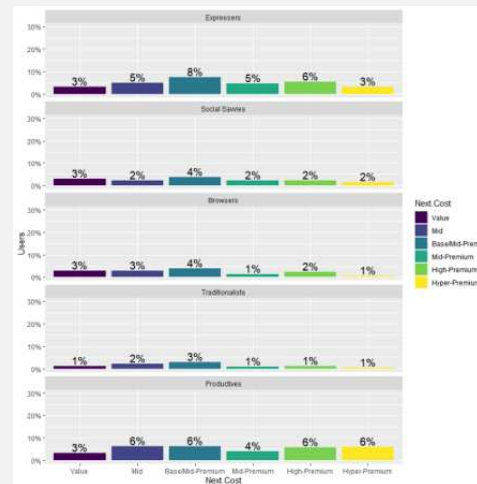
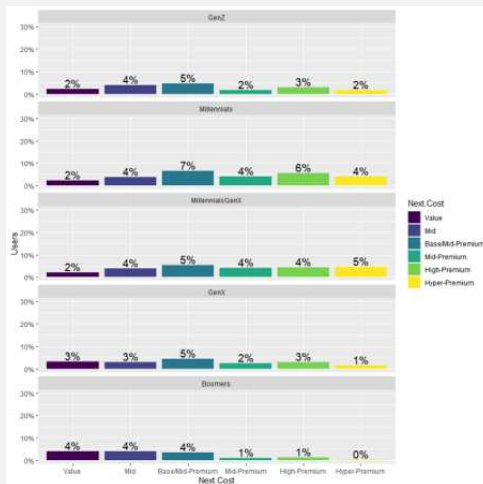
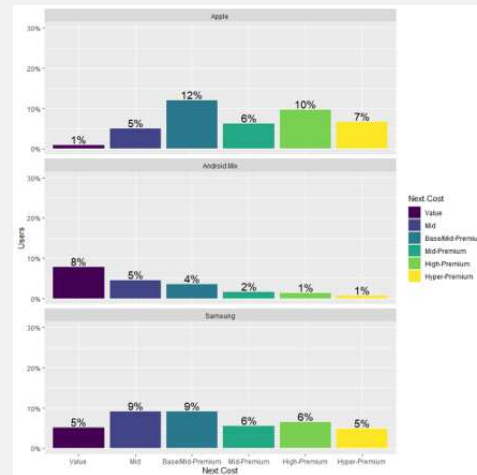
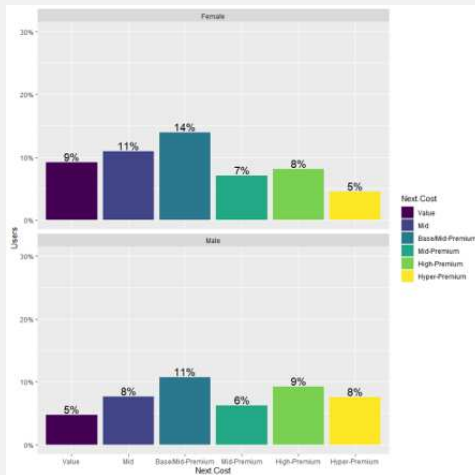
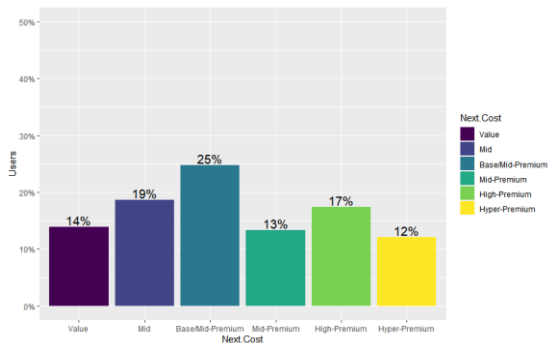
Age

- Up to 24 GenZ
- 25 to 34 Millennials
- 35 to 44 Millennials/GenX
- 45 to 54 GenX
- 55+ Boomers



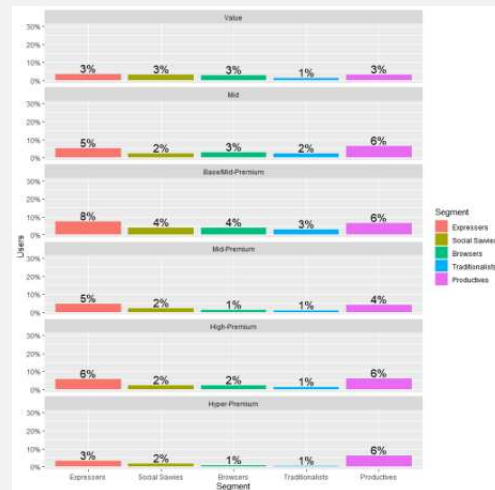
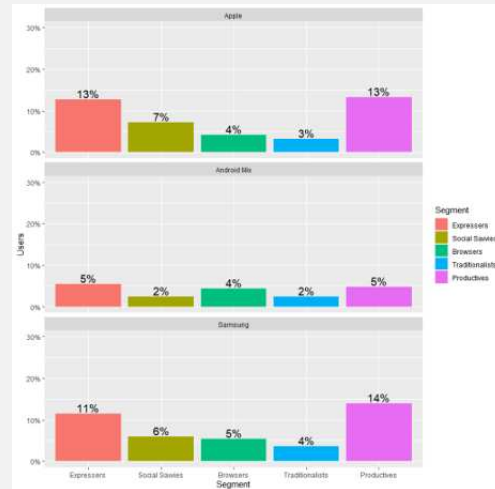
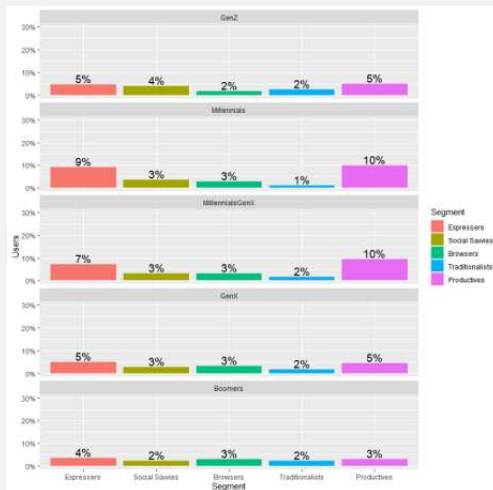
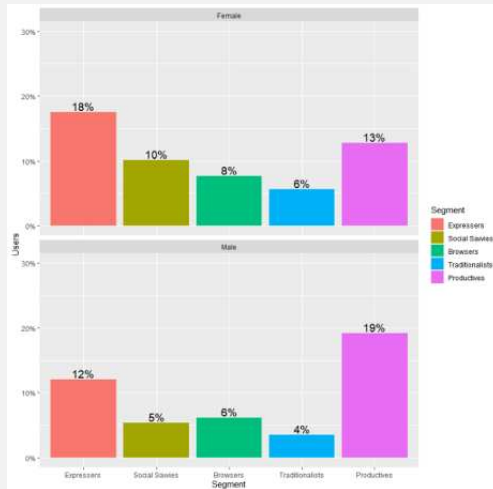
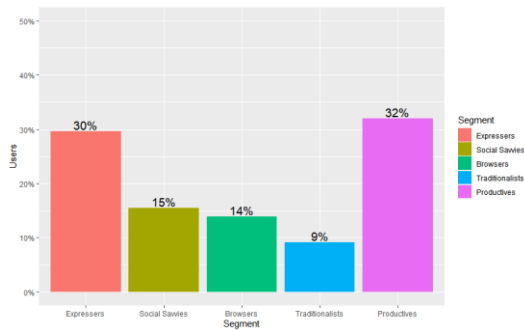
Tier* (next device)

- Value under \$300
- Mid \$300 to \$599
- Premium-Base & Mid \$600 to \$899
- Premium-Mid \$900 to \$999
- Premium-High \$1,000 to \$1,249
- Hyper-Premium \$1250+



User Segment

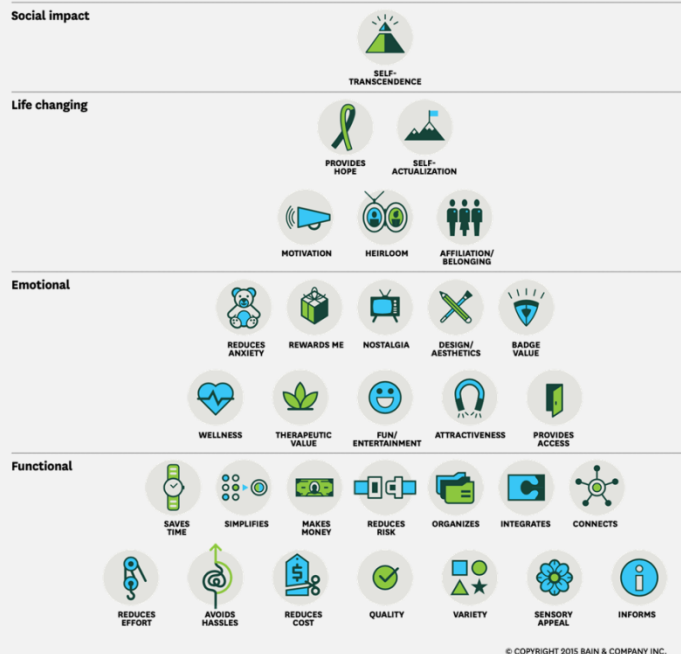
- Productives
- Expressers
- Social Savvies
- Traditionalists
- Browserers



Background Information (Phase 3)

Data reported in Phase 3 were collected in Sept through mid-Oct 2020. The online survey has 109 questions and the median time for completion of this survey was about 37 minutes with 991 were used in this analysis after cleaning up the data set (9 data points with missing gender and segmentation classification)

The purpose of the survey is to test 11 product concepts driven by the results of Phase 1 and 2. A secondary purpose is to explore purchase intent of foldables as well as further explore the relationship of smartphone satisfaction to brand loyalty. This report focuses on the secondary purpose with exploratory analysis.



Brand	LO	Android Mix	Samsung
F	249	99	151
M	276	79	137

Background Information (Phase 3)

The survey also was originally intended to use user segmentation from HQ Segmentation Model ("Productives", "Expressers", "Social Savvies", "Traditionalists" and "Browsers") as basis of analysis. Due to changes of HQ direction after the survey and the report was completed, the data set was re-analyzed with basic demographic breakouts, product tier (for next upgrade) and reported in this series of data library reports. However, the breakout by segments will still be retained because SEA is still using the segmentation scheme for current products (and upcoming products in Q1 2021).

Age	Gen Z (up to 24)	Millennials (25 – 39)	GenX (40– 55)	Boomers (55+)
F	198	166	70	65
M	80	245	136	31

Tier	Value (300-)	Mid (300 -599)	Base/Mid Premium (699-899)	Mid- Premium (900 – 999)	High Premium (1000 - 1249)	Hyper- Premium (1250+)
F	96	119	140	57	54	33
M	48	72	84	75	98	115

Seg	Expressers	Social Savvies	Browsers	Traditionalists	Productives
F	112	61	84	154	88
M	179	38	55	166	54

Appendix

Overall Satisfaction vs. Apps/Ecosystem Categories and Hardware Features

Overall results and interpretations:

The results showed that there are 2 components of the relationship:

The table on the right side shows the "loadings" of the variables contributing to the overall satisfaction. Note that the entry to "Preload" is empty as the effect of preloads on the first component is negligible.

The first component alone can explain about 40% of the total variation of satisfaction. Most of the variables contribute to overall satisfaction

The second component is interesting as there are a number of negative values, implying an "inverse" relationship with total satisfaction (the higher the score, the lower the satisfaction). Note that there are also a number of positive values associated with a number of variables. A possible interpretation of what happens is that for this particular "component". The positive responses can counter the negative experiences.

"Hypothesis"

The chart on the right shows the combined effect of both components (1 and 2) on each of the variable. The green boxes identify those variables contributing most to satisfaction.

The red boxes can be interpreted as current experience with such features may not contribute as much to the overall satisfaction even they are "improved" substantially.

The uncolored boxes can be interpreted as improvement of those features/apps will contribute to satisfaction.

Disclaimer: The results presented here is based on preliminary data analysis and both the interpretations and results are subject to changes. This section is provided for informative purpose and should not be taken for recommendations

Source: Elements of Value Phase 1 & 2 Survey (N=1105)

Features	Overall
ARVR	-0.97
Aspect.Ratio	-0.17
Basic.Communication	4.70
Battery.Capacity	2.15
Buttons	1.27
Camera	7.25
Chassis.Material	1.88
Cloud	-0.01
Color.Choices	3.90
Connectivity.with.PCLaptop..Tablet	-1.50
Consistent.usability	6.67
Display.Color	4.61
EarbudsHeadphones	-0.65
Fast.Charging	1.90
General.User.Interface	3.30
HealthFitness	-1.54
Internet.Browsing	3.04
Look.and.Feel	4.94
Mobile.Wallet..Pay	-0.91
Navigation.Maps	3.13
one.hand.use	-1.50
Packaging	2.92
Personalization	-0.01
Playing.Games	1.18
Preloads	-0.67
Productivity.tools	-0.57
Refreshrate	3.15
Responsiveness	4.23
Security	4.64
Shape	1.30
Shopping..Coupon.Apps	-0.93
Size	3.25
Smart.Home.Control	-0.73
Smart.Watches..Fitness.Bands	-1.41
Social.Media	3.15
Storage.Capacity	7.98
Streaming.Music	-1.66
Streaming.Video	0.79
Taking.Pictures.or.Videos	4.52
Tracking	-0.47
Voice.Assistant	-0.19
Waterproof..Dust.proof	0.23
Wireless.Charging	0.04

Data: X dimension: 1105 43

Y dimension: 1105 1

Fit method: kernelppls

Number of components considered: 2

TRAINING: % variance explained

X 1 comps 2 comps

Current.Satisfaction 46.81 50.74

NULL 38.67 42.43

Loadings:

	Comp 1	Comp 2
Basic.Communication.rate	0.165	
Taking.Pictures.or.Videos.rate	0.163	0.162
Streaming.Video.rate	0.165	
Streaming.Music.rate	0.154	-0.181
Playing.Games.rate	0.148	-0.116
Productivity.tools.rate	0.153	-0.212
HealthFitness.rate	0.141	-0.303
Smart.Home.Control.rate	0.137	-0.236
Internet.Browsing.rate	0.161	
Shopping..Coupon.Apps.rate	0.138	-0.250
Mobile.Wallet..Pay.rate	0.135	-0.270
Voice.Assistant.rate	0.142	-0.206
Navigation.Maps.rate	0.154	
Tracking.rate	0.147	-0.220
Security.rate	0.157	
Social.Media.rate	0.162	
Preloads.rate	0.103	-0.276
Connectivity.with.PCLaptop..Tablet.rate	0.145	-0.231
EarbudsHeadphones.rate	0.144	-0.145
Smart.Watches..Fitness.Bands.rate	0.138	-0.321
ARVR.rate	0.113	-0.310
Cloud.rate	0.142	-0.118
Personalization.rate	0.157	-0.126
Size.rate	0.161	0.165
Shape.rate	0.156	
Buttons.rate	0.164	
Chassis.Material.rate	0.156	
Color.Choices.rate	0.150	0.138
Look.and.Feel.rate	0.175	0.230
Aspect.Ratio.rate	0.166	
Display.Color.rate	0.172	0.225
Refreshrate.rate	0.176	0.132
Camera.rate	0.165	0.303
Battery.Capacity.rate	0.142	0.121
Waterproof..Dust.proof.rate	0.122	
Wireless.Charging.rate	0.141	
Fast.Charging.rate	0.148	0.129
Responsiveness.rate	0.168	0.189
Storage.Capacity.rate	0.153	0.259
General.User.Interface.rate	0.171	0.114
one.hand.use.rate	0.148	
Consistent.usability.rate	0.171	0.237
Packaging.rate	0.174	0.112

SS loadings Comp 1 Comp 2

Proportion Var 0.023 0.032

Cumulative Var 0.023 0.055

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The results suggest that both females and females have different underlying “latent” trends.

A cursory review of their model parameters confirmed that they have 2 different components.

Features	Female	Male
ARVR	-2.98	2.32
Aspect.Ratio	-1.27	1.30
Basic.Communication	3.84	4.34
Battery.Capacity	3.73	-1.99
Buttons	0.30	1.77
Camera	7.82	3.24
Chassis.Material	1.74	1.55
Cloud	1.53	-3.30
Color.Choices	3.40	2.67
Connectivity.with.PCLaptop..Tablet	-2.66	0.89
Consistent.usability	6.38	4.45
Display.Color	2.40	6.24
Earbuds.Headphones	-1.14	-0.06
Fast.Charging	4.07	-2.73
General.User.Interface	4.13	0.49
Health.Fitness	-2.42	0.24
Internet.Browsing	3.38	0.60
Look.and.Feel	2.52	6.94
Mobile.Wallet..Pay	0.36	-2.81
Navigation.Maps	1.58	4.06
one.hand.use	-0.82	-2.41
Packaging	1.06	4.27
Personalization	-0.35	0.10
Playing.Games	1.06	1.03
Preloads	1.16	-3.37
Productivity.tools	-1.00	0.18
Refreshrate	3.42	0.73
Responsiveness	4.13	2.73
Security	2.64	5.97
Shape	0.51	2.03
Shopping.Coupon.Apps	1.07	-4.44
Size	2.36	3.25
Smart.Home.Control	-0.27	-1.48
Smart.Watches..Fitness.Bands	-2.43	0.53
Social.Media	2.81	1.95
Storage.Capacity	7.51	5.38
Streaming.Music	-1.89	-1.18
Streaming.Video	0.65	0.53
Taking.Pictures.or.Videos	4.91	1.69
Tracking	-0.59	-0.58
Voice.Assistant	0.96	-2.47
Waterproof..Dust.proof	-0.29	0.75
Wireless.Charging	0.45	-0.96

[illegible]

Disclaimer: The results presented here is based on preliminary data analysis and both the interpretations and results ae subject to changes. This section is provided for informative purpose and should not be taken for recommendations

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The results suggest that LO users have a much more even distribution of satisfaction drivers compared to Android and Samsung smartphone owners.

A cursory review of their model parameters confirmed that both Android and Samsung owners have 2 different components (trends) compared to 1 component for LO.

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The results suggest that Gen X have different underlying trends. Again, a detailed review of the underlying model identified the trend.

Besides, specific marketing messages can also be crafted to specific groups with “green boxes” to maximize impact.

[illegible]

Gen Z

Millennials

Millennials / Gen Z

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 50.08
NMI

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 43.80
NMI

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 50.08
NMI

Loadings

Basic Communication rate	0.379
Taking Pictures on Videos rate	0.386
Streaming Videos rate	0.386
Streaming Music rate	0.386
Playing Games rate	0.386
Productivity Apps rate	0.386
Health/fitness rate	0.386
Smart Home Control rate	0.386
Internet Browsing rate	0.386
Shopping, Clothing Apps rate	0.386
Mobile Assist. Pay rate	0.386
Voice Assistant rate	0.386
Navigation Apps rate	0.386
Tracking rate	0.386
Security rate	0.386
Social Media rate	0.386
Connectivity with PC/Laptop, Tablet rate	0.386
Fast Charging rate	0.386
Smart watches, fitness Bands rate	0.386
ADBB rate	0.386
Cloud rate	0.386
Personalization rate	0.386
Size rate	0.386
Buttons rate	0.386
Chemical Material rate	0.386
Color Choices rate	0.386
Look and Feel rate	0.386
Aspect Ratio rate	0.386
Display Color rate	0.386
Refresh rate	0.386
Battery Capacity rate	0.386
Wireless charging rate	0.386
Storage Capacity rate	0.386
Response time rate	0.386
Storage Capacity rate	0.386
General User Interface rate	0.386
One-hand-use rate	0.386
Consistent usability rate	0.386
Packaging rate	0.386

SS Storages 0.800
Propriety Var = 0.823

Loadings

Basic Communication rate	0.386
Taking Pictures on Videos rate	0.386
Streaming Videos rate	0.379
Streaming Music rate	0.379
Playing Games rate	0.386
Productivity Apps rate	0.386
Health/fitness rate	0.386
Smart Home Control rate	0.386
Internet Browsing rate	0.386
Shopping, Clothing Apps rate	0.386
Mobile Assist. Pay rate	0.386
Voice Assistant rate	0.386
Navigation Apps rate	0.386
Tracking rate	0.386
Security rate	0.386
Previews rate	0.386
Connectivity with PC/Laptop, Tablet rate	0.386
Fast Charging rate	0.386
Smart watches, fitness Bands rate	0.386
ADBB rate	0.386
Cloud rate	0.386
Personalization rate	0.386
Size rate	0.386
Buttons rate	0.386
Chemical Material rate	0.386
Color Choices rate	0.386
Look and Feel rate	0.386
Aspect Ratio rate	0.386
Display Color rate	0.386
Refresh rate	0.386
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Wireless charging rate	0.386
Storage Capacity rate	0.386
Response time rate	0.386
Storage Capacity rate	0.386
General User Interface rate	0.386
One-hand-use rate	0.386
Consistent usability rate	0.386
Packaging rate	0.386

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Propriety Var = 0.823

Loadings

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Smart Home Control rate	0.385
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Voice Assistant rate	0.385
Navigation Apps rate	0.385
Tracking rate	0.385
Security rate	0.385
Social Media rate	0.385
Connectivity with PC/Laptop, Tablet rate	0.385
Fast Charging rate	0.385
Smart watches, fitness Bands rate	0.385
ADBB rate	0.385
Cloud rate	0.385
Personalization rate	0.385
Size rate	0.385
Buttons rate	0.385
Chemical Material rate	0.385
Color Choices rate	0.385
Look and Feel rate	0.385
Aspect Ratio rate	0.385
Display Color rate	0.385
Refresh rate	0.385
Battery Capacity rate	0.385
Wireless charging rate	0.385
Storage Capacity rate	0.385
Response time rate	0.385
Storage Capacity rate	0.385
General User Interface rate	0.385
One-hand-use rate	0.385
Consistent usability rate	0.385
Packaging rate	0.385

SS Storages 0.800
Propriety Var = 0.824

Gen X

Boomers

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 50.08
NMI

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 43.81
NMI

Data: x dimension 105.43
y dimension 105.1
Number of components considered: 1
TRAINING: x variance explained 5.70ms
y 48.05ms
Current Satisfaction 50.08
NMI

Loadings

Basic Communication rate	0.386
Taking Pictures on Videos rate	0.386
Streaming Videos rate	0.386
Streaming Music rate	0.386
Playing Games rate	0.386
Productivity Apps rate	0.386
Health/fitness rate	0.386
Smart Home Control rate	0.386
Internet Browsing rate	0.386
Shopping, Clothing Apps rate	0.386
Mobile Assist. Pay rate	0.386
Voice Assistant rate	0.386
Navigation Apps rate	0.386
Tracking rate	0.386
Security rate	0.386
Social Media rate	0.386
Connectivity with PC/Laptop, Tablet rate	0.386
Fast Charging rate	0.386
Smart watches, fitness Bands rate	0.386
ADBB rate	0.386
Cloud rate	0.386
Personalization rate	0.386
Size rate	0.386
Buttons rate	0.386
Chemical Material rate	0.386
Color Choices rate	0.386
Look and Feel rate	0.386
Aspect Ratio rate	0.386
Display Color rate	0.386
Refresh rate	0.386
Battery Capacity rate	0.386
Wireless charging rate	0.386
Storage Capacity rate	0.386
Response time rate	0.386
Storage Capacity rate	0.386
General User Interface rate	0.386
One-hand-use rate	0.386
Consistent usability rate	0.386
Packaging rate	0.386

SS Storages 0.800
Propriety Var = 0.823

Loadings

Basic Communication rate	0.386
Taking Pictures on Videos rate	0.386
Streaming Videos rate	0.386
Streaming Music rate	0.386
Playing Games rate	0.386
Productivity Apps rate	0.386
Health/fitness rate	0.386
Smart Home Control rate	0.386
Internet Browsing rate	0.386
Shopping, Clothing Apps rate	0.386
Mobile Assist. Pay rate	0.386
Voice Assistant rate	0.386
Navigation Apps rate	0.386
Tracking rate	0.386
Security rate	0.386
Previews rate	0.386
Connectivity with PC/Laptop, Tablet rate	0.386
Fast Charging rate	0.386
Smart watches, fitness Bands rate	0.386
ADBB rate	0.386
Cloud rate	0.386
Personalization rate	0.386
Size rate	0.386
Buttons rate	0.386
Chemical Material rate	0.386
Color Choices rate	0.386
Look and Feel rate	0.386
Aspect Ratio rate	0.386
Display Color rate	0.386
Refresh rate	0.386
Battery Capacity rate	0.386
Wireless charging rate	0.386
Storage Capacity rate	0.386
Response time rate	0.386
Storage Capacity rate	0.386
General User Interface rate	0.386
One-hand-use rate	0.386
Consistent usability rate	0.386
Packaging rate	0.386

SS Storages 0.800
Propriety Var = 0.823

Loadings

Basic Communication rate	0.386
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Smart Home Control rate	0.385
Internet Browsing rate	0.385
Shopping, Clothing Apps rate	0.385
Mobile Assist. Pay rate	0.385
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Navigation Apps rate	0.385
Tracking rate	0.385
Security rate	0.385
Social Media rate	0.385
Connectivity with PC/Laptop, Tablet rate	0.385
Fast Charging rate	0.385
Smart watches, fitness Bands rate	0.385
ADBB rate	0.385
Cloud rate	0.385
Personalization rate	0.385
Size rate	0.385
Buttons rate	0.385
Chemical Material rate	0.385
Color Choices rate	0.385
Look and Feel rate	0.385
Aspect Ratio rate	0.385
Display Color rate	0.385
Refresh rate	0.385
Battery Capacity rate	0.385
Wireless charging rate	0.385
Storage Capacity rate	0.385
Response time rate	0.385
Storage Capacity rate	0.385
General User Interface rate	0.385
One-hand-use rate	0.385
Consistent usability rate	0.385
Packaging rate	0.385

SS Storages 0.800
Propriety Var = 0.824

Overall Satisfaction vs. Apps/Ecosystem Categories and Hardware Features by Tier

Overall results and interpretations:

The results suggest that we will need to look more carefully into the feature/apps trade-off with mid tier and hyper-premium users

A cursory review of their model parameters confirmed that as males have 2 different components (trends) .

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Features	Value						Mid						Premium - Base/Mid					
	Value	Mid	Base/Mid	Premium	Premium	Premium	Value	Mid	Base/Mid	Premium	Premium	Premium	Value	Mid	Base/Mid	Premium	Premium	Premium
ARVR	1.52	-0.99	1.04	1.92	1.42	6.05	1.52	-0.99	1.04	1.92	1.42	6.05	1.52	-0.99	1.04	1.92	1.42	6.05
Aspect.Ratio	2.24	-0.11	2.51	2.58	2.66	-2.29	2.24	-0.11	2.51	2.58	2.66	-2.29	2.24	-0.11	2.51	2.58	2.66	-2.29
Basic.Communication	2.76	-1.41	3.49	2.68	2.56	-5.81	2.76	-1.41	3.49	2.68	2.56	-5.81	2.76	-1.41	3.49	2.68	2.56	-5.81
Battery.Capacity	2.36	2.18	1.72	2.68	2.23	-2.68	2.36	2.18	1.72	2.68	2.23	-2.68	2.36	2.18	1.72	2.68	2.23	-2.68
Buttons	1.93	2.96	2.99	2.29	2.87	-2.41	1.93	2.96	2.99	2.29	2.87	-2.41	1.93	2.96	2.99	2.29	2.87	-2.41
Camera	2.55	5.18	3.07	3.06	2.90	3.00	2.55	5.18	3.07	3.06	2.90	3.00	2.55	5.18	3.07	3.06	2.90	3.00
Chassis.Material	2.15	3.61	2.33	2.27	2.68	-1.35	2.15	3.61	2.33	2.27	2.68	-1.35	2.15	3.61	2.33	2.27	2.68	-1.35
Cloud	1.94	5.20	1.64	2.17	1.47	-3.19	1.94	5.20	1.64	2.17	1.47	-3.19	1.94	5.20	1.64	2.17	1.47	-3.19
Color.Choices	2.08	4.64	2.61	2.51	2.37	1.26	2.08	4.64	2.61	2.51	2.37	1.26	2.08	4.64	2.61	2.51	2.37	1.26
Connectivity.with.PCLaptop..Tablet	2.09	-1.11	1.84	2.25	1.70	-1.50	2.09	-1.11	1.84	2.25	1.70	-1.50	2.09	-1.11	1.84	2.25	1.70	-1.50
Consistent.usability	2.81	6.93	2.95	2.80	2.87	3.78	2.81	6.93	2.95	2.80	2.87	3.78	2.81	6.93	2.95	2.80	2.87	3.78
Display.Color	3.02	3.88	2.64	2.35	2.95	0.62	3.02	3.88	2.64	2.35	2.95	0.62	3.02	3.88	2.64	2.35	2.95	0.62
EarbudsHeadphones	2.35	-2.82	2.03	2.23	1.92	2.43	2.35	-2.82	2.03	2.23	1.92	2.43	2.35	-2.82	2.03	2.23	1.92	2.43
Fast.Charging	2.24	2.44	1.92	2.20	2.23	-1.41	2.24	2.44	1.92	2.20	2.23	-1.41	2.24	2.44	1.92	2.20	2.23	-1.41
General.User.Interface	2.94	1.32	2.60	2.35	3.43	1.23	2.94	1.32	2.60	2.35	3.43	1.23	2.94	1.32	2.60	2.35	3.43	1.23
HealthFitness	2.46	-0.24	1.27	2.09	1.77	-1.20	2.46	-0.24	1.27	2.09	1.77	-1.20	2.46	-0.24	1.27	2.09	1.77	-1.20
Internet.Browsing	2.34	2.61	3.01	2.65	1.64	2.61	2.34	2.61	3.01	2.65	1.64	2.61	2.34	2.61	3.01	2.65	1.64	2.61
Look.and.Feel	2.83	3.18	2.68	3.14	3.42	1.69	2.83	3.18	2.68	3.14	3.42	1.69	2.83	3.18	2.68	3.14	3.42	1.69
Mobile.Wallet..Pay	2.14	-3.85	1.86	1.82	2.33	2.17	2.14	-3.85	1.86	1.82	2.33	2.17	2.14	-3.85	1.86	1.82	2.33	2.17
Navigation.Maps	2.40	2.47	2.26	2.35	2.83	4.65	2.40	2.47	2.26	2.35	2.83	4.65	2.40	2.47	2.26	2.35	2.83	4.65
one.hand.use	2.16	-0.82	2.09	2.03	2.08	-2.63	2.16	-0.82	2.09	2.03	2.08	-2.63	2.16	-0.82	2.09	2.03	2.08	-2.63
Packaging	2.68	1.61	2.57	2.87	2.48	3.29	2.68	1.61	2.57	2.87	2.48	3.29	2.68	1.61	2.57	2.87	2.48	3.29
Personalization	2.37	0.19	2.53	1.65	1.81	2.87	2.37	0.19	2.53	1.65	1.81	2.87	2.37	0.19	2.53	1.65	1.81	2.87
Playing.Games	2.15	-0.59	2.06	2.76	2.61	0.94	2.15	-0.59	2.06	2.76	2.61	0.94	2.15	-0.59	2.06	2.76	2.61	0.94
Preloads	1.25	1.05	1.97	1.25	0.83	-2.00	1.25	1.05	1.97	1.25	0.83	-2.00	1.25	1.05	1.97	1.25	0.83	-2.00
Productivity.tools	2.00	-0.91	2.40	2.18	2.17	2.19	2.00	-0.91	2.40	2.18	2.17	2.19	2.00	-0.91	2.40	2.18	2.17	2.19
Refreshrate	2.78	4.45	2.69	2.86	2.69	-0.14	2.78	4.45	2.69	2.86	2.69	-0.14	2.78	4.45	2.69	2.86	2.69	-0.14
Responsiveness	2.78	3.01	3.02	2.45	2.90	1.40	2.78	3.01	3.02	2.45	2.90	1.40	2.78	3.01	3.02	2.45	2.90	1.40
Security	2.74	0.60	3.09	2.73	2.13	3.98	2.74	0.60	3.09	2.73	2.13	3.98	2.74	0.60	3.09	2.73	2.13	3.98
Shape	2.42	-3.18	3.15	2.72	2.83	-0.76	2.42	-3.18	3.15	2.72	2.83	-0.76	2.42	-3.18	3.15	2.72	2.83	-0.76
Shopping..Coupon.Apps	1.84	-2.58	2.16	2.33	1.58	0.20	1.84	-2.58	2.16	2.33	1.58	0.20	1.84	-2.58	2.16	2.33	1.58	0.20
Size	2.59	1.33	2.38	2.94	2.30	1.57	2.59	1.33	2.38	2.94	2.30	1.57	2.59	1.33	2.38	2.94	2.30	1.57
Smart.Home.Control	1.89	-1.10	2.33	1.08	2.62	0.70	1.89	-1.10	2.33	1.08	2.62	0.70	1.89	-1.10	2.33	1.08	2.62	0.70
Smart.Watches..Fitness.Bands	1.34	0.58	2.11	1.29	2.54	0.15	1.34	0.58	2.11	1.29	2.54	0.15	1.34	0.58	2.11	1.29	2.54	0.15
Social.Media	3.31	-0.22	2.37	2.34	2.60	-0.02	3.31	-0.22	2.37	2.34	2.60	-0.02	3.31	-0.22	2.37	2.34	2.60	-0.02
Storage.Capacity	2.56	10.31	2.42	2.42	2.02	3.25	2.56	10.31	2.42	2.42	2.02	3.25	2.56	10.31	2.42	2.42	2.02	3.25
Streaming.Music	2.30	-4.39	2.25	2.37	1.89	2.89	2.30	-4.39	2.25	2.37	1.89	2.89	2.30	-4.39	2.25	2.37	1.89	2.89
Streaming.Video	2.37	-0.71	2.27	2.69	2.65	3.21	2.37	-0.71	2.27	2.69	2.65	3.21	2.37	-0.71	2.27	2.69	2.65	3.21
Taking.Pictures.or.Videos	2.55	1.27	2.60	3.16	2.50	7.16	2.55	1.27	2.60	3.16	2.50	7.16	2.55	1.27	2.60	3.16	2.50	7.16
Tracking	2.10	0.87	2.03	2.09	2.15	0.49	2.10	0.87	2.03	2.09	2.15	0.49	2.10	0.87	2.03	2.09	2.15	0.49
Voice.Assistant	1.91	0.95	1.92	1.88	1.99	3.43	1.91	0.95	1.92	1.88	1.99	3.43	1.91	0.95	1.92	1.88	1.99	3.43
Waterproof..Dust.proof	2.13	0.45	1.16	2.16	1.68	-2.89	2.13	0.45	1.16	2.16	1.68	-2.89	2.13	0.45	1.16	2.16	1.68	-2.89
Wireless.Charging	2.61	-1.69	1.99	1.36	2.72	-2.46	2.61	-1.69	1.99	1.36	2.72	-2.46	2.61	-1.69	1.99	1.36	2.72	-2.46

Overall Satisfaction vs. Apps/Ecosystem Categories and Hardware Features by Segment

Overall results and interpretations:

The results suggest that we will need to look more carefully into the feature/apps trade-off with traditionalists users

A cursory review of their model parameters confirmed that as males have 2 different components (trends) .

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Productives						Expressers						SocialSavvies					
Features	Productives	Expressers	SocialSavvies	Traditionalists	Browsers												
ARVR	2.11	1.56	1.80	-4.05	0.70												
Aspect.Ratio	2.24	2.48	2.14	0.54	2.51												
Basic.Communication	2.72	2.54	2.75	4.96	2.29												
Battery.Capacity	1.98	2.75	1.82	1.26	2.23												
Buttons	2.39	2.49	2.45	2.09	2.29												
Camera	2.71	3.25	2.62	5.18	2.89												
Chassis.Material	2.54	2.69	1.73	-0.06	2.37												
Cloud	2.05	1.91	2.11	1.21	2.27												
Color.Choices	2.34	2.76	1.74	3.20	2.99												
Connectivity.with.PCLaptop.Tablet	2.08	1.89	1.91	-2.53	2.44												
Consistent.usability	2.81	2.83	3.26	4.19	2.86												
Display.Color	2.90	2.49	2.75	2.63	3.00												
Earbuds.Headphones	2.31	2.13	1.43	-1.84	2.10												
Fast.Charging	2.13	2.73	1.75	-0.81	2.80												
General.User.Interface	2.36	2.91	3.06	-0.53	2.97												
HealthFitness	2.12	1.60	1.97	0.39	1.61												
Internet.Browsing	2.22	2.51	3.10	4.76	2.61												
Look.and.Feel	2.81	2.76	2.76	2.41	3.22												
Mobile.Wallet.Pay	2.09	1.06	2.59	-1.21	1.96												
Navigation.Maps	2.68	2.23	2.71	2.90	2.06												
one.hand.use	1.98	2.42	1.49	-2.36	2.55												
Packaging	2.90	2.68	2.32	3.39	2.38												
Personalization	2.17	2.11	2.38	5.19	1.96												
Playing.Games	2.16	2.28	2.46	-2.61	2.53												
Preloads	1.42	1.73	1.88	-0.86	1.14												
Productivity.tools	2.38	1.83	2.61	0.51	1.65												
Refreshrate	2.53	2.81	2.80	2.05	3.03												
Responsiveness	2.33	2.87	2.80	1.17	3.92												
Security	2.69	2.49	2.63	2.75	2.51												
Shape	2.52	2.58	2.41	-3.66	2.57												
Shopping.CouponApps	1.92	1.49	2.69	-0.96	3.11												
Size	2.57	2.72	2.06	0.14	2.00												
Smart.Home.Control	2.04	1.92	2.01	0.73	1.39												
Smart.Watches.Fitness.Bands	2.55	1.76	2.08	-4.29	2.29												
Social.Media	2.30	2.66	2.68	2.16	2.72												
Storage.Capacity	2.57	3.16	2.41	5.49	2.99												
Streaming.Video	2.04	1.88	2.48	-0.57	1.90												
Streaming.Vision	2.34	2.48	2.45	-0.20	2.61												
Taking.Pictures.or.Videos	2.46	2.47	2.87	5.80	2.53												
Tracking	2.07	2.05	2.05	-1.43	2.36												
Voice.Assistant	2.24	2.05	2.53	-2.79	1.66												
Waterproof.Dust.proof	2.05	1.99	1.17	2.00	1.50												
Wireless.Charging	2.16	2.00	2.28	-2.64	1.98												

Data: 4 dimension: 111.43 File format: xlsx Number of comments considered: 1 TASDMS: 0 variance explained X: 43.90 Y: 43.90 Current.Satisfaction: 34.35 Null:						Data: 4 dimension: 111.43 File format: xlsx Number of comments considered: 1 TASDMS: 0 variance explained X: 43.90 Y: 43.90 Current.Satisfaction: 34.35 Null:						Data: 4 dimension: 111.43 File format: xlsx Number of comments considered: 1 TASDMS: 0 variance explained X: 43.90 Y: 43.90 Current.Satisfaction: 34.35 Null:					
Loadings:						Loadings:						Loadings:					
Basic.Communication.rate						Basic.Communication.rate						Basic.Communication.rate					
Taking.Pictures.or.Videos.rate						Taking.Pictures.or.Videos.rate						Taking.Pictures.or.Videos.rate					
Streaming.Video.rate						Streaming.Video.rate						Streaming.Video.rate					
Streaming.Music.rate						Streaming.Music.rate						Streaming.Music.rate					
Productivity.Tools.rate						Productivity.Tools.rate						Productivity.Tools.rate					
Smart.Home.Control.rate						Smart.Home.Control.rate						Smart.Home.Control.rate					
Internet.Browsing.rate						Internet.Browsing.rate						Internet.Browsing.rate					
Shopping.Coupon.Apps.rate						Shopping.Coupon.Apps.rate						Shopping.Coupon.Apps.rate					
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Voice.Assistant.rate						Voice.Assistant.rate						Voice.Assistant.rate					
Navigation.Maps.rate						Navigation.Maps.rate						Navigation.Maps.rate					
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Please report data or chart errors and direct questions, comments, feedback or analysis request to

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shung@sea.samsung.com

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